

**UpToDate®**

Official reprint from UpToDate®

www.uptodate.com © 2026 UpToDate, Inc. and/or its affiliates. All Rights Reserved.

Vancomycin: Drug information

2026© UpToDate, Inc. and its affiliates and/or licensors. All Rights Reserved.

Contributor Disclosures

For additional information see "[Vancomycin: Patient drug information](#)" and "[Vancomycin: Pediatric drug information](#)"

For abbreviations, symbols, and age group definitions [show table](#)

ALERT: US Boxed Warning

Risk of embryo-fetal toxicity due to excipients:



Formulations of vancomycin injection contain the excipients polyethylene glycol (PEG 400) and N-acetyl D-alanine (NADA), which have resulted in fetal malformations in animal reproduction studies at dose exposures approximately 8 and 32 times, respectively, higher than the exposures at the human equivalent dose. If use of vancomycin is needed during the first or second trimester of pregnancy, use other available formulations of vancomycin.

Brand Names: US

Firvanq; Tyzavan; Vancocin

Brand Names: Canada

JAMP Vancomycin; JAMP-Vancomycin; PMS-Vancomycin; PMSC-Vancomycin [DSC]; Vancocin; Vancomycin HCl

Pharmacologic Category

Glycopeptide

Dosing: Adult

Dosage guidance:

Safety: Risk of toxicity (eg, acute kidney injury) increases as a function of trough concentration, especially when trough is maintained above 15 to 20 mg/L; recent

data suggest risk increases along the vancomycin AUC continuum, especially when daily AUC exceeds 650 to 1,300 mg•h/L ([Ref](#)).

Dosing: Initial IV dosing in patients without obesity should be based on actual body weight; subsequent dosing should generally be adjusted based on therapeutic monitoring.

Clinical considerations: The oral route is ineffective for treating systemic infections; the IV route is ineffective for treating *C. difficile* infection. Trough monitoring has traditionally been used for therapeutic monitoring; however, for serious methicillin-resistant *S. aureus* (MRSA) infections (eg, bacteremia, infective endocarditis, meningitis, osteomyelitis, pneumonia, sepsis), AUC monitoring is preferred ([Ref](#)). For patients with uncomplicated skin and soft tissue infections who are not obese and have normal renal function, therapeutic monitoring is generally not needed ([Ref](#)).

Usual dosage range:

Oral: 125 to 500 mg 4 times daily.

IV:

Intermittent infusion: 15 to 20 mg/kg/dose (rounded to the nearest 250 mg) every 8 to 12 hours initially; for serious MRSA infections (eg, bacteremia, infective endocarditis, meningitis, osteomyelitis, pneumonia, sepsis), adjust based on therapeutic monitoring to achieve a target AUC/minimum inhibitory concentration (MIC) determined by broth microdilution (MIC_{BMD}) ratio of 400 to 600 (assuming a vancomycin MIC_{BMD} of 1 mg/L). Trough-only monitoring (target trough: 15 to 20 mg/L) is no longer recommended in patients with serious MRSA infections ([Ref](#)), but may be needed in nonserious MRSA or non-MRSA infections. Early and frequent monitoring for dosage adjustments is recommended, especially when empiric doses exceed 4 g/day ([Ref](#)).

Loading dose: Seriously ill patients with documented/suspected MRSA infection: A loading dose of 20 to 35 mg/kg (based on actual body weight; maximum: 3 g/dose) may be considered to rapidly achieve target concentrations. After administration of the loading dose, the initiation of the maintenance dose should occur at the next dosing interval (eg, for a prescribed interval of every 8 hours, initiate the maintenance dose 8 hours after the start of the loading dose) ([Ref](#)).

Continuous infusion: **Note:** May be considered for critically ill patients who are unable to achieve AUC target with intermittent infusion dosing. Loading dose: 15 to 20 mg/kg, followed by a maintenance continuous infusion dose of 30 to 40

mg/kg/day (up to 60 mg/kg/day) to achieve a target steady state concentration of 20 to 25 mg/L ([Ref](#)).

[Collapse All](#)

Bloodstream infection

Bloodstream infection:

Empiric therapy or pathogen-specific therapy for ampicillin-susceptible enterococci (alternative agent):

Note: Reserve for patients unable to receive first-line agents ([Ref](#)).

IV: 15 to 20 mg/kg/dose every 8 to 12 hours initially; adjust based on therapeutic monitoring. A loading dose may be considered in seriously ill patients ([Ref](#)). Duration of therapy is 7 to 14 days for patients at low risk (ie, fever resolution within 72 hours and absence of metastatic focus of infection or endovascular hardware) with longer courses warranted for endocarditis or metastatic sites of infection ([Ref](#)). Some experts recommend a duration of 5 to 7 days for low-risk bacteremia with rapid blood culture clearance (within 24 hours) ([Ref](#)).

Empiric therapy or pathogen-specific therapy for ampicillin-resistant enterococci: **IV:** 15 to 20 mg/kg/dose every 8 to 12 hours initially; adjust based on therapeutic monitoring ([Ref](#)). May use in combination with an aminoglycoside ([Ref](#)). A loading dose may be considered in seriously ill patients ([Ref](#)). Duration of therapy is 7 to 14 days for patients at low risk (ie, fever resolution within 72 hours and absence of metastatic focus of infection or endovascular hardware) with longer courses warranted for endocarditis or metastatic sites of infection ([Ref](#)). Some experts recommend a duration of 5 to 7 days for low-risk bacteremia with rapid blood culture clearance (within 24 hours) ([Ref](#)).

*Empiric therapy or pathogen-specific therapy for methicillin-resistant *S. aureus*:* **IV:** 15 to 20 mg/kg/dose every 8 to 12 hours initially; adjust based on therapeutic monitoring ([Ref](#)). A loading dose may be considered in seriously ill patients ([Ref](#)). Treat uncomplicated *S. aureus* infection for ≥ 14 days from first negative blood culture, with longer courses warranted for endocarditis or metastatic sites of infection ([Ref](#)).

Empiric therapy or pathogen-specific therapy for methicillin-resistant coagulase-negative staphylococci: **IV:** 15 to 20 mg/kg/dose every 8 to 12 hours initially; adjust based on therapeutic monitoring. Treat uncomplicated bacteremia for 5 to 7 days from day of first negative blood culture, with longer courses

warranted for endocarditis or metastatic sites of infection ([Ref](#)). For catheter-related bloodstream infections, consider antibiotic lock therapy for catheter salvage, in addition to systemic therapy ([Ref](#)).

Antibiotic lock technique (catheter-salvage strategy) (off-label use): **Note:** For infections caused by susceptible organisms when the catheter cannot be removed; use in addition to systemic antibiotics. Catheter salvage is **not** recommended for *S. aureus* ([Ref](#)).

Intracatheter: Prepare lock solution to final concentration of vancomycin 2.5 to 5 mg/mL; may be combined with heparin. Instill into each lumen of the catheter access port using a volume sufficient to fill the catheter (2 to 5 mL) with a dwell time of up to 72 hours, depending on frequency of catheter use. Withdraw lock solution prior to catheter use; replace with fresh vancomycin lock solution after catheter use. Antibiotic lock therapy is given for the same duration as systemic antibiotics ([Ref](#)).

CNS infection

CNS infection:

Cerebrospinal fluid shunt infection (off-label use): As a component of empiric therapy or pathogen-specific therapy (eg, methicillin-resistant *S. aureus* or coagulase-negative staphylococci):

IV: 15 to 20 mg/kg/dose every 8 to 12 hours initially; adjust based on therapeutic monitoring ([Ref](#)). A loading dose may be considered in seriously ill patients ([Ref](#)).

Intraventricular (adjunct to systemic therapy; use a preservative-free preparation): 5 to 20 mg/day; some experts recommend adjusting dosage and administration interval based on cerebrospinal fluid (CSF) vancomycin concentrations (goal: 10 to 20 times MIC of causative organism), ventricular size, and daily output from ventricular drain ([Ref](#)); data for monitoring are limited ([Ref](#)). When intraventricular vancomycin is administered via a ventricular drain, clamp drain for 15 to 60 minutes after administration (allows solution to equilibrate in CSF) ([Ref](#)). **Note:** Intraventricular administration is generally reserved for use in patients who fail parenteral therapy despite removal of CSF shunt or when CSF shunt cannot be removed ([Ref](#)).

Intracranial abscess (brain abscess, intracranial epidural abscess) (off-label use): As a component of empiric therapy or pathogen-specific therapy for methicillin-resistant

S. aureus: **IV**: 15 to 20 mg/kg/dose every 8 to 12 hours initially; adjust based on therapeutic monitoring. A loading dose may be considered in seriously ill patients ([Ref](#)). Duration generally ranges from 6 to 8 weeks depending on clinical management and response to therapy ([Ref](#)).

Meningitis, bacterial (off-label use): As a component of empiric therapy or pathogen-specific therapy (eg, methicillin-resistant S. aureus or penicillin- and cephalosporin-resistant S. pneumoniae): IV: 15 to 20 mg/kg/dose every 8 to 12 hours initially; adjust based on therapeutic monitoring ([Ref](#)). A loading dose may be considered in seriously ill patients ([Ref](#)).

Spinal epidural abscess (off-label use): As a component of empiric therapy or pathogen-specific therapy for methicillin-resistant S. aureus: IV: 15 to 20 mg/kg/dose every 8 to 12 hours initially; adjust based on therapeutic monitoring. A loading dose may be considered in seriously ill patients ([Ref](#)). Duration generally ranges from 4 to 8 weeks depending on clinical management and response to therapy ([Ref](#)).

***Clostridioides difficile* infection, prophylaxis**



***Clostridioides difficile* infection, prophylaxis (off-label use):**

Note: For patients with a recent history of *C. difficile* infection (CDI) who subsequently require systemic antibiotics and are at high risk for CDI recurrence (eg, patients ≥ 65 years of age or with significant immunocompromise who have been hospitalized with severe CDI in the past 3 months) ([Ref](#)).

Oral: 125 mg once daily; continue for 5 to 7 days after completion of systemic antibiotics ([Ref](#)).

***Clostridioides difficile* infection, treatment**



***Clostridioides difficile* infection, treatment:**

Note: Criteria for disease severity is based on expert opinion and should not replace clinical judgment. There is no role for vancomycin doses other than 125 mg and 500 mg ([Ref](#)).

Initial, nonfulminant infection (alternative agent): Oral: 125 mg 4 times daily ([Ref](#)). Usual duration is 10 days; if delayed response to treatment, a longer duration (eg, up to 14 days) may be considered ([Ref](#)). If antibiotic(s) for a

primary infection are essential, some experts extend CDI treatment 1 week beyond other antibiotic(s) ([Ref](#)).

Recurrent, nonfulminant infection (alternative agent): **Note:** Regimen selection depends on prior treatment ([Ref](#)).

Standard regimen: **Note:** Some experts reserve for patients who did not receive vancomycin for the initial episode ([Ref](#)).

Oral: 125 mg 4 times daily ([Ref](#)). Usual duration is 10 days; if delayed response to treatment, a longer duration (eg, up to 14 days) may be considered ([Ref](#)). If antibiotic(s) for a primary infection are essential, some experts extend CDI treatment 1 week beyond other antibiotic(s) ([Ref](#)).

Pulsed-tapered regimen: **Oral:** 125 mg 4 times daily for 10 to 14 days, then 125 mg twice daily for 7 days, then 125 mg once daily for 7 days, then 125 mg every 2 or 3 days for 2 to 8 weeks ([Ref](#)).

Combination regimen with rifaximin: **Note:** Rifaximin resistance may be a concern; some experts avoid in patients who have previously received rifamycins, and others do not routinely recommend this regimen ([Ref](#)).

Oral: 125 mg 4 times daily for 10 days followed by rifaximin ([Ref](#)).

Fulminant infection (ie, ileus, megacolon, and/or hypotension/shock):

Oral: 500 mg 4 times daily with IV metronidazole; if ileus is present, may consider vancomycin retention enema ([Ref](#)). For patients who clinically improve, some experts reduce dose (eg, 125 mg 4 times daily) and discontinue IV metronidazole ([Ref](#)). Usual duration is 10 days; if delayed response to treatment, a longer duration (eg, up to 14 days) may be considered ([Ref](#)). If antibiotic(s) for a primary infection are essential, some experts extend CDI treatment 1 week beyond other antibiotic(s) ([Ref](#)).

Rectal retention enema (off-label route):

Note: Optimal regimen not established. Reserve for patients with fulminant infection with ileus who have not responded to standard therapy and performed by individuals with expertise in administration, as there is risk of colonic perforation ([Ref](#)).

Rectal retention enema: 500 mg in 100 mL NS; retained for as long as possible and replaced every 6 hours. Use in combination with oral vancomycin (if the ileus is partial) or in place of oral vancomycin (if the

ileus is complete) **plus** IV metronidazole ([Ref](#)). Usual duration is 10 days; if delayed response to treatment, a longer duration (eg, up to 14 days) may be considered ([Ref](#)). If antibiotic(s) for a primary infection are essential, some experts extend CDI treatment 1 week beyond other antibiotic(s) ([Ref](#)).

Cystic fibrosis, acute pulmonary exacerbation, moderate to severe

Cystic fibrosis, acute pulmonary exacerbation, moderate to severe (off-label use):

Empiric or pathogen-directed therapy for methicillin-resistant *S. aureus*: **IV**: 15 to 20 mg/kg/dose every 8 hours initially; adjust based on therapeutic monitoring ([Ref](#)). Duration is usually 10 to 14 days depending on clinical response ([Ref](#)).

Diabetic foot infection, moderate to severe

Diabetic foot infection, moderate to severe:

Note: For empiric or pathogen-directed therapy for methicillin-resistant *S. aureus* ([Ref](#)).

IV: 15 to 20 mg/kg/dose every 8 to 12 hours initially; adjust based on therapeutic monitoring ([Ref](#)). Duration (which may include appropriate oral step-down therapy) is usually 2 to 4 weeks in the absence of osteomyelitis ([Ref](#)).

Endocarditis, treatment

Endocarditis, treatment:

Enterococcus (native or prosthetic valve): **Note:** For penicillin-resistant strains or those unable to receive beta-lactams ([Ref](#)).

IV: 15 mg/kg/dose every 12 hours initially; adjust based on therapeutic monitoring ([Ref](#)). Administer in combination with gentamicin. Duration of therapy is 6 weeks ([Ref](#)).

Staphylococcus (alternative agent):

Note: For methicillin-resistant isolates or methicillin-susceptible isolates in patients unable to receive beta-lactams ([Ref](#)).

Native valve: **IV**: 15 to 20 mg/kg/dose every 8 to 12 hours initially; adjust based on therapeutic monitoring ([Ref](#)). A loading dose may be considered in seriously ill patients ([Ref](#)). Duration of therapy is 6 weeks ([Ref](#)). For

endocarditis caused by methicillin-resistant *Staphylococcus aureus*, some experts recommend 4 to 6 weeks ([Ref](#)).

Prosthetic valve: **IV:** 15 to 20 mg/kg/dose every 8 to 12 hours initially; adjust based on therapeutic monitoring ([Ref](#)). A loading dose may be considered in seriously ill patients ([Ref](#)). Use as part of an appropriate combination regimen. Duration of therapy is ≥ 6 weeks ([Ref](#)).

Viridans group streptococci and *S. bovis* (native or prosthetic valve): **Note:** For patients unable to receive penicillin or ceftriaxone ([Ref](#)).

IV: 15 mg/kg/dose every 12 hours initially; adjust based on therapeutic monitoring. Duration of therapy is 4 weeks (native valve) or 6 weeks (prosthetic valve) ([Ref](#)). For prosthetic valve endocarditis caused by isolates with risk of or confirmed penicillin resistance, some experts use for 6 weeks in combination with gentamicin for the first 2 weeks of therapy ([Ref](#)).

Endophthalmitis, treatment

Endophthalmitis, treatment (off-label use): Intravitreal: Usual dose: 1 mg per 0.1 mL NS or sterile water injected into vitreum, usually in combination with ceftazidime ([Ref](#)). A repeat dose(s) may be considered at 24 to 48 hours based on culture result, severity of the infection, and response to treatment ([Ref](#)).

Intra-abdominal infection, health care associated

Intra-abdominal infection, health care associated (off-label use): *Empiric or pathogen-directed therapy for Enterococcus spp. in high-risk patients (eg, postoperative infection or health care-associated infection in patients with prior use of antibiotics that select for Enterococcus, immunocompromising condition, valvular heart disease, or prosthetic intravascular material):* **IV:** 15 to 20 mg/kg/dose every 8 to 12 hours initially; adjust based on therapeutic monitoring ([Ref](#)). Use as part of an appropriate combination regimen ([Ref](#)). Total duration of therapy (which may include transition to oral antibiotics) is 4 to 5 days following adequate source control ([Ref](#)).

Osteomyelitis and/or discitis

Osteomyelitis and/or discitis: As a component of empiric therapy or pathogen-specific therapy (eg, methicillin-resistant *S. aureus*): **IV:** 15 to 20 mg/kg/dose

every 8 to 12 hours initially ([Ref](#)); adjust based on therapeutic monitoring. A loading dose may be considered in seriously ill patients ([Ref](#)). Duration of therapy is generally 6 weeks but depends on clinical response and presence of residual infection or hardware. Some patients who receive initial IV antibiotics can be transitioned to oral therapy to complete the course. Shorter courses are appropriate if the affected bone is completely resected by amputation ([Ref](#)).

Peritonitis, treatment



Peritonitis, treatment (peritoneal dialysis) (off-label use) (off-label route):

Note: Intraperitoneal administration is preferred to IV administration unless the patient has sepsis. Supplemental doses and more frequent monitoring of serum levels may be needed for patients receiving automated peritoneal dialysis ([Ref](#)).

Intermittent (automated peritoneal dialysis): **Intraperitoneal:** 15 mg/kg added to one dialysate exchange every 4 days; allow to dwell for ≥ 6 hours.

Intermittent (CAPD): **Intraperitoneal:** 15 to 30 mg/kg added to one dialysate exchange every 5 to 7 days; allow to dwell for ≥ 6 hours ([Ref](#)).

Continuous (with every CAPD exchange): **Intraperitoneal:** Loading dose: 20 to 25 mg/kg added to first dialysate exchange; maintenance dose: 25 mg/L for each subsequent dialysate exchange ([Ref](#)).

Duration: For patients with adequate clinical response, duration of therapy is ≥ 2 to 3 weeks depending on pathogen. For patients with no improvement after 5 days, remove catheter and treat with appropriate systemic antibiotics for 14 days after catheter removal ([Ref](#)).

Pneumonia



Pneumonia, as a component of empiric therapy or pathogen-specific therapy for methicillin-resistant *S. aureus*: **IV:** 15 to 20 mg/kg/dose every 8 to 12 hours initially; adjust based on therapeutic monitoring ([Ref](#)). A loading dose may be considered in seriously ill patients ([Ref](#)). **Note:** Duration of therapy varies based on disease severity and response to therapy; treatment is typically given for 7 days. When used for empiric therapy, give as part of an appropriate combination regimen ([Ref](#)).

Prosthetic joint infection



Prosthetic joint infection (off-label use): IV:

Pathogen-specific therapy for methicillin-resistant or susceptible S. aureus (alternative agent in beta-lactam intolerance): 15 to 20 mg/kg/dose every 8 to 12 hours initially ([Ref](#)); adjust based on therapeutic monitoring. A loading dose may be considered in seriously ill patients ([Ref](#)). Duration ranges from 2 to 6 weeks depending on prosthesis management, use of rifampin, and other patient-specific factors ([Ref](#)).

Pathogen-specific therapy for Enterococcus spp (penicillin susceptible [alternative agent] or penicillin resistant): 15 mg/kg/dose every 12 hours initially; adjust based on therapeutic monitoring. Duration: 4 to 6 weeks ([Ref](#)).

Note: In select cases (eg, debridement and retention of prosthesis or one-stage arthroplasty), give oral suppressive antibiotic therapy with an appropriate regimen following completion of initial treatment ([Ref](#)).

Sepsis/septic shock



Sepsis/septic shock: As a component of empiric therapy or pathogen-specific therapy for methicillin-resistant S. aureus : IV: 15 to 20 mg/kg/dose every 8 to 12 hours; adjust based on therapeutic monitoring ([Ref](#)). A loading dose is recommended; administer within 1 hour of suspected or confirmed sepsis ([Ref](#)). Duration is dependent on underlying source and patient response; short courses are preferred, when appropriate ([Ref](#)).

Septic arthritis



Septic arthritis:

Note: For use as a component of empiric therapy or pathogen-specific therapy for methicillin-resistant S. aureus. May also be used for methicillin-susceptible infections in patients who cannot receive beta-lactams ([Ref](#)).

IV: 15 to 20 mg/kg/dose every 8 to 12 hours initially; adjust based on therapeutic monitoring. A loading dose may be considered in seriously ill patients ([Ref](#)). Total treatment duration is 3 to 4 weeks (in the absence of osteomyelitis, bacteremia, or other complicating conditions), including appropriate oral step-down therapy ([Ref](#)).

Skin and soft tissue infection

Skin and soft tissue infection (hospitalized patient): As a component of empiric therapy or pathogen-specific therapy for methicillin-resistant *S. aureus* : IV: 15 mg/kg/dose every 12 hours initially ([Ref](#)). For patients with uncomplicated skin and soft tissue infections who are not obese and have normal renal function, therapeutic monitoring is generally not needed; for complicated or severe infections, adjust based on therapeutic monitoring ([Ref](#)). **Note:** For empiric therapy of necrotizing infection, must be used in combination with other agents ([Ref](#)).

Spontaneous bacterial peritonitis

Spontaneous bacterial peritonitis (off-label use):

Note: Reserve for patients who are critically ill or with risk for multidrug-resistance (eg, methicillin-resistant *S. aureus*) ([Ref](#)).

IV: 15 to 20 mg/kg/dose every 8 to 12 hours initially; adjust based on therapeutic monitoring. A loading dose may be considered in seriously ill patients ([Ref](#)). For empiric therapy, use as part of an appropriate combination. Duration of therapy is 5 to 7 days as long as fever and pain have resolved ([Ref](#)).

Streptococcus, maternal prophylaxis for prevention of neonatal disease

***Streptococcus* (group B), maternal prophylaxis for prevention of neonatal disease (alternative agent) (off-label use):**

IV: 20 mg/kg at the onset of labor or prelabor rupture of membranes, then every 8 hours until delivery; maximum single dose: 2 g ([Ref](#)). Some experts prefer vancomycin 2 g initially and then 1 g every 12 hours thereafter until delivery ([Ref](#)). **Note:** Vancomycin is reserved for patients with penicillin allergy at high risk for anaphylaxis that have documented clindamycin-resistant group B *Streptococcus* or no available susceptibility data ([Ref](#)).

Surgical prophylaxis

Surgical prophylaxis (in combination with other appropriate agents when coverage for methicillin-resistant *S. aureus* is indicated or for gram-positive coverage in patients unable to tolerate beta-lactams) (off-label use): IV: 15 mg/kg (usual maximum: 2 g/dose initially ([Ref](#))) started within 60 to 120 minutes prior to initial surgical incision. Vancomycin doses may be repeated

intraoperatively in 2 half-lives (approximately 8 to 12 hours in patients with normal renal function) if procedure is lengthy or if there is excessive blood loss ([Ref](#)). In cases where an extension of prophylaxis is warranted postoperatively, total duration should be ≤ 24 hours ([Ref](#)). **Postoperative** prophylaxis is not recommended in clean and clean-contaminated surgeries ([Ref](#)).

Surgical site infection

Surgical site infection: As a component of empiric therapy or pathogen-specific therapy for methicillin-resistant *S. aureus* : IV: 15 mg/kg/dose every 12 hours initially; adjust based on therapeutic monitoring ([Ref](#)).

Toxic shock syndrome, staphylococcal

Toxic shock syndrome, staphylococcal: As a component of empiric therapy or pathogen-specific therapy for methicillin-resistant *S. aureus* : IV: 15 to 20 mg/kg/dose every 8 to 12 hours initially; adjust based on therapeutic monitoring ([Ref](#)). A loading dose may be considered in seriously ill patients ([Ref](#)). Duration varies based on underlying etiology; 10 to 14 days of treatment is recommended in the absence of bacteremia or other distinct focus of infection ([Ref](#)).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Note: Initial IV dosing in nonobese patients should be based on actual body weight; subsequent dosing should generally be adjusted based on therapeutic monitoring. Trough monitoring has traditionally been used for therapeutic monitoring; however, for serious methicillin-resistant *S. aureus* (MRSA) infections (eg, bacteremia, infective endocarditis, meningitis, osteomyelitis, pneumonia, sepsis), AUC monitoring is preferred ([Ref](#)). A ratio of AUC over 24 hours to minimum inhibitory concentration (AUC/MIC) of ≥ 400 is the primary pharmacokinetic/pharmacodynamic predictor of vancomycin efficacy in serious MRSA infections ([Ref](#)). Serum concentration monitoring should be conducted within the first 48 hours of therapy for patients with suspected or documented serious infections due to MRSA, with subsequent dosing adjusted to maintain AUC/MIC between

400 to 600 in order to maximize efficacy and minimize risk of vancomycin nephrotoxicity (Ref).

Altered kidney function:

Oral: There are no dosage adjustments provided in the manufacturer's labeling. However, dosage adjustment unlikely due to low systemic absorption.

IV:

Intermittent infusion: **Note:** The following table provides general recommendations (expert opinion derived from ASHP/IDSA/PIDS/SIDP [Rybak 2020]; Golightly 2013). Refer to institution-specific policies and procedures for more detailed guidance.

Vancomycin IV Initial Dose Adjustments in Altered Kidney Function

CrCl (mL/minute)	Suggested loading dose (when applicable) ^a	Suggested initial maintenance dose	Suggested dosing interval
>90 to <130	25 to 30 mg/kg	15 to 20 mg/kg	8 to 12 hours
50 to 90	20 to 25 mg/kg	15 to 20 mg/kg	12 hours
15 to <50	20 to 25 mg/kg	10 to 15 mg/kg	24 hours
<15 ^b	20 to 25 mg/kg	10 to 15 mg/kg	48 to 72 hours

^aLoading doses recommended in critically ill patients with suspected/documentated serious MRSA infections. A loading dose of up to 35 mg/kg may be considered in critically ill patients with sepsis. Obese patients usually require 20 to 25 mg/kg loading doses. Maximum recommended loading dose is 3 g (ASHP/IDSA/PIDS/SIDP [Rybak 2020]).

^bMonitor vancomycin serum concentrations more frequently, especially early on in therapy, to achieve target concentrations as these patients may have unstable or less predictable drug clearance. Care should be taken not to administer maintenance doses when serum concentrations remain >20 mg/L (Golightly 2013; expert opinion).

Continuous infusion:

Loading dose: Administer an appropriate loading dose (eg, 15 to 20 mg/kg) (Ref); higher loading doses may be considered in critically ill patients with sepsis (Ref); also refer to institution-specific policies and procedures.

Maintenance dose: Various protocols have been developed (Ref); recommendations may vary based on the population studied. The following is an example protocol (Ref), and doses should be adjusted to achieve a target steady state concentration of 20 to 25 mg/L (Ref); also refer to institution-specific policies and procedures.

CrCl >80 to 119 mL/minute: 30 mg/kg administered over 24 hours.

CrCl >50 to 80 mL/minute: 25 mg/kg administered over 24 hours.

CrCl 25 to 50 mL/minute: 14 mg/kg administered over 24 hours.

CrCl <25 mL/minute: 7 mg/kg administered over 24 hours.

Augmented renal clearance (measured urinary CrCl $\geq$ 130 mL/minute/1.73 m²):

Augmented renal clearance (ARC) is a condition that occurs in certain critically ill patients without organ dysfunction and with normal serum creatinine concentrations. Young patients (<55 years of age) admitted post-trauma or major surgery are at highest risk for ARC, as well as those with sepsis, burns, or hematologic malignancies. An 8- to 24-hour measured urinary CrCl is necessary to identify these patients ([Ref](#)).

Oral: There are no dosage adjustments provided in the manufacturer's labeling. However, dosage adjustment unlikely due to low systemic absorption

IV:

Intermittent infusion: Loading dose (when applicable): 25 to 35 mg/kg ([Ref](#)) followed by 15 to 20 mg/kg every 8 hours depending on degree of augmented kidney function; some patients may require more frequent dosing (eg, 15 mg/kg every 6 hours) to attain target concentrations ([Ref](#)); utilize frequent serum concentration monitoring.

Continuous infusion:

Loading dose: Administer an appropriate loading dose (eg, 15 to 20 mg/kg) ([Ref](#)); higher loading doses (eg, 25 mg/kg) have been used in some protocols and may vary based on population studied ([Ref](#)); also refer to institution-specific policies and procedures.

Maintenance dose: 40 to 60 mg/kg/day depending on degree of augmented kidney function with frequent serum concentration monitoring; adjust to achieve a target steady state concentration of 20 to 25 mg/L ([Ref](#)).

Hemodialysis, intermittent (thrice weekly): Dialyzable (25% to 40% depending on dialyzer permeability) ([Ref](#)).

Oral: There are no dosage adjustments provided in the manufacturer's labeling. However, dosage adjustment unlikely due to low systemic absorption.

IV:

Vancomycin IV Dosing Depending on Dose Timing and Dialyzer Permeability^a

Dose timing and dialyzer permeability	Vancomycin dose ^b
Dose given after dialysis ends	
Low permeability (low flux)	Loading dose: 25 mg/kg Maintenance dose: 7.5 mg/kg ^c
High permeability (high flux)	Loading dose: 25 mg/kg Maintenance dose: 10 mg/kg ^c
Dose given during last hours of dialysis (intradialytic)	
Low permeability (low flux)	Loading dose: 30 mg/kg Maintenance dose: 7.5 to 10 mg/kg ^c
High permeability (high flux)	Loading dose: 35 mg/kg Maintenance dose: 10 to 15 mg/kg ^c

^aASHP/IDSA/PIDS/SIDP [Rybak 2020]

^bInitial recommended loading/maintenance doses. The optimal pharmacokinetic/pharmacodynamic target in this population is unknown, but targeting predialysis concentrations of 15 to 20 mg/L are likely to achieve AUCs of 400 to 600 mg•hour/L (ASHP/IDSA/PIDS/SIDP [Rybak 2020]; Crew 2015). Predialysis serum concentrations should be obtained no less than weekly and should determine subsequent dosing (ASHP/IDSA/PIDS/SIDP [Rybak 2020]).

^cThrice-weekly dose administration. Typically, patients may require ~25% larger doses for the 3-day interdialytic period (eg, Friday to Monday) to maintain sufficient vancomycin exposure on the third day.

Peritoneal dialysis:

Oral: There are no dosage adjustments provided in the manufacturer's labeling. However, dosage adjustment unlikely due to low systemic absorption.

IV:

Loading dose: 20 to 25 mg/kg ([Ref](#)). A vancomycin serum concentration should be obtained ~48 to 72 hours after the loading dose, and subsequent doses (usually 10 to 15 mg/kg) should be administered based on attainment of goal serum concentrations ([Ref](#)). Doses may vary based on infection site and severity, as well as the presence or absence of residual renal function. Some experts use maintenance doses of up to 20 mg/kg/dose ([Ref](#)).

CRRT: Drug clearance is dependent on the effluent flow rate, filter type, and method of renal replacement. Recommendations are based on high-flux dialyzers and effluent flow rates of 20 to 25 mL/kg/hour (or ~1,500 to 3,000 mL/hour) unless otherwise noted. Close monitoring of response and adverse reactions due to drug accumulation is important.

Oral: There are no dosage adjustments provided in the manufacturer's labeling. However, dosage adjustment unlikely due to low systemic absorption.

IV:

Loading dose: 20 to 25 mg/kg followed by 7.5 to 10 mg/kg every 12 hours with more frequent serum concentration monitoring ([Ref](#)). In patients with suspected or confirmed serious MRSA infections, dose adjustments should be made based on AUC monitoring occurring in the first 24 to 48 hours of therapy ([Ref](#)).

PIRRT (eg, sustained, low-efficiency diafiltration): Drug clearance is dependent on the effluent flow rate, filter type, and method of renal replacement. Close monitoring of response and adverse reactions due to drug accumulation is important.

Oral: There are no dosage adjustments provided in the manufacturer's labeling. However, dosage adjustment unlikely due to low systemic absorption.

IV:

Loading dose (administer even if PIRRT is occurring): 20 to 25 mg/kg, followed by 15 mg/kg after each PIRRT session ends (or during the final 60 to 90 minutes of the session) with more frequent serum concentration monitoring ([Ref](#)). In patients with suspected or confirmed serious MRSA infections, dose adjustments should be made based on AUC monitoring occurring in the first 24 to 48 hours of therapy ([Ref](#)).

Dosing: Liver Impairment: Adult

The liver dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

Note: In patients with cirrhosis, the creatinine-based equations overestimate kidney function due to loss of muscle mass and reduced liver production of creatine ([Ref](#)). Compared to patients with normal liver function or Child-Turcotte-Pugh class A, incidences of elevated AUC/minimum inhibitory concentration ratio as well as trough concentrations and vancomycin nephrotoxicity are higher in patients with Child-Turcotte-Pugh class B or C treated with IV vancomycin ([Ref](#)).

Liver impairment prior to treatment initiation

Initial or dose adjustment in patients with preexisting liver cirrhosis:

Oral:

Child-Turcotte-Pugh class A to C: Initial: No dosage adjustment necessary due to low systemic absorption ([Ref](#)).

IV:

Child-Turcotte-Pugh class A: Initial: No dosage adjustment necessary ([Ref](#)).

Child-Turcotte-Pugh class B and C: Initial: No dosage adjustment necessary ([Ref](#)). Use caution as creatinine-based equations may overestimate kidney function in these patients; consider a conservative starting dose. If concurrent kidney dysfunction exists, an empiric reduction in dose or frequency (in addition to kidney impairment dosing adjustments) may be necessary to achieve clinical goals ([Ref](#)).

Liver impairment developing in patient *already receiving vancomycin*:

Acute worsening of liver function (eg, requiring hospitalization):

Child-Turcotte-Pugh class A to C: No dosage adjustment necessary; intensify therapeutic drug monitoring to guide dosing ([Ref](#)).

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Example of dosing regimen:

Class 1, 2, or 3 obesity (BMI ≥ 30 kg/m²):

Loading dose: Note: Consider utilizing a loading dose when rapid attainment of target concentrations is necessary (eg, sepsis, documented/suspected methicillin-resistant *S. aureus* infection) ([Ref](#)).

Initial: IV: 20 to 25 mg/kg using **actual body weight**; maximum loading dose: 3 g. After administration of the loading dose, initiate maintenance dose at the next dosing interval ([Ref](#)). In critically ill patients, may consider loading doses of 20 to 35 mg/kg using **actual body weight**; maximum loading dose: 3 g ([Ref](#)).

Maintenance dose: IV: Use **actual body weight** and the following clearance (CL) equations to calculate a maintenance dose; empiric maintenance doses >4.5 g/day are unlikely to be necessary ([Ref](#)). **Note:** If vancomycin therapy is continued, individualize vancomycin dose using early Bayesian approach (ie, 2 serum concentrations within first 24 to 48 hours) to achieve target AUC ([Ref](#)).

1. Calculate estimated vancomycin CL ([Ref](#)).

Estimate CL (L/hour): $9.656 - [0.078 \times \text{age}] - [2.009 \times \text{SCr}] + [0.04 \times \text{actual body weight}^{0.75}] + [1.09 \times \text{sex}]$.

Where adult age is in years; SCr is serum creatinine in mg/dL; actual body weight in kg scaled to an exponent of 0.75; and sex is 1 if male and 0 if female.

2. Calculate empiric vancomycin maintenance regimen ([Ref](#)).

Estimate daily dose (rounded to nearest 250 mg): Estimated CL (L/hour) × 500 mg•hour/L.

Where 500 mg•hour/L is the mid-range AUC target selected for a minimum inhibitory concentration (MIC) of 1 mg/L. When the vancomycin CL is estimated to be ≤3 L/hour, administer the dose every 24 hours or divide the daily dose every 12 hours. If CL is estimated to be >3 L/hour, divide the daily dose and administer every 6 to 12 hours ([Ref](#)).

Rationale for recommendations:

Vancomycin is a hydrophilic antimicrobial with a low to medium V_d and CL that is proportional to glomerular filtration rate. Vancomycin V_d correlates with actual body weight but does not increase proportionately with increasing body size ([Ref](#)) and has been reported to be lower in patients with obesity compared to individuals who are not obese (0.3 to 0.5 L/kg vs 0.68 L/kg) ([Ref](#)). Because vancomycin loading doses are dependent on V_d , lower loading doses are generally needed in patients with obesity and the dose can be capped once a threshold dose is reached ([Ref](#)). Vancomycin CL can be estimated by variables used to estimate kidney function. Estimated vancomycin CL in patients with obesity is ~6 L/hour and rarely exceeds 9 L/hour. The empiric maintenance daily dose can be calculated by an estimate of CL and the AUC/MIC target of 400 to 600 ([Ref](#)). Using a targeted AUC approach may reduce the total daily dose (TDD) of vancomycin versus a target trough concentration; lower TDD may reduce the risk of acute kidney injury ([Ref](#)). Optimal dosing and monitoring regimens are uncertain; refer to institutional protocol.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "[Vancomycin: Pediatric drug information](#)")

Dosage guidance:

Dosing: For IV dosing, initial dosage recommendations for patients with normal kidney function presented; doses should be adjusted based on serum concentration monitoring. Dosing presented in mg/kg/dose and mg/kg/**day**; routes of administration may vary (eg, IV, oral, intrathecal, intracatheter, intraperitoneal, rectal); use caution.

Clinical considerations: Serum concentration monitoring and assurance of adequate hydration status is recommended; utilize local antibiogram and protocols for further guidance.

General dosing:

Infants, Children, and Adolescents: IV: Initial: 45 to 60 mg/kg/**day** divided every 6 to 8 hours; dose and frequency should be individualized based on serum concentrations ([Ref](#)). **Note:** Based on adult data, an AUC₂₄ target of 400 mg•hour/L is recommended in patients with serious methicillin-resistant *S. aureus* (MRSA) infections; specific dosing recommendations may be higher when targeting this range ([Ref](#)). See "MRSA infection, serious".

[Collapse All](#)

Anthrax

Anthrax: Limited data available:

Note: Consult public health officials for event-specific recommendations; do not delay therapy.

Cutaneous, without meningitis, treatment (alternative agent): Infants, Children, and Adolescents: Initial: IV: 20 mg/kg/dose every 8 hours; adjust dose to target AUC₂₄ of 400 mg•hour/L (or a trough concentration of 15 to 20 mg/L if AUC monitoring is not available). Treat for 7 to 10 days or until patient is clinically stable. If known or suspected aerosol exposure, transition to an appropriate oral agent for postexposure prophylaxis for a total of 60 days from exposure ([Ref](#)).

Systemic, including meningitis, treatment (alternative agent): Infants, Children, and Adolescents: Initial: IV: 20 mg/kg/dose every 8 hours; adjust dose to target AUC₂₄ of 400 mg•hour/L (or a trough concentration of 15 to 20 mg/L if AUC monitoring is not available). Use as part of an appropriate combination regimen for ≥2 weeks; duration may be shortened and patient transitioned to appropriate oral therapy based on clinical response. Following completion of initial therapy, patients with a known or suspected aerosol exposure who are

immunocompromised should transition to an appropriate oral agent for postexposure prophylaxis for a total of 60 days from onset of illness ([Ref](#)).

Antibiotic lock therapy; catheter salvage

Antibiotic lock therapy; catheter salvage: Limited data available: Optimal dose not established:

Note: For infections caused by susceptible organisms when the vascular catheter cannot be removed; use in addition to systemic antibiotics. Catheter salvage not effective in all cases; removal of catheter is recommended for infections with *S. aureus* ([Ref](#)). Dosing regimens variable; consider age and size of patient and catheter size (including number of lumens) when determining dose due to potential for lock to be delivered intravenously.

Infants, Children, and Adolescents: Intracatheter: Usual concentrations of lock solution: 2 to 5 mg/mL of vancomycin with or without heparin additive; most common concentrations reported: vancomycin 2 mg/mL, 2.5 mg/mL, or 5 mg/mL; refer to institutional protocol if available ([Ref](#)). Concentrations described in literature range from 0.025 to 10 mg/mL with or without heparin or citrate ([Ref](#)); a vancomycin concentration of 5 mg/mL has been shown to be more efficacious than 1 mg/mL when biofilm present ([Ref](#)). Instill into each lumen of the catheter access port using a volume sufficient to fill the catheter, with a dwell time of ideally ≥ 8 to 12 hours and up to 72 hours, depending on frequency of catheter use. Withdraw lock solution prior to catheter use; replace with fresh vancomycin lock solution after catheter use. Antibiotic lock therapy is given for the same duration as systemic antibiotics ([Ref](#)). **Note:** If heparin is utilized in the lock solution, the dose used should not approach therapeutic unit/kg dose.

Clostridioides difficile infection, prophylaxis

***Clostridioides difficile* infection, prophylaxis:**

Note: For patients with a recent history of *C. difficile* infection (eg, within the last 6 months) who subsequently require systemic antibiotics ([Ref](#)).

Children and Adolescents: Oral: 10 mg/kg/dose every 12 hours during concomitant antibiotic use and for up to 5 days following antibiotic completion; maximum dose: 125 mg/dose ([Ref](#)).

***Clostridioides difficile* infection, treatment**



***Clostridioides difficile* infection, treatment:**

Manufacturer's labeling: Infants, Children, and Adolescents: Oral: 40 mg/kg/**day** divided every 6 to 8 hours for 7 to 10 days; maximum daily dose: 2,000 mg/day.

Guideline recommendations:

Non-severe infection, initial or first recurrence: Children and Adolescents: Oral: 10 mg/kg/dose 4 times daily for 10 days; maximum dose: 125 mg/dose ([Ref](#)).

Severe/fulminant infection, initial: Children and Adolescents:

Oral: 10 mg/kg/dose 4 times daily for 10 days; maximum dose: 500 mg/dose; may consider adding IV metronidazole in critically ill patients ([Ref](#)).

Rectal: **Note:** Consider use when ileus is present. Limited data available: Rectal enema: 500 mg in 100 mL NS; dose volume is determined by age ([Ref](#)); the optimal doses have not been established in pediatric patients; suggested volumes for children: 1 to 3 years: 50 mL; 4 to 9 years: 75 mL; >10 years: 100 mL ([Ref](#)); administer 4 times daily with or without IV metronidazole ([Ref](#)).

Second or subsequent recurrence: Children and Adolescents: Pulsed-tapered regimen: Oral: 10 mg/kg/dose 4 times daily for 10 to 14 days; then 10 mg/kg/dose twice daily for 7 days, then 10 mg/kg/dose once daily for 7 days, then 10 mg/kg/dose every 2 or 3 days for 2 to 8 weeks; maximum dose: 125 mg/dose ([Ref](#)).

Endocarditis, treatment



Endocarditis, treatment:

Note: Dosage adjustment to target trough serum concentrations of 10 to 15 mg/L is recommended in pediatric endocarditis by the AHA ([Ref](#)). Dosage adjustment to target AUC₂₄ of 400 mg•hour/L has been recommended in the treatment of proven or suspected MRSA infections based on adult data ([Ref](#)).

Empiric therapy/culture negative: Children and Adolescents: IV: Initial: 60 mg/kg/**day** divided every 6 hours; initial maximum daily dose: 2,000 mg/**day**; use in combination with other antibiotics for at least 4 to 6 weeks; longer duration

may be required if prosthetic material is present or in cases of recurrent endocarditis ([Ref](#)).

Streptococcus (including enterococcus): Children and Adolescents: IV: Initial: 40 mg/kg/**day** divided every 8 to 12 hours; initial maximum daily dose: 2,000 mg/**day**; treat for at least 4 to 6 weeks; a longer duration and additional antibiotics may be required depending on organism and presence of prosthetic material ([Ref](#)).

S. aureus:

Non-methicillin resistant: Children and Adolescents: IV: Initial: 40 mg/kg/**day** divided every 8 to 12 hours; initial maximum daily dose: 2,000 mg/**day**; treat for at least 4 to 6 weeks; a longer duration and additional antibiotics may be required depending on organism and presence of prosthetic material ([Ref](#)).

Methicillin-resistant: Children and Adolescents: IV: Initial: 40 mg/kg/**day** divided every 8 to 12 hours for at least 6 weeks; usual initial maximum daily dose: 2,000 mg/**day** ([Ref](#)); however, higher initial doses have been recommended for patients with serious MRSA infection with normal renal function, though dosing based on studies and models that were not specific to endocarditis ([Ref](#)). See "MRSA infection, serious".

Enterocolitis

Enterocolitis (*S. aureus*): Infants, Children, and Adolescents: Oral: 40 mg/kg/**day** divided every 6 to 8 hours for 7 to 10 days; maximum daily dose: 2,000 mg/**day**.

Meningitis, including health care-associated meningitis

Meningitis, including health care-associated meningitis: Limited data available: Infants, Children, and Adolescents: IV: Initial: 15 mg/kg/dose every 6 hours ([Ref](#)). Higher initial doses have been recommended for patients with serious MRSA infection with normal renal function, though dosing based on studies and models that were not specific to meningitis ([Ref](#)). See "MRSA infection, serious".

Methicillin-resistant *Staphylococcus aureus* infection, serious

Methicillin-resistant *Staphylococcus aureus* infection, serious:

Note: Doses should be adjusted based on patient-specific serum concentrations to a target AUC_{24} of 400 mg•hour/L, but potentially up to 600 mg•hour/L, based on adult data. In pediatric patients, an AUC_{24} of ≥ 400 mg•hour/L has been associated with trough concentrations of 7 to 10 mg/L, though trough concentrations do not clearly predict AUC on an individual level and trough-only monitoring is not recommended ([Ref](#)). Some studies have indicated that doses on the lower end of the range (ie, 60 mg/kg/**day** divided every 6 hours) will achieve target AUC_{24} in most children ([Ref](#)). To minimize risk of acute kidney injury, maintain AUC_{24} <800 mg•hour/L and trough <15 mg/L. For patients with obesity, start with a one-time loading dose of 20 mg/kg (based on total body weight), then start maintenance dosing ([Ref](#)).

Infants ≥ 3 months and Children <12 years: IV: Initial: 60 to 80 mg/kg/**day** in divided doses every 6 hours; initial maximum daily dose: 3,600 mg/**day**.

Children ≥ 12 years and Adolescents: IV: Initial: 60 to 70 mg/kg/**day** in divided doses every 6 to 8 hours; initial maximum daily dose: 3,600 mg/**day**.

Osteoarticular infection, acute



Osteoarticular infection (eg, bacterial arthritis, osteomyelitis), acute:

Infants, Children, and Adolescents: IV: Initial: 40 to 60 mg/kg/**day** in divided doses every 6 to 8 hours ([Ref](#)). Higher initial doses have been recommended for patients with serious MRSA infection with normal kidney function ([Ref](#)). See "MRSA infection, serious."

Duration of therapy: Minimum total duration is 10 to 14 days for bacterial arthritis and 21 to 28 days for osteomyelitis; however, duration for infections caused by MRSA commonly require longer durations; individualize based on several factors, including causative pathogen, response to therapy, and normalization of inflammatory markers ([Ref](#)).

Peritonitis



Peritonitis (peritoneal dialysis [PD]): Limited data available:

Prophylaxis:

Touch contamination of PD line: **Note:** Use should be reserved for patients with known MRSA colonization; may be considered for patients with a

history of methicillin-resistant coagulase-negative *Staphylococcus* infection ([Ref](#)).

Infants, Children, and Adolescents: Intraperitoneal: 25 mg per liter ([Ref](#)).

High-risk gastrostomy placement: **Note:** Use should be reserved for patients at high risk for MRSA ([Ref](#)).

Infants, Children, and Adolescents: IV: 10 mg/kg as a single dose administered 60 to 90 minutes before procedure (complete infusion within 60 minutes prior to procedure); maximum dose: 1,000 mg ([Ref](#)).

Treatment: Infants, Children, and Adolescents:

Intermittent: Intraperitoneal: Initial dose: 30 mg/kg in the long dwell; subsequent doses: 15 mg/kg/dose every 3 to 5 days during the long dwell; **Note:** Increased clearance may occur in patients with residual renal function; subsequent doses should be based on serum concentration monitoring; redosing should occur when serum concentration <15 mg/L ([Ref](#)).

Continuous: Intraperitoneal: Loading dose: 500 mg per liter of dialysate to dwell for 6 hours; maintenance dose: 25 mg per liter ([Ref](#)).

Pneumonia, community acquired

Pneumonia, community acquired: Infants >3 months, Children, and Adolescents: IV: Initial: 40 to 60 mg/kg/**day** in divided doses every 6 to 8 hours; ([Ref](#)). **Note:** Higher doses may be necessary when treating MRSA infections; doses should be adjusted based on patient-specific serum concentrations to a target AUC₂₄ of 400 mg•hour/L ([Ref](#)). See "MRSA infection, serious".

Skin and soft tissue infection, complicated

Skin and soft tissue infection, complicated: Note: Duration of treatment should be individualized and is dependent on severity of infection, adequacy of source control, and clinical improvement. For necrotizing fasciitis, continue treatment until further debridement is not necessary, patient has clinically improved, and patient is afebrile for 48 to 72 hours.

Necrotizing infections, mixed (non-MRSA): Infants, Children, and Adolescents: IV: Initial: 10 to 13 mg/kg/dose every 8 hours ([Ref](#)).

Serious MRSA infection, including necrotizing infection and pyomyositis: **Note:**

Dosage adjustment to target AUC₂₄ of 400 mg•hour/L recommended for serious MRSA infections based on adult data. A loading dose of 20 mg/kg (based on total body weight) is recommended in patients with obesity ([Ref](#)).

Infants ≥3 months and Children <12 years: IV: Initial: 60 mg/kg/**day** in divided doses every 6 hours; maximum daily dose: 3,600 mg/**day** ([Ref](#)). Based on pharmacokinetic modeling studies, doses up to 80 mg/kg/**day** may be necessary to achieve target AUC₂₄ ([Ref](#)).

Children ≥12 years and Adolescents: IV: Initial: 60 mg/kg/**day** in divided doses every 6 to 8 hours; maximum daily dose: 3,600 mg/**day** ([Ref](#)). Based on pharmacokinetic modeling studies, doses up to 70 mg/kg/**day** may be necessary to achieve target AUC₂₄ ([Ref](#)).

Surgical prophylaxis



Surgical (perioperative) prophylaxis: Infants, Children, and Adolescents: IV: 15 mg/kg/dose within 120 minutes prior to surgical incision. May be administered in combination with other antibiotics depending upon the surgical procedure ([Ref](#)).

Ventriculitis



Ventriculitis (including health care-associated ventriculitis and cerebrospinal fluid [CSF] shunt infections):

Infants, Children, and Adolescents: Limited data available: Intraventricular or intrathecal: **Use a preservative-free preparation:** 5 to 20 mg/**day**; usual dose: 10 or 20 mg/**day** ([Ref](#)); due to the smaller CSF volume in infants, some guidelines recommend decreasing the infant dose; adult dosage recommendations are based on ventricle size ([Ref](#)).

Continuous infusion dosing: Limited data available; optimal dosing unknown:

Infants, Children, and Adolescents: IV: Loading dose: 10 to 15 mg/kg/dose administered over 1 to 2 hours, followed by maintenance infusion of 40 to 60 mg/kg/**day**; adjust dose to achieve target serum concentration ([Ref](#)). **Note:** Required dose to achieve target concentration varies significantly between patients and depending on age, renal function, and target concentration; total daily doses of 30 to 110 mg/kg/**day** have been reported ([Ref](#)). Pediatric patients with cancer or who are critically ill may require higher doses to achieve target concentrations ([Ref](#)); initial loading doses of

15 to 20 mg/kg have also been described for patients with cancer ([Ref](#)). **Note:** When transitioning from intermittent to continuous infusion, an initial loading dose may not be required; the total daily dose will likely need reduced depending on patient-specific factors, concentrations achieved during intermittent dosing, and clinical considerations ([Ref](#)).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Oral: There are no dosage adjustments provided in manufacturer's labeling; however, dosage adjustment unlikely due to low systemic absorption.

IV: **Note:** Vancomycin concentrations should be monitored in patients with any renal impairment:

Infants, Children, and Adolescents: The following adjustments have been recommended ([Ref](#)): **Note:** Renally-adjusted dose recommendations are based on intravenous doses of 10 mg/kg/dose every 6 hours or 15 mg/kg/dose every 8 hours.

GFR 30 to 50 mL/minute/1.73 m²: 10 mg/kg/dose every 12 hours.

GFR 10 to 29 mL/minute/1.73 m²: 10 mg/kg/dose every 18 to 24 hours.

GFR <10 mL/minute/1.73 m²: 10 mg/kg/dose; redose based on serum concentrations.

Intermittent hemodialysis: 10 mg/kg/dose; redose based on serum concentrations.

Peritoneal dialysis (PD): 10 mg/kg/dose; redose based on serum concentrations.

CRRT: 10 mg/kg/dose every 12 to 24 hours; monitor serum concentrations.

Dosing: Liver Impairment: Pediatric

Oral: There are no dosage adjustments provided in the manufacturer's labeling; however, dosage adjustment unlikely needed due to low systemic absorption.

IV: There are no dosage adjustments provided in the manufacturer's labeling; however, degrees of hepatic dysfunction do not affect the pharmacokinetics of vancomycin ([Ref](#)).

Adverse Reactions (Significant): Considerations

[Collapse All](#)

Anaphylaxis



Vancomycin may rarely cause life-threatening immune-mediated **anaphylaxis**, which may present as generalized and extensive **pruritus** and/or **erythema of skin**, respiratory distress, bronchospasm, hypoxia, and **hypotension**. Clinical presentation is similar to vancomycin infusion reaction (a nonimmune-mediated anaphylactoid **infusion-related reaction**; formerly called “red man syndrome”), making it difficult for clinicians to distinguish between the 2 reactions ([Ref](#)).

Mechanism: Non-dose-related; immunologic; IgE-mediated with specific antibodies formed against a drug allergen following initial exposure (immunologically mediated) or result in direct mast cell stimulation ([Ref](#)). IgE binding and cross-linking of the high affinity IgE receptor (FcεRI) on the surface of mast cells causes release of histamine and other mediators that can result in urticaria, flushing, airway obstruction, hypotension, and tachycardia ([Ref](#)).

Onset: Rapid; IgE-mediated reactions generally occur within 1 hour of administration but may occur up to 6 hours after exposure ([Ref](#)). In a systematic review analyzing 7 case reports of vancomycin-induced anaphylaxis, the median time to onset of signs/symptoms was 2 minutes (range: 1 to 35 minutes) ([Ref](#)).

Risk factors:

- Previous exposure to vancomycin (necessary for IgE-mediated anaphylaxis) ([Ref](#))

Clostridioides difficile infection



Although oral vancomycin is used for the treatment of *Clostridioides difficile* infection (CDI), ***Clostridioides difficile*-associated diarrhea** and ***Clostridioides difficile* colitis** have been reported with intravenous vancomycin ([Ref](#)). Clinical symptoms range from mild diarrhea to life-threatening colitis, toxic megacolon, and sepsis. In patients with severe CDI, frequent symptoms include watery diarrhea, abdominal pain, fever, nausea, anorexia, and malaise ([Ref](#)).

Mechanism: Non-dose-related; antibiotics disrupt the indigenous gut microbiota which promotes *C. difficile* spore germination, growth, and toxin production, leading to epithelial damage and colitis ([Ref](#)).

Onset: Varied; may start on the first day of antibiotic therapy or up to 3 months postantibiotic ([Ref](#)).

Risk factors:

- Antibiotic exposure (highest risk factor); antibiotics most frequently associated with *C. difficile* include clindamycin, fluoroquinolones, and third-/fourth-generation cephalosporins ([Ref](#))
- Long durations in a hospital or other health care setting (recent or current) ([Ref](#))
- Advanced age ([Ref](#))
- Immunocompromised conditions or a serious underlying condition ([Ref](#))
- GI surgery/manipulation ([Ref](#))
- Antiulcer medications, such as proton pump inhibitors and H2 blockers (suggested risk factors) ([Ref](#))
- Chemotherapy (suggested risk factor) ([Ref](#))

Drug-induced immune thrombocytopenia



Drug-induced **immune thrombocytopenia** (DITP) has been associated with use.

Vancomycin-induced ITP has been associated with severe bleeding characterized by petechial hemorrhages, ecchymoses, and oozing from the buccal mucosa.

Rarely, acutely-ill patients have experienced gross hematuria, lower GI hemorrhage, intrapulmonary hemorrhage, and excessive bleeding from venipuncture sites ([Ref](#)).

Mechanism: Non-dose-related; immunologic; platelet-reactive antibodies of the IgG class, the IgM class, or both, have been detected in patients with thrombocytopenia while receiving vancomycin. These antibodies reacted with platelets only in the presence of vancomycin, suggesting that the mechanism is similar to quinine-induced thrombocytopenia rather than a hapten-specific antibody ([Ref](#)).

Onset: Varied; DITP; typically occurs within 1 to 2 weeks after initiating therapy or longer in patients with intermittent exposure ([Ref](#)). In case reports, the mean time to platelet nadir count was ~8 days following first exposure. However, there are rare case reports describing a rapid onset (within 24 hours) of acute severe thrombocytopenia, primarily in settings of reexposure to vancomycin ([Ref](#)).

Hypersensitivity reactions (delayed)



Maculopapular rash and severe cutaneous adverse reactions (SCARs), including **drug rash with eosinophilia and systemic symptoms (DRESS)**, **toxic epidermal necrosis (TEN)**, **Stevens-Johnson syndrome (SJS)**, and **acute generalized exanthematous pustulosis (AGEP)**, have occurred rarely with use and may be life-threatening ([Ref](#)). In addition, vancomycin-induced **dermatologic disorder** (linear IgA bullous dermatosis [LABD]) has been reported rarely and clinical presentation may mimic TEN, making it difficult to distinguish ([Ref](#)). Other reactions include **erythema multiforme**, **exfoliative dermatitis**, and **hypersensitivity angitis** ([Ref](#)).

Mechanism: Non-dose-related; immunologic. Delayed hypersensitivity reactions are mediated by T-cells or antibodies other than IgE (eg, IgG-mediated, such as some cytopenias) ([Ref](#)). SCARs are delayed type IV hypersensitivity reactions involving a T-cell mediated drug-specific immune response ([Ref](#)). The mechanism behind vancomycin-induced LABD is unknown; LABD is a rare immune-mediated blistering disorder resulting in linear deposition of IgA at the basement membrane zone ([Ref](#)).

Onset: Delayed; type IV reactions are delayed hypersensitivity reactions that typically occur days to weeks after drug exposure but may occur more rapidly (usually within 1 to 4 days) upon reexposure ([Ref](#)). DRESS usually does not develop until after 2 weeks of administration ([Ref](#)). In a systematic case review, a median onset of 9 days and 21 days was observed for SJS/TEN and DRESS, respectively ([Ref](#)). In vancomycin-induced LABD, lesions typically appear 1 to 15 days after the first vancomycin dose ([Ref](#)); a median latency of 7 days was observed in a systematic review ([Ref](#)).

Risk factors:

- Patients with end-stage renal disease (suggested risk factor) ([Ref](#))
- In DRESS, a strong association was observed for patients with the HLA-A*32:01 allele in a study involving predominantly European ancestry ([Ref](#))
- Cross-reactivity with teicoplanin ([Ref](#))

Nephrotoxicity



Systemic exposure is associated with **nephrotoxicity** (usually reversible), which may result in **kidney impairment** (or **kidney failure**), predominantly occurring in patients with multiple risk factors ([Ref](#)). Cases of systemic absorption and nephrotoxicity with oral vancomycin have been reported ([Ref](#)).

Mechanism: Non-dose-related; most commonly attributed to acute tubular necrosis (or **renal tubular necrosis**), resulting from direct oxidative stress on proximal tubule cells or obstructive tubular cast formation. In addition, acute **interstitial nephritis** has also been described, characterized by tubular and interstitial inflammation, resulting from an immunologically mediated (non-IgE) process ([Ref](#)).

Onset: Intermediate; usually occurs 5 to 7 days and up to 14 days following monotherapy ([Ref](#)). Acute interstitial nephritis was observed at a median onset of 26 days in a systematic case review ([Ref](#)).

Risk factors:

- Vancomycin exposure (trough levels ≥ 15 mg/L, larger AUC [>650 to $1,300$ mg·h/L], high daily doses [>4 g/day]) ([Ref](#))
- Duration of therapy >7 days ([Ref](#))
- Obesity ([Ref](#))
- Preexisting kidney dysfunction ([Ref](#))
- Critical illness ([Ref](#))
- Concurrent nephrotoxin therapy or concurrent prolonged use of piperacillin/tazobactam ([Ref](#))
- Older adults >65 years:
 - Parenteral: Less commonly associated risk factor ([Ref](#))
 - Oral: Increases the risk of systemic absorption from oral vancomycin

Neutropenia/pancytopenia

Neutropenia (severe) and **agranulocytosis** have been observed in numerous case reports and case series; in some cases, **drug fever** also accompanied the neutropenia ([Ref](#)). Reversible **pancytopenia** has also been reported in case reports ([Ref](#)).

Mechanism: Non-dose-related; available data suggest a peripheral mechanism mediated by antibodies and direct toxicity to the bone marrow ([Ref](#)).

Onset: Varied; usually occurs after 7 to 12 days of treatment, with most cases occurring after 20 days ([Ref](#)). However, 1 case report described an onset of 8

weeks following discontinuation of a 3-week course, and then upon rechallenge, neutropenia recurred 3 days following reinitiation ([Ref](#)).

Risk factors:

- Prolonged exposure (ie, >7 days) ([Ref](#))
- Teicoplanin: In patients experiencing neutropenia who were switched to teicoplanin (another glycopeptide), 50% of these patients also developed teicoplanin-induced neutropenia ([Ref](#))

Ototoxicity



Vancomycin is infrequently associated with **ototoxicity**, manifested as **tinnitus**, sensorineural **hearing loss**, **dizziness**, or **vertigo**; some cases have reported irreversible hearing loss ([Ref](#)). Of note, vancomycin has not been found to be ototoxic in animal models ([Ref](#)).

Mechanism: Non-dose-related; proposed to be via direct damage to the auditory branch of the eighth cranial nerve, although data are conflicting and unclear if ototoxicity is directly attributable to vancomycin or to other confounding factors ([Ref](#)).

Risk factors:

- Older adults ([Ref](#))
- Coadministration with ototoxic agents (eg, aminoglycosides) ([Ref](#))
- Kidney dysfunction (potential risk factor) ([Ref](#))

Vancomycin infusion reaction



Vancomycin infusion reaction, a non-IgE-mediated drug reaction most often characterized by an erythematous rash, generalized **flushing**, and **pruritus**, may occur. Severe reactions, which are uncommon, may also include **hypotension**, **chest pain**, and **dyspnea**. Rarely, vancomycin infusion reaction may be life-threatening and cause severe hypotension and cardiac arrest or cardiovascular collapse. Clinical presentation can be similar to IgE-mediated anaphylaxis making it difficult for clinicians to distinguish between the 2 reactions ([Ref](#)). Reactions usually cease promptly after infusion is stopped.

Mechanism: Non-IgE-mediated drug reaction caused by histamine release from mast cells and basophils found in the skin, lung, GI tract, myocardium, and

vascular system ([Ref](#)). The mast cell receptor MRGPRX2 has also been identified as a cause of non-IgE-mediated drug reactions ([Ref](#)).

Onset: Rapid; usually occurs 4 to 10 minutes after the start of the infusion with the first dose but may also occur at any time ([Ref](#)).

Risk factors:

- Typically caused by rapid IV infusion (<1 hour) of large doses ([Ref](#))
- Concomitant medications that also induce histamine release including ciprofloxacin; barbiturates; opioids (except fentanyl which rarely induces histamine); certain neuromuscular antagonists (atracurium, cisatracurium, doxacurium, mivacurium, succinylcholine, tubocurarine); propofol; plasma expanders (dextran, polygeline); and radiocontrast agents ([Ref](#)).

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Reported adverse reactions are for oral administration, unless otherwise indicated.

>10%:

Endocrine & metabolic: Hypokalemia (13%)

Gastrointestinal: Abdominal pain (15%), nausea (17%)

1% to 10%:

Cardiovascular: Peripheral edema (6%)

Gastrointestinal: Diarrhea (9%), flatulence (8%), vomiting (9%)

Genitourinary: Urinary tract infection (8%)

Nervous system: Fatigue (5%), headache (7%)

Neuromuscular & skeletal: Back pain (6%)

Renal: Nephrotoxicity (5%; including increased serum creatinine, kidney failure, kidney impairment)

Miscellaneous: Fever (9%)

Frequency not defined (any route of administration [IV or oral]):

Cardiovascular: Chest pain, hypotension, shock

Dermatologic: Bullous dermatitis, erythema of skin

Gastrointestinal: Constipation

Hematologic & oncologic: Anemia, leukopenia

Local: Injection-site phlebitis, irritation at injection site, pain at injection site

Nervous system: Depression, insomnia, malaise, vertigo

Neuromuscular & skeletal: Myalgia

Postmarketing (any route of administration [IV or oral]):

Cardiovascular: Flushing ([Ref](#)), vasculitis

Dermatologic: Acute generalized exanthematous pustulosis ([Ref](#)), dermatologic disorder (linear IgA bullous dermatosis) ([Ref](#)), erythema multiforme ([Ref](#)), exfoliative dermatitis ([Ref](#)), fixed drug eruption ([Ref](#)), maculopapular rash ([Ref](#)), pruritus, skin rash, Stevens-Johnson syndrome ([Ref](#)), toxic epidermal necrolysis ([Ref](#)), urticaria

Gastrointestinal: *Clostridioides difficile*-associated diarrhea ([Ref](#)), *Clostridioides difficile* colitis ([Ref](#)), peritonitis (following intraperitoneal administration during CAPD) ([Ref](#))

Hematologic & oncologic: Agranulocytosis ([Ref](#)), eosinophilia, Henoch-Schönlein purpura ([Ref](#)), immune thrombocytopenia ([Ref](#)), neutropenia (reversible) (literature suggests an incidence ranging from 2% to 12%) ([Ref](#)), pancytopenia ([Ref](#)), thrombocytopenia

Hypersensitivity: Anaphylaxis ([Ref](#)), drug reaction with eosinophilia and systemic symptoms ([Ref](#)), hypersensitivity angitis ([Ref](#)), hypersensitivity reaction ([Ref](#)), nonimmune anaphylaxis, vancomycin infusion reaction (literature suggests an incidence ranging from 4% to as high as 47%; can occur with oral administration) ([Ref](#))

Local: Inflammation at injection site

Nervous system: Chills, dizziness, pain, vertigo

Neuromuscular & skeletal: Muscle spasm (chest and back)

Otic: Hearing loss ([Ref](#)), ototoxicity ([Ref](#)), tinnitus ([Ref](#))

Renal: Acute kidney injury ([Ref](#)), increased blood urea nitrogen ([Ref](#)), interstitial nephritis ([Ref](#)), renal tubular necrosis ([Ref](#))

Respiratory: Dyspnea, wheezing

Miscellaneous: Drug fever

Contraindications

Hypersensitivity to vancomycin or any component of the formulation; solutions containing dextrose may be contraindicated in patients with allergy to corn or corn products.

Warnings/Precautions

Concerns related to adverse effects:

- Extravasation and thrombophlebitis: IV vancomycin is an irritant; ensure proper needle or catheter placement prior to and during infusion; avoid extravasation. Pain, tenderness, and necrosis may occur with extravasation. If thrombophlebitis occurs, slow infusion rates, dilute solution (eg, 2.5 to 5 g/L) and rotate infusion sites.
- Superinfection: Prolonged use may result in fungal or bacterial superinfection.

Disease-related concerns:

- Inflammatory bowel disease: Clinically significant serum concentrations have been reported in patients with inflammatory disorders of the intestinal mucosa who have taken oral vancomycin (multiple doses) for the treatment of *C. difficile*-associated diarrhea. Although use may be warranted, the risk for adverse reactions may be higher in this situation; consider monitoring serum trough concentrations in patients with renal insufficiency, severe colitis, and a prolonged course (IDSA/SHEA [McDonald 2018]; Pettit 2015).
- Renal impairment: Use with caution in patients with renal impairment or those receiving other nephrotoxic drugs; dosage modification required and close monitoring is recommended in patients with preexisting renal impairment and those at high risk for renal impairment. Accumulation may occur after multiple oral doses of vancomycin in patients with renal impairment; consider monitoring serum concentrations in this circumstance.

Other warnings/precautions:

- Appropriate use: Oral vancomycin is only indicated for the treatment of CDI or enterocolitis due to *S. aureus* and is not effective for systemic infections; parenteral vancomycin is not effective for the treatment of enterocolitis.
- Intraocular administration (off-label route): Hemorrhagic occlusive retinal vasculitis (HORV), including permanent visual loss, has been reported in patients receiving intracameral or intravitreal administration of vancomycin during or after cataract surgery.

- Intraperitoneal administration (off-label route): Use caution when administering intraperitoneally (IP); in some continuous ambulatory peritoneal dialysis (CAPD) patients, chemical peritonitis (cloudy dialysate, fever, severe abdominal pain) has occurred. Symptoms are self-limited and usually clear after vancomycin discontinuation.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Capsule, Oral, as hydrochloride:

Vancocin: 125 mg, 250 mg [contains fd&c blue #2 (indigotine, indigo carmine)]

Generic: 125 mg, 250 mg

Solution, Intravenous [preservative free]:

Generic: 1.25 g/250 mL in Dextrose 5% (250 mL); 1.5 g/300 mL in Dextrose 5% (300 mL)

Solution, Intravenous, as hydrochloride:

Generic: 750 mg/150 mL (150 mL); 1000 mg/200 mL (200 mL); 1250 mg/250 mL (250 mL); 1500 mg/300 mL (300 mL); 1750 mg/350 mL (350 mL)

Solution, Intravenous, as hydrochloride [preservative free]:

Tyzavan: 500 mg/100 mL (100 mL); 750 mg/150 mL (150 mL); 1000 mg/200 mL (200 mL); 1250 mg/250 mL (250 mL); 1500 mg/300 mL (300 mL); 1750 mg/350 mL (350 mL); 2000 mg/400 mL (400 mL)

Generic: 500 mg/100 mL (100 mL); 2000 mg/400 mL (400 mL); 1 g/200 mL in Dextrose 5% (200 mL); 1 g/200 mL in NaCl 0.9% (200 mL); 500 mg/100 mL in Dextrose 5% (100 mL); 750 mg/150 mL in Dextrose 5% (150 mL)

Solution Reconstituted, Intravenous [preservative free]:

Generic: 1.25 g (1 ea)

Solution Reconstituted, Intravenous, as hydrochloride:

Generic: 500 mg (1 ea); 1 g (1 ea); 1.5 g (1 ea); 10 g (1 ea)

Solution Reconstituted, Intravenous, as hydrochloride [preservative free]:

Generic: 500 mg (1 ea); 750 mg (1 ea); 1 g (1 ea); 1.5 g (1 ea); 5 g (1 ea); 10 g (1 ea); 100 g (1 ea)

Solution Reconstituted, Intravenous, as hydrochloride [strength expressed as base, preservative free]:

Generic: 1.75 g (1 ea); 2 g (1 ea)

Solution Reconstituted, Oral, as hydrochloride:

Firvanq: 25 mg/mL (150 mL, 300 mL); 50 mg/mL (150 mL, 300 mL) [contains fd&c red #40 (allura red ac dye), quinoline yellow (d&c yellow #10), sodium benzoate; grape flavor]

Firvanq: 50 mg/mL (150 mL [DSC], 300 mL [DSC]) [contains fd&c red #40 (allura red ac dye), quinoline yellow (d&c yellow #10), sodium benzoate; white grape flavor]

Generic: 25 mg/mL (150 mL, 300 mL); 50 mg/mL (150 mL, 300 mL); 250 mg/5 mL (80 mL, 150 mL, 300 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Capsules (Vancocin Oral)

125 mg (per each): \$103.35

250 mg (per each): \$190.55

Capsules (Vancomycin HCl Oral)

125 mg (per each): \$3.06 - \$31.31

250 mg (per each): \$6.00 - \$57.72

Solution (Tyzavan Intravenous)

500 mg/100 mL (per mL): \$0.11

750 mg/150 mL (per mL): \$0.11

1000 mg/200 mL (per mL): \$0.11

1250 mg/250 mL (per mL): \$0.11

1500MG/300ML (per mL): \$0.11

1750MG/350ML (per mL): \$0.11

2000MG/400ML (per mL): \$0.11

Solution (Vancomycin HCl in Dextrose Intravenous)

1GM/200ML 5% (per mL): \$0.16

1.25GM/250ML 5% (per mL): \$0.22

1.5GM/300ML 5% (per mL): \$0.19

500 mg/100 mL 5% (per mL): \$0.09

750MG/150ML 5% (per mL): \$0.10

Solution (Vancomycin HCl in NaCl Intravenous)

1GM/200ML 0.9% (per mL): \$0.14

500 mg/100 mL 0.9% (per mL): \$0.09

750MG/150ML 0.9% (per mL): \$0.10

Solution (Vancomycin HCl Intravenous)

500 mg/100 mL (per mL): \$0.11

750 mg/150 mL (per mL): \$0.10

1000 mg/200 mL (per mL): \$0.10

1250 mg/250 mL (per mL): \$0.09

1500MG/300ML (per mL): \$0.09

1750MG/350ML (per mL): \$0.09

2000MG/400ML (per mL): \$0.08

Solution (reconstituted) (Firvanq Oral)

25 mg/mL (per mL): \$1.11

50 mg/mL (per mL): \$1.47

Solution (reconstituted) (Vancomycin HCl Intravenous)

1 g (per each): \$3.24 - \$19.54

1.25 g (per each): \$14.40 - \$24.12

1.5 g (per each): \$12.50 - \$29.22

1.75 g (per each): \$44.94

2 g (per each): \$51.36

5 g (per each): \$19.72 - \$108.31

10 g (per each): \$31.31 - \$260.68

100 g (per each): \$600.00

500 mg (per each): \$2.96 - \$9.79

750 mg (per each): \$6.70 - \$11.81

Solution (reconstituted) (Vancomycin HCl Oral)

25 mg/mL (per mL): \$0.83 - \$0.90

50 mg/mL (per mL): \$1.09 - \$1.19

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Capsule, Oral, as hydrochloride:

Vancocin: 125 mg, 250 mg [contains fd&c blue #2 (indigotine, indigo carmine)]

Generic: 125 mg, 250 mg

Solution, Intravenous, as hydrochloride:

Generic: 1 g/200 mL in NaCl 0.9% (200 mL)

Solution Reconstituted, Intravenous, as hydrochloride:

Generic: 500 mg (1 ea); 1000 mg (1 ea); 1 g (1 ea); 5 g (1 ea); 10 g (1 ea)

Extemporaneous Preparations

Note: Vancomycin 25 mg/mL and 50 mg/mL oral solutions are commercially available.

Oral Solution

Using a vial of vancomycin powder for injection (reconstituted to 50 mg/mL), add the appropriate volume for the dose to 30 mL of water. For oral administration, common flavoring syrups may be added to improve taste ([Ref](#)).

25 mg/mL Oral Solution

A vancomycin 25 mg/mL solution in Ora-Sweet and water (1:1) may be prepared by reconstituting vancomycin for injection with sterile water, then dilute with a 1:1 mixture of Ora-Sweet and distilled water to a final concentration of 25 mg/mL; transfer to amber prescription bottle. Stable for 75 days refrigerated or for 26 days at room temperature ([Ref](#)).

Administration: Adult

Intravenous: Administer vancomycin with a final concentration not to exceed 5 mg/mL by IV intermittent infusion over at least 60 minutes and not to exceed a rate of 10 to 15 mg/minute (recommended additional infusion period of ≥ 30 minutes for every 500 mg over 1 g administered ([Ref](#)); in adult patients in need of fluid restriction, a concentration up to 10 mg/mL may be used, but risk of infusion-related reactions is increased. Not for IM administration.

If a maculopapular rash appears on the face, neck, trunk, and/or upper extremities (vancomycin infusion reaction [formerly “red man syndrome”]), slow the infusion rate to over 1½ to 2 hours and increase the dilution volume ([Ref](#)). Hypotension, shock, and cardiac arrest (rare) have also been reported with too rapid of infusion. Administration of antihistamines prior to infusion may prevent or minimize this reaction ([Ref](#)).

Irritant; ensure proper needle or catheter placement prior to and during infusion. Avoid extravasation.

Extravasation management: If extravasation occurs, stop infusion immediately; leave cannula/needle in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula; elevate extremity. Information conflicts regarding the use of dry cold or dry warm compresses ([Ref](#)); however, dry warm compresses may be of benefit in increasing local blood flow to enhance drug removal from the extravasation site. Hyaluronidase may be considered for refractory cases ([Ref](#)).

Hyaluronidase: **Intradermal or SUBQ:** Inject a total of 1 mL (15 units/mL) as 5 separate 0.2 mL injections (using a tuberculin syringe) around the site of

extravasation; if IV catheter remains in place, administer IV through the infiltrated catheter; may repeat in 30 to 60 minutes if no resolution ([Ref](#)).

Antibiotic lock technique (off-label use): Instill prepared vancomycin lock solution into each lumen of the catheter access port using a volume sufficient to fill the catheter (2 to 5 mL) with a dwell time of 48 to 72 hours (dependent on frequency of catheter use). Withdraw lock solution prior to catheter use; replace with fresh vancomycin lock solution after catheter use ([Ref](#)).

Intraperitoneal (off-label route): May administer intermittently (1 exchange daily; allow to dwell ≥ 6 hours) or continuously (with every exchange) ([Ref](#)).

Intraventricular (off-label route): Use preservative-free preparations only. May be administered intraventricularly with a final concentration of 2.5 to 10 mg/mL for the treatment of CSF shunt infections. When administered through a ventricular drain, clamp drain for 15 to 60 minutes before opening the drain to allow vancomycin solution to equilibrate in the CSF ([Ref](#)).

Intravitreal (off-label route): May administer vancomycin intravitreally with a final concentration of 1 mg/0.1 mL NS or sterile water ([Ref](#)).

Oral:

Solution: Shake reconstituted oral solution well before each use.

Injection: Reconstituted powder for injection (not premixed solution) may be diluted and used for oral administration; common flavoring syrups may be added to improve taste.

Enteral feeding tube:

The following recommendations are based upon the best available evidence and clinical expertise. Senior editorial team: Joseph I. Boullata, PharmD, RPh, CNS-S, FASPEN, FACN; Peggi A. Guenter PhD, RN, FASPEN; Kathleen Gura, PharmD, BCNSP, FASHP, FASPEN, FPPA, FMSHP; Mark G. Klang MS, RPh, BCNSP, PhD, FASPEN; Linda Lord, NP, ACNP-BC, CNSC, FASPEN.

Oral capsule: Enteral feeding tube administration utilizing vancomycin oral capsules is not recommended, as capsule content properties may increase the risk of clogging the tube ([Ref](#)).

Oral solution (commercially available):

Gastric tubes (eg, NG, G-tube): Shake well before use. Dilute dose with 30 mL purified water; draw up diluted solution into enteral dosing syringe and administer via feeding tube ([Ref](#)).

Dosage form information: Oral solutions with manufacturer-provided diluents (eg, Firvanq) do not include instructions for enteral feeding tube administration.

General guidance: Hold enteral nutrition during vancomycin administration ([Ref](#)). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) before administration ([Ref](#)). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose ([Ref](#)). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) and restart enteral nutrition ([Ref](#)).

Parenteral formulation for enteral administration:

Note: Use vancomycin powder for injection reconstituted to 50 mg/mL.

Gastric (eg, NG, G-tube) or post-pyloric (eg, J-tube) tubes: Dilute dose in 30 mL purified water; draw up diluted solution into enteral dosing syringe and administer via feeding tube ([Ref](#)).

General guidance: Hold enteral nutrition during vancomycin administration ([Ref](#)). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) before administration ([Ref](#)). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose ([Ref](#)). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) and restart enteral nutrition ([Ref](#)).

Not e: Recommendations may not account for differences in inactive ingredients, osmolality, or other formulation properties that may vary among products from different manufacturers.

Rectal (off-label route): May be administered as a retention enema per rectum ([Ref](#)); 500 mg in 100 to 500 mL of NS, volume may depend on length of segment being treated. If sodium chloride causes hyperchloremia could use solution with lower chloride concentration (eg, LR) ([Ref](#)).

Preparation for Administration: Adult

IV: Reconstitute 500 mg, 750 mg, 1 g, 1.25 g, 1.5 g, 1.75 g, and 2 g vials with SWFI to a final concentration of 50 mg/mL; reconstitute 5 g bulk vials with SWFI to a final concentration of 50 mg/mL or 100 mg/mL; reconstitute 10 g bulk vials with SWFI to a final concentration of 100 mg/mL (**Note:** Volume of SWFI required for dilution of bulk vials may vary by product; refer to manufacturer's labeling for detailed recommendations). Reconstituted solution must be further diluted with at least 100 mL of a compatible diluent (eg, D5W,

NS) per 500 mg of vancomycin prior to parenteral administration. Final concentration should not exceed 5 mg/mL; however, a concentration up to 10 mg/mL may be used in fluid-restricted patients, but risk of infusion-related reactions is increased. **Note:** Premixed flexible bags and galaxy containers are also available.

Intrathecal, intraventricular (off-label route): Vancomycin powder for injection may be diluted to a 2.5, 5, or 10 mg/mL concentration in preservative free 0.9% sodium chloride for administration into the CSF ([Ref](#)).

Oral:

Oral solution: Recommendations vary based on formulation; refer to manufacturer's product labeling for reconstitution instructions.

Powder for injection: Reconstitute powder for injection; flavoring syrups may be added to improve taste. See Extemporaneous Preparations section for details. **Note:** Multiple concentrations described (25 mg/mL, 50 mg/mL); use caution when determining dose volume.

Administration: Pediatric

The following feeding tube recommendations are based upon the best available evidence and clinical expertise. Senior editor panel: Joseph I. Boullata, PharmD, RPh, CNS-S, FASPEN, FACN; Peggi A. Guenter, PhD, RN, FASPEN; Kathleen Gura, PharmD, BCNSP, FASHP, FASPEN, FPPA, FMSHP; Mark G. Klang, MS, RPh, BCNSP, PhD, FASPEN; Linda Lord, NP, ACNP-BC, CNSC, FASPEN.

Note: Recommendations may not account for differences in inactive ingredients, osmolality, or other formulation properties that may vary among products from different manufacturers.

Oral:

Oral solution (commercially available): Shake reconstituted oral solution well before each use. Administer with an accurate measuring device (calibrated oral syringe or measuring cup); do not use a household teaspoon or tablespoon to measure dose (overdosage may occur).

Administration via feeding tube:

Gastric tubes (eg, NG, G-tube): Shake well before use. Dilute dose with 30 mL purified water; draw up diluted solution into enteral dosing syringe and administer via feeding tube ([Ref](#)).

Dosage form information: Oral solutions with manufacturer-provided diluents (eg, Firvanq) do not include instructions for enteral feeding tube administration.

General guidance: Hold enteral nutrition during vancomycin administration ([Ref](#)). Flush feeding tube with the lowest volume of purified water necessary to clear the tube prior to administration based on size of patient and/or feeding tube (eg, neonates: 1 to 3 mL; infants and children: 2 to 5 mL; adolescents: 15 mL); refer to institutional policies and procedures ([Ref](#)). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose ([Ref](#)). Flush feeding tube with an appropriate volume of purified water and restart enteral nutrition ([Ref](#)).

Parenteral formulation for enteral administration: Note: Use vancomycin powder for injection reconstituted to 50 mg/mL.

Oral administration: Reconstituted powder for injection (not premixed solution) may be diluted and used for oral administration; common flavoring syrups may be added to improve taste.

Administration via feeding tube:

Gastric (eg, NG, G-tube) or post-pyloric (eg, J-tube) tubes: Dilute dose in 30 mL purified water; draw up diluted solution into enteral dosing syringe and administer via feeding tube ([Ref](#)).

General guidance: Hold enteral nutrition during vancomycin administration ([Ref](#)). Flush feeding tube with the lowest volume of purified water necessary to clear the tube prior to administration based on size of patient and/or feeding tube (eg, neonates: 1 to 3 mL; infants and children: 2 to 5 mL; adolescents: 15 mL); refer to institutional policies and procedures ([Ref](#)). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose ([Ref](#)). Flush feeding tube with an appropriate volume of purified water and restart enteral nutrition ([Ref](#)).

Capsules:

Administration via feeding tube: NOT recommended: Enteral feeding tube administration utilizing vancomycin oral capsules is not recommended as capsule content properties may increase the risk of clogging the tube ([Ref](#)).

Parenteral:

IV:

Intermittent infusion: Administer intermittent IV infusion over 60 minutes. Vancomycin infusion reaction (formerly "red man syndrome") may occur if the infusion is too rapid. It is not an allergic reaction, but may be characterized by hypotension and/or a maculopapular rash appearing on the face, neck, trunk, and/or upper extremities; if this should occur, slow the infusion rate to administer dose over 90 to 120 minutes ([Ref](#)) and increase the dilution volume; the reaction usually dissipates in 30 to 60 minutes; administration of antihistamines just before the infusion may also prevent or minimize this reaction.

Continuous infusion: After loading dose, administer over 24 hours ([Ref](#)).

Vancomycin is an irritant; ensure proper needle or catheter placement prior to and during infusion. Avoid extravasation. If extravasation occurs, stop infusion immediately; leave cannula/needle in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula; elevate extremity. Information varies regarding the use of dry cold or dry warm compresses; however, dry warm compresses may be of benefit in increasing local blood flow to enhance drug removal from the extravasation site ([Ref](#)). Hyaluronidase may be considered for refractory cases ([Ref](#)).

Intrathecal/Intraventricular: Use preservative-free preparations only. Administer as diluted solution (1 to 10 mg/mL) over 1 to 2 minutes ([Ref](#)). When administered through a ventricular drain, clamp drain for 15 to 60 minutes to allow vancomycin solution to equilibrate in the cerebrospinal fluid (CSF) ([Ref](#)).

Intracatheter (vascular); antibiotic lock technique: Instill prepared vancomycin lock solution into each lumen of the catheter access port using a volume sufficient to fill the catheter with a dwell time of ≥ 8 to 12 hours and up to 72 hours (dependent on frequency of catheter use). Withdraw lock solution prior to catheter use; replace with fresh vancomycin lock solution after catheter use ([Ref](#)).

Rectal: Instill vancomycin enema solution via rectal foley; retain for 1 hour. In pediatric patients the optimal doses have not been established; suggested volumes for pediatric patients: 1 to 3 years of age: 50 mL; 4 to 9 years of age: 75 mL; >10 years of age: 100 mL ([Ref](#)).

Preparation for Administration: Pediatric**Oral:**

Oral solution (Firvanq): Tap bottom of bottle containing powder for oral solution to loosen powder; shake provided grape-flavored diluent. Add about half of the

provided diluent to the bottle containing the powder for oral solution, recap and shake for ~45 seconds. Add rest of diluent and shake for ~30 seconds. Consult product labeling for additional information.

Powder for injection: Reconstitute powder for injection; flavoring syrups may be added to improve taste. See "Extemporaneous Preparations" section for details. **Note:** Multiple concentrations described (25 mg/mL, 50 mg/mL); use caution when determining dose volume.

Parenteral:

Reconstitution:

Single- dose vials (500 to 2,000 mg): Reconstitute with SWFI to a final concentration of 50 mg/mL. Further dilution is needed prior to administration.

Pharmacy bulk vials:

5 g bulk vial: Reconstitute vial with SWFI to a final concentration of 50 or 100 mg/mL; final concentration varies based on manufacturer; see product-specific labeling for reconstitution details. Following reconstitution, further dilution is needed prior to administration.

10 or 100 g bulk vial: Reconstitute vials with SWFI to a final concentration of 100 mg/mL; diluent volume varies by product; refer to product-specific labeling for details. Following reconstitution, further dilution is needed prior to administration. **Note:** 100 g pharmacy bulk vials should not be used for doses <500 mg.

IV infusion: Further dilute the reconstituted solution in a compatible diluent (eg, D5W, NS) to a final concentration ≤ 5 mg/mL. In fluid restricted patients, a concentration of 10 mg/mL may be used, but the risk of infusion reactions increases. For neonatal patients, Vermont Oxford Network recommends a concentration of 5 mg/mL ([Ref](#)). **Note:** Premixed flexible bags and galaxy containers are also available.

Intrathecal/intraventricular: Dilute to 1 to 5 mg/mL concentration in preservative-free NS for administration into the cerebrospinal fluid ([Ref](#)); concentrations of 10 mg/mL have been reported in premature neonates and adults ([Ref](#)).

Rectal enema: Reconstitute powder for injection; further dilution of dose in NS to prepare either a 1 mg/mL or 5 mg/mL enema solution has been used in adults (eg, 500 mg in 100 mL or 500 mL NS) ([Ref](#)); the more dilute solutions (1 mg/mL) have been suggested to ensure delivery to the ascending and transverse colon ([Ref](#)). If sodium chloride causes hyperchloremia, a solution with lower chloride concentration (eg, LR) could be considered ([Ref](#)).

Vesicant/Extravasation Risk

Irritant

Storage/Stability

Capsules: Store at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F).

Flexible bags: Store below 25°C (77°F) in the original package. Use within 28 days of removal from aluminum overpouch. Stable at room temperature for 28 days.

Galaxy containers: Store Galaxy containers at or below -20°C (-4°F). Handle frozen product containers with care; may be fragile in the frozen state. Thaw frozen containers at 25°C (77°F) or 5°C (41°F). Do not immerse in water bath or microwave. Thawed solution in remains chemically stable for 72 hours at 25°C (77°F) or for 30 days when stored at 5°C (41°F). Do not refreeze thawed antibiotics.

Oral solution: Store at 2°C to 8°C (36°F to 46°F) prior to and following reconstitution; discard reconstituted solution after 14 days or if appears hazy or contains particulates.

Vials: Store intact vials at 20°C to 25°C (68°F to 77°F). Reconstitute vial using an appropriate diluent; recommendations may vary by product; refer to manufacturer's labeling for choice of diluent and for appropriate storage conditions and timeframes. Prior to administration, further dilution in a compatible solution is required; recommendations may vary by product; refer to manufacturer's labeling for list of compatible solutions and appropriate storage conditions and timeframes.

Pharmacy bulk packages: Store at 20°C to 25°C (68°F to 77°F). Discard pharmacy bulk packages no later than 4 hours after initial closure puncture.

Use: Labeled Indications

***Clostridioides difficile* infection, treatment (oral):** Treatment of *C. difficile* infection (CDI) in adults and pediatric patients <18 years of age.

Endocarditis (injection):

Corynebacteria (diphtheroids): Treatment of diphtheroid endocarditis in combination with either rifampin, an aminoglycoside, or both in early-onset prosthetic valve endocarditis caused by diphtheroids.

Enterococcal: Treatment of endocarditis caused by enterococci (eg, *Enterococcus faecalis*), in combination with an aminoglycoside.

Staphylococcal: Treatment of staphylococcal endocarditis.

Streptococcal: Treatment of endocarditis due to *Streptococcus viridans* or *Streptococcus bovis*, as monotherapy or in combination with an aminoglycoside.

Enterocolitis (oral): Treatment of enterocolitis caused by *Staphylococcus aureus* (including methicillin-resistant strains) in adults and pediatric patients <18 years of age. **Note:** Staphylococcal enterocolitis is uncommon; the disease and treatment are not well described in the literature (Iwata 2014; Lin 2010).

Staphylococcal infections (injection): Treatment of serious or severe infections (eg, bloodstream infections, bone infections, lower respiratory tract infections, skin and skin structure infections) caused by susceptible strains of methicillin-resistant (beta-lactam-resistant) staphylococci; empiric therapy of infections when methicillin-resistant staphylococci are suspected.

Use: Off-Label: Adult

Catheter-related bloodstream infection, antibiotic lock technique (catheter-salvage therapy); Cerebrospinal fluid shunt infection; Clostridioides difficile infection, prophylaxis; Clostridioides difficile infection, treatment (rectal administration); Cystic fibrosis, acute pulmonary exacerbation; Endophthalmitis, treatment; Intra-abdominal infection, health care-associated; Intracranial abscess (brain abscess, intracranial epidural abscess); Meningitis, bacterial; Peritonitis, treatment (peritoneal dialysis); Prosthetic joint infection; Spinal epidural abscess; Spontaneous bacterial peritonitis; Streptococcus (group B), maternal prophylaxis for prevention of neonatal disease; Surgical prophylaxis

Medication Safety Issues

[Collapse All](#)

Sound-alike/look-alike issues:

Vancomycin may be confused with clindamycin, gentamicin, tobramycin, valACYclovir, vecuronium, Vibramycin

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (epidural and intrathecal medications) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Other safety concerns:

IV vanc is an error-prone stemmed/coined drug name (mistaken as Invanz)

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: [Launch drug interactions program](#))

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Aescin: Antibiotics may increase serum concentrations of Aescin. Management: Consider alternatives to this combination as aescin product labeling states coadministration with antibiotics is not recommended. *Risk D: Consider Therapy Modification*

Aminoglycosides: Vancomycin may increase nephrotoxic effects of Aminoglycosides. Vancomycin may increase neurotoxic effects of Aminoglycosides. Management: Consider avoiding coadministration of aminoglycosides and vancomycin unless clinically indicated. If coadministered, monitor closely for signs of nephrotoxicity and neurotoxicity. *Risk D: Consider Therapy Modification*

Amphotericin B: Vancomycin may increase nephrotoxic effects of Amphotericin B. *Risk C: Monitor*

Bacillus Calmette Guerin (BCG) (Intravesical): Antibiotics may decrease therapeutic effects of Bacillus Calmette Guerin (BCG) (Intravesical). *Risk X: Avoid*

Bacillus Calmette Guerin (BCG) (Percutaneous): Antibiotics may decrease therapeutic effects of Bacillus Calmette Guerin (BCG) (Percutaneous). *Risk C: Monitor*

Bacillus clausii: Antibiotics may decrease therapeutic effects of Bacillus clausii. Management: Bacillus clausii should be taken in between antibiotic doses during concomitant therapy. *Risk D: Consider Therapy Modification*

Cholera Vaccine: Antibiotics may decrease therapeutic effects of Cholera Vaccine. Management: Avoid cholera vaccine in patients receiving systemic antibiotics, and within 14 days following the use of oral or parenteral antibiotics. *Risk X: Avoid*

Cloxacillin: May increase nephrotoxic effects of Vancomycin. *Risk C: Monitor*

Colistimethate: Vancomycin may increase nephrotoxic effects of Colistimethate.

Management: Avoid coadministration of colistimethate and vancomycin whenever possible due to the potential for additive or synergistic nephrotoxicity. If coadministration cannot be avoided, closely monitor renal function. *Risk D: Consider Therapy Modification*

Fecal Microbiota (Live) (Oral): Antibiotics may decrease therapeutic effects of Fecal Microbiota (Live) (Oral). *Risk X: Avoid*

Fecal Microbiota (Live) (Rectal): Antibiotics may decrease therapeutic effects of Fecal Microbiota (Live) (Rectal). *Risk X: Avoid*

Flucloxacillin: May increase nephrotoxic effects of Vancomycin. *Risk C: Monitor*

Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies): Antibiotics may decrease therapeutic effects of Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies). *Risk C: Monitor*

Lactobacillus and Estriol: Antibiotics may decrease therapeutic effects of Lactobacillus and Estriol. *Risk C: Monitor*

Loop Diuretics: May increase nephrotoxic effects of Vancomycin. *Risk C: Monitor*

Mycophenolate: Antibiotics may decrease active metabolite exposure of Mycophenolate. Specifically, concentrations of mycophenolic acid (MPA) may be reduced. *Risk C: Monitor*

Neuromuscular-Blocking Agents: Vancomycin may increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents: May increase serum concentrations of Vancomycin. *Risk C: Monitor*

Piperacillin: May increase nephrotoxic effects of Vancomycin. *Risk C: Monitor*

Sodium Picosulfate: Antibiotics may decrease therapeutic effects of Sodium Picosulfate. Management: Consider using an alternative product for bowel cleansing prior to a colonoscopy in patients who have recently used or are concurrently using an antibiotic. *Risk D: Consider Therapy Modification*

Taurolidine: Vancomycin may decrease therapeutic effects of Taurolidine. *Risk C: Monitor*

Typhoid Vaccine: Antibiotics may decrease therapeutic effects of Typhoid Vaccine. Only the live attenuated Ty21a strain is affected. Management: Avoid use of live attenuated typhoid vaccine (Ty21a) in patients being treated with systemic antibacterial agents. Postpone vaccination until 3 days after cessation of antibiotics and avoid starting antibiotics within 3 days of last vaccine dose. *Risk D: Consider Therapy Modification*

Reproductive Considerations

Pregnancy status should be evaluated in patients who may become pregnant prior to using IV formulations containing the excipients polyethylene glycol (PEG 400) or N-acetyl D-alanine (NADA).

Pregnancy Considerations

Vancomycin crosses the placenta and can be detected in fetal serum, amniotic fluid, and cord blood (Bourget 1991; Reyes 1989). Adverse fetal effects, including sensorineural hearing loss or nephrotoxicity, have not been reported following maternal use during the second or third trimesters of pregnancy.

The safety of IV vancomycin in pregnancy varies by drug product formulation due to excipients and trimester. Formulations containing the excipients polyethylene glycol (PEG 400) or N-acetyl D-alanine (NADA) have caused fetal malformations in animal reproduction studies. During the first or second trimesters of pregnancy, use other available formulations of vancomycin.

The pharmacokinetics of vancomycin may be altered during pregnancy and pregnant patients may need a higher dose of vancomycin. Maternal half-life is unchanged, but the volume of distribution and the total plasma clearance may be increased (Bourget 1991). Individualization of therapy through serum concentration monitoring may be warranted.

Oral vancomycin is recommended for the treatment of *Clostridioides difficile* infections in pregnant patients (ACG [Kelly 2021]).

Vancomycin is recommended as an alternative option to prevent the transmission of group B streptococcal (GBS) disease from mothers to newborns. Untreated asymptomatic GBS disease can result in maternal urinary tract infection, intraamniotic infection, endometritis, preterm labor, and/or stillbirth. Vertical transmission from the mother can cause sepsis, pneumonia, or meningitis in the newborn. Vancomycin IV is recommended for use in women who are at high risk for anaphylaxis to penicillin (or whose risk is unknown), and the GBS isolate is resistant to clindamycin. Dose and rate of infusion should be based on maternal weight and renal function; similar to nonpregnant patients (ACOG 2020).

In patients known to be colonized with methicillin-resistant *S. aureus* (MRSA), a single dose of vancomycin is recommended as part of the antibiotic regimen for prophylactic use prior to cesarean delivery. Monotherapy with vancomycin does not provide sufficient coverage for cesarean delivery surgical prophylaxis (ACOG 2018).

Based on limited data, vancomycin is considered likely compatible with pregnancy when used for the treatment of airway diseases, such as cystic fibrosis (ERS/TSANZ [Middleton 2020]).

Breastfeeding Considerations

Vancomycin is present in human milk following IV administration.

Data related to the presence of vancomycin in human milk are available from a mother who received vancomycin 1 g IV every 12 hours during pregnancy and for at least 1 week prior to sampling. Vancomycin 12.7 mcg/mL was detected in human milk 4 hours after a maternal dose (Reyes 1989).

Vancomycin exhibits minimal oral absorption; therefore, the amount available to pass into milk would be limited following oral administration and unlikely to provide clinically relevant exposure to an infant exposed via human milk.

In general, antibiotics that are present in human milk may cause non-dose-related modification of bowel flora. Monitor infants for GI disturbances, such as thrush or diarrhea (WHO 2002).

Vancomycin is recommended for the treatment of *Clostridioides difficile* infections in breastfeeding patients (ACG [Kelly 2021]) and is considered compatible with breastfeeding when used for the treatment of airway diseases, such as cystic fibrosis (ERS/TSANZ [Middleton 2020]). According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and benefits of treatment to the mother.

Dietary Considerations

May be taken with food.

Monitoring Parameters

IV: Periodic renal function tests, CBC, pregnancy test prior to use for formulation containing PEG 400 and NADA excipients, serial auditory function testing may be helpful to minimize risk of ototoxicity, serum trough vancomycin concentrations in select patients (eg, aggressive dosing, life-threatening infection, seriously ill, unstable renal function, concurrent nephrotoxins, prolonged courses).

AUC monitoring: Frequency of AUC monitoring should be based on clinical judgement; frequent or daily monitoring may be appropriate for hemodynamically unstable patients; hemodynamically stable patients may only require once-weekly monitoring (ASHP/IDSA/PIDS/SIDP [Rybak 2020]).

Trough monitoring:

Hemodynamically stable patients: Draw trough concentrations at least once weekly (ASHP/IDSA/SIDP [Rybak 2009]).

Hemodynamically unstable patients: Draw trough concentrations more frequently or in some instances daily (ASHP/IDSA/SIDP [Rybak 2009]).

Prolonged courses (>3 to 5 days): Draw at least one steady-state trough concentration; repeat as clinically appropriate (ASHP/IDSA/SIDP [Rybak 2009]).

Note: Drawing >1 trough concentration prior to the fourth dose for short course (<3 days) or lower intensity dosing (target trough concentrations <15 mg/L) is not recommended. For patients with uncomplicated skin and soft tissue infections who are not obese and have normal renal function, serum trough monitoring is generally not needed (IDSA [Liu 2011]).

Oral/rectal therapy: Serum sample monitoring is not typically required; systemic absorption of enteral vancomycin may occur in patients with mucosal disruption due to colitis, especially in patients with renal failure. Monitoring serum vancomycin levels may be considered for patients with renal failure who have severe colitis and require a prolonged course of enteral vancomycin (IDSA/SHEA [McDonald 2018]; Pettit 2015).

Reference Range

IV:

Timing of serum samples:

First-order pharmacokinetic analytic equations or Bayesian software to estimate AUC: Requires collection of 2 serum concentrations, postdistributional peak concentration ($C_{\max}$) drawn 1 to 2 hours after infusion and trough concentration ($C_{\min}$) drawn at the end of the dosing interval. It is preferable that a near steady-state postdistributional peak and trough concentration within the same dosing interval (if possible) are used with the equation-based method. Bayesian-derived AUC monitoring does not require steady-state serum concentrations (ASHP/IDSA/PIDS/SIDP [Rybak 2020]).

Trough monitoring: Draw trough concentration just before the administration of a dose at steady-state conditions. Steady-state conditions generally occur approximately after the third dose; therefore, may begin monitoring vancomycin trough concentrations before the fourth dose (usually within 1 hour of administration). More specific recommendations for timing of serum samples may be found in "Monitoring Parameters" (Alvarez 2016; ASHP/IDSA/SIDP [Rybak 2009]).

Target concentrations:

Intermittent infusion:

AUC/minimum inhibitory concentration determined by broth microdilution (MIC_{BMD}): 400 to 600, assuming MIC_{BMD} of 1 mg/L. When MIC_{BMD} is >1 mg/L, probability of attaining an AUC/MIC target of ≥ 400 is low with conventional dosing; higher doses may risk unnecessary toxicity. When MIC_{BMD} is <1 mg/L, decreasing the dose to achieve the AUC/MIC target is not recommended (ASHP/IDSA/PIDS/SIDP [Rybak 2020]).

Trough: 10 to 20 mg/L; target within this range depends on site and severity of infection, as well as clinical response. For pathogens with a minimum inhibitory concentration (MIC) ≤ 1 mg/L, the minimum trough concentration should be 15 mg/L to meet target AUC/MIC of ≥ 400 . For complicated infections (eg, bacteremia, endocarditis, osteomyelitis, meningitis, and hospital-acquired pneumonia caused by *S. aureus*) or infections associated with severe sepsis or septic shock, trough concentrations of 15 to 20 mg/L are recommended to improve penetration and improve clinical outcomes (ASHP/IDSA/SIDP [Rybak 2009]; Liu 2011). The American Thoracic Society (ATS)/Infectious Diseases Society of America (IDSA) guidelines for hospital-acquired pneumonia and the IDSA meningitis guidelines also recommend trough concentrations of 15 to 20 mg/L (Kalil 2016; Tunkel 2004; Tunkel 2017).

Continuous infusion: Target steady-state concentration: 20 to 25 mg/L (ASHP/IDSA/PIDS/SIDP [Rybak 2020]).

Concentrations associated with toxicity: Serum concentration >80 mg/L.

Intraventricular (off-label route): Limited data available (IDSA [Tunkel 2004; Tunkel 2017]; Smetana 2018): Prior to administration of the next intraventricular dose, withdraw a sample of cerebrospinal fluid (CSF). This trough CSF concentration divided by the vancomycin MIC for the isolated bacterial pathogen (inhibitory quotient) should exceed 10 to 20.

Mechanism of Action

Inhibits bacterial cell wall synthesis by blocking glycopeptide polymerization through binding tightly to D-alanyl-D-alanine portion of cell wall precursor

Pharmacokinetics (Adult Data Unless Noted)

Absorption: Oral: Poor; Rectal: significant absorption through inflamed colonic mucosa may occur; Intraperitoneal (IP): 60% of an IP dose absorbed in 6 hours.

Distribution: Distributes widely in body tissue and fluids, except for cerebrospinal fluid (CSF).

V_d :

Neonates, term: 0.57 to 0.69 L/kg (de Hoog 2004).

Infants: 0.56 L/kg (Rainkie 2015).

Children ≤ 6 years of age: 0.61 L/kg (Rainkie 2015).

Children > 6 years of age: 0.47 L/kg (Rainkie 2015).

Adolescents: 0.49 L/kg (Rainkie 2015).

Adults: 0.4 to 1 L/kg (ASHP/IDSA/SIDP [Rybak 2009]); 0.3 to 0.5 L/kg in patients who are morbidly obese (Adane 2015; Bauer 1998; Hong 2015).

Relative diffusion from blood into CSF: Good only with inflammation (exceeds usual MICs).

Children:

CSF concentrations: 0.2 to 17.3 mg/L (de Hoog 2004).

CSF:blood level ratio: Normal meninges: Nil; Inflamed meninges: 7.1% to 68% (de Hoog 2004).

Adults:

Uninflamed meninges: 0 to 4 mg/L; serum concentration dependent (ASHP/IDSA/SIDP [Rybak 2009]).

Inflamed meninges: 6 to 11 mg/L; serum concentration dependent (ASHP/IDSA/SIDP [Rybak 2009]).

CSF:serum level ratio: Normal meninges: Nil; Inflamed meninges: ~80% (Shokouhi 2014).

Protein binding: ~55%.

Metabolism: No apparent metabolism.

Half-life elimination: Biphasic: Terminal:

Preterm neonates (GA: 32 to 34 weeks); PNA ~3 to 5 days: 5.9 to 9.8 hours (Schaad 1980).

Term neonates; PNA ~2 to 3 days: 6.7 hours (Schaad 1980).

Infants: 2.8 hours (Rainkie 2015).

Children < 6 years of age: 2.4 hours (Rainkie 2015).

Children ≥ 6 years of age: 2.9 hours (Rainkie 2015).

Adolescents: 3.2 hours (Rainkie 2015).

Adults: 4 to 6 hours; significantly prolonged with renal impairment.

End-stage renal disease: 7.5 days.

Time to peak, serum: IV: Immediately after completion of infusion.

Excretion: Primarily via glomerular filtration; IV: Urine (75% as unchanged drug in the first 24 hours); Oral: Primarily feces.

Clearance: presence of malignancy in children is associated with an increase in vancomycin clearance.

Neonates: 0.63 to 1.5 mL/minute/kg; dependent on GA and/or PMA (de Hoog 2004).

Pediatric patients: Median: 1.1 mL/minute/kg (range: 0.33 to 1.87 mL/minute/kg) (Marsot 2012).

Adults: 1.6 to 6.2 L/hour (Matzke 1984); patients who are obese: ~ 6 L/hour (rarely exceeds 9 L/hour) (ASHP/IDSA/PIDS/SIDP [Rybak 2020]).

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Pediatric: Extracorporeal membrane oxygenation (ECMO): Reported pharmacokinetic parameters in pediatric patients receiving ECMO vary widely based on ECMO circuitry/filters, age, weight, kidney function, and underlying diseases. In general, volume of distribution may be increased and clearance may be increased or decreased; reported parameters vary significantly; elimination half-life appears to be dependent upon renal function (Amaker 1996; Buck 1998; Cies 2017; Moffett 2018; Mulla 2005; Zylbersztajn 2018).

Older adult: Total systemic and renal clearance may be reduced.

Anti-infective considerations:

Parameters associated with efficacy: **Note:** Ratios, including the minimum inhibitory concentration (MIC), depend upon the methodology used; MIC determined by E-test is typically 1.5 to 2 times MIC determined by broth microdilution (ASHP/IDSA/PIDS/SIDP [Rybak 2020]).

Staphylococcus aureus: $AUC/MIC_{BMD} \geq 400$ mg•hour/L (ASHP/IDSA/PIDS/SIDP [Rybak 2020]; Kullar 2011; Lodise 2014; Moise-Broder 2004); specific cutoff for efficacy has varied slightly between studies. **Note:** The AUC/MIC threshold associated with efficacy has not been clearly established for other pathogens (eg,

Enterococcus spp. and coagulase-negative *Staphylococci*) (Chenhsu 2025; Gwee 2022; Jumah 2018; Kim 2025; Limelette 2025; Sel 2025).

Parameters associated with toxicity: Nephrotoxicity: AUC ≥ 600 to 650 mg•hour/L (adults); ≥ 800 mg•hour/L (pediatrics); risk continues to increase along AUC continuum (Aljefri 2019; ASHP/IDSA/PIDS/SIDP [Rybak 2020]; Fiorito 2018; Kim 2025; Le 2015; Lodise 2020; Sel 2025); $C_{\min} \geq 15$ mg/L (ASHP/IDSA/PIDS/SIDP [Rybak 2020]; Sel 2025; van Hal 2013).

Postantibiotic effect: A short postantibiotic effect has been observed in *E. faecalis* (0.5 to 1 hour) and *S. aureus* (0.6 to 2 hours); slightly longer in *S. epidermidis* (4.3 to 6.5 hours) (Hanberger 1991; Löwdin 1998).

Patient Counseling Points

(For additional information see "[Vancomycin: Patient drug information](#)")

What is this drug used for?

Capsules and oral solution:

- It is used to treat certain types of bowel infections. This includes an infection called C diff.

Injection:

- It is used to treat or prevent bacterial infections.
- If taken by mouth, some products can be used to treat certain types of bowel infections. This includes C diff.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

All products:

- Diarrhea, stomach pain, upset stomach, or throwing up

Capsules and oral solution:

- Gas
- Feeling tired or weak
- Back pain
- Headache

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

All products:

- Kidney problems like unable to pass urine, change in how much urine is passed, blood in the urine, or a big weight gain
- Change in hearing or hearing loss
- Severe skin reactions; red, swollen, blistered, or peeling skin; red or irritated eyes; sores in the mouth, throat, nose, eyes, genitals, or any areas of skin; fever; chills; body aches; shortness of breath; or swollen glands
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Capsules and oral solution:

- Low potassium like muscle pain or weakness, muscle cramps, or a heartbeat that does not feel normal
- Urinary tract infection like blood in the urine, burning or pain when passing urine, feeling the need to pass urine often or right away, fever, lower stomach pain, or pelvic pain
- Swelling of arms or legs

Injection (if given in the vein):

- Fever, chills, or sore throat
- Any unexplained bruising or bleeding
- Pain when passing urine
- Change in balance
- Flushing
- During the infusion, a rash on the face, neck, trunk, and arms

- Shortness of breath or wheezing, itching, muscle pain, chest pain, and signs of low blood pressure like dizziness or passing out
- Redness, burning, pain, swelling, blisters, skin sores, or leaking of fluid where the drug is going into your body
- C diff-associated diarrhea (CDAD) like stomach pain, cramps, or very loose, watery, or bloody stools

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations ([show table](#))

(AE) United Arab Emirates: Cinavan | Vancocin | Vancolon | Vancomicina | Vancomycin | Vancomycin HCL | Vantico; (AR) Argentina: Rivervan | Vancocin | Vancoled | Vancomicina | Vancomicina celtyc | Vancomicina drawer | Vancomicina fabra | Vancomicina filaxis | Vancomicina northia | Vancotenk | Varedet; (AT) Austria: Vancomycin | Vancomycin Abbott | Vancomycin alpharma | Vancomycin AptaPharma | Vancomycin Astro | Vancomycin eberth | Vancomycin eli lilly | Vancomycin hikma | Vancomycin mip | Vancomycin pfizer; (AU) Australia: Dbl vancomycin | Vancocin | Vancomycin | Vancomycin alphapharm | Vancomycin an | Vancomycin baxter | Vancomycin bnm | Vancomycin comp | Vancomycin sandoz | Vancomycin viatris | Vycin; (BD) Bangladesh: Vancin | Vancobac | Vancomin | Vanmycin | Vanogut; (BE) Belgium: Norimko | Vamysin | Vancocin | Vancomycin actavis | Vancomycin aurobindo | Vancomycine Billev pharma | Vancomycine bristol-m. squibb | Vancomycine fresenius kabi | Vancomycine mayne pharma (ben) | Vancomycine mylan; (BG) Bulgaria: Acviscin | Edicin | Vancocin | Vancomycin | Vancomycin mip | Vancomycin rompharm; (BR) Brazil: Celovan | Cloridrato de vancomicina | Hicovan | Novamicin | Vanclomin | Vancocid | Vancocina | Vancocina cp | Vancomicina | Vancoson; (CH) Switzerland:

Vancocin | Vancomycin fairmed | Vancomycin labatec | Vancomycin pfizer | Vancomycin sandoz;
 (CI) Côte d'Ivoire: Edicin; (CL) Chile: Kovan | Vancomicina; (CN) China: Fang ke lin | Hai zheng mei
 neng | Lai ke xin | Vancocin | Vancocin cp | Vanconal | Vancorin; (CO) Colombia: Alvanco |
 Enterocaps | Injovan | Sumicina | Tancofeto | Vanaurus | Vanbiotic | Vancocin cp | Vancocina hcl |
 Vancomicina | Vancomicina clorhidrato | Vancosala | Vaneotik | Vankina | Vanoxat | Varedet; (CR)
 Costa Rica: Cloridrato de vancomicina | Vanaurus | Vancomicina | Vancomicina clorhidrato |
 Vancomicina normon | Vancomycin | Vanconor | Varedet; (CZ) Czech Republic: Edicin | Lyphocin |
 Vancocin | Vancoled | Vancomycin actavis | Vancomycin AptAPharma | Vancomycin hikma |
 Vancomycin Kabi | Vancomycin mylan | Vancomycin olikla | Vancomycin Pharmaswiss |
 Vankomycin Pfizer; (DE) Germany: Ab vancomycin | Vanco | Vanco 500 ar | Vanco cell | Vanco saar
 | Vancomycin | Vancomycin Abbott | Vancomycin Enterocaps | Vancomycin Kabi | Vancomycin
 lyomark | Vancomycin noridem | Vancomycin Ratiopharm | Vancosan; (DK) Denmark: Vancomycin
 alpharma; (DO) Dominican Republic: Megamicin | Valepta | Vanaurus | Vancocina cp |
 Vancomicina; (EC) Ecuador: Clorhidrato de vancomicina polvo para inyeccion | Kup vancomycin hcl
 | Vanbiotic | Vancomax | Vancomicina | Vancomicina clorhidrato | Vancomicina hcl | Vancomicina
 richet | Vanin | Varedet; (EE) Estonia: Edicin | Vancocin | Vancocin matrigel | Vancomycin |
 Vancomycin Kabi | Vancomycin sandoz | Vancomycin teva | Vancosan | Vanmixan; (EG) Egypt:
 Edicin | Vancocin cp | Vancogut | Vancolon | Vancomix | Vancomycin | Vetovancin; (ES) Spain:
 Diatracin | Vancomicina abbott | Vancomicina chiesi | Vancomicina Combino Pharm | Vancomicina
 normon | Vancomicina Pfizer | Vancomicina Sandoz; (ET) Ethiopia: Vancolon | Vancomycin |
 Vancomycin aurobindo | Vancomycin sala | Vancoten | Vansafe | Vansafe cp; (FI) Finland: Orivan |
 Vancocin | Vancomycin Abbott | Vancomycin alpharma | Vancomycin orion | Vancomycin strides |
 Vancomycin xellia | Vancosan; (FR) France: Vancocine | Vancomycine dakota | Vancomycine hikma
 | Vancomycine kabi | Vancomycine lilly | Vancomycine merck | Vancomycine Sandoz; (GB) United
 Kingdom: Vancocin | Vancomycin; (GR) Greece: Vancomycin | Vancomycin HCL | Vancomycin Kabi |
 Vancomycin/generics | Vancomycin/noridem | Vancomycin/Norma | Vancomycin/Vianex |
 Vancomycin/Vocate | Vancoten | Voncon | Vondem | Voxin; (GT) Guatemala: Vancomicina |
 Vancorean; (HK) Hong Kong: Vancocin | Vancoled | Vancomycin | Vancotex; (HR) Croatia: Edicin |
 Vancomycin Kabi | Vankomicin mip | Vankomicin Pharmaswiss | Vankomicin viatris; (HU) Hungary:
 Edicin | Vanco ratiopharm | Vancoled | Vancomicina normon | Vancomycin | Vancomycin actavis |
 Vancomycin AptAPharma | Vancomycin human | Vancomycin Kabi | Vancomycin rompharm |
 Vancomycin-b; (ID) Indonesia: Ledervan | Vancep | Vancocin | Vancodex | Vancomycin | Vancosan
 | Vantocil; (IE) Ireland: Vancocin | Vancomycin | Vancomycin mylan; (IL) Israel: Vancoavenir; (IN)
 India: Celovan | Cp van | Cytovan | E staph | Forstaf | G vanc | Ledovanc | Uvanco | Vanco-l |
 Vancocin | Vancocin cp | Vancogen | Vancogram | Vancoled | Vancolym | Vancomate |
 Vancomycin | Vancopil | Vancorid CP | Vancorin | Vancosa | Vancotech cp | Vancotroy | Vancowar
 cp | Vanking | Vanlid | Vansafe cp | Vantech | Vantox cp | Venbruta | Vmycin; (IQ) Iraq: Vancoject |
 Voxin; (IT) Italy: Copovan | Farmaciclin | Levovanox | Maxivanil | Vancocina | Vancomicina actavis |
 Vancomicina kabi | Vancomicina mylan | Zengac; (JO) Jordan: Tq vanc | Vanco | Vancolon |
 Vancomycin | Vinkamine | Zermacin; (JP) Japan: Solrein | Storacin | Vancomycin HCL | Vancomycin
 hcl shionogi | Vancomycin hydrochloride Hospira | Vancomycin hydrochloride meiji | Vancomycin
 hydrochloride merck hoei | Vancomycin hydrochloride nig | Vancomycin Hydrochloride Sandoz |
 Vancomycin hydrochloride Sawai | Vancomycin hydrochloride Taiyo | Vancomycin kobayashi |
 Vancomycin meiji | Vancomycin towa | Vanmycin; (KE) Kenya: Vanbact | Vanco | Vancocin cp |
 Vancolon | Vancomicina azevedos | Vancomycin hikma | Vinkamine | Zermacin; (KR) Korea,
 Republic of: Allvanco | Auskovancomycin | Bc vancomycin hcl | Daewoong vancomycin | Dong a
 vacomycin hydrochloride | Dongkwang vancomycin | Hanomycin | Inno.n vancomycin hcl | Kukje
 vancomycin hcl | Medica vancomycin hydrochloride | Pfizer vancomycin | Samsung vancomycin hci
 | Union vancomycin | Vanco kit | Vancocin | Vancocin cp | Vancoled | Vancomycin | Vancomycin
 hcl korea united | Vancomycin hcl myungmoon | Vancostacin | Vancotreacin | Vancotrin |

Vancozin | Varicin | Yooyoung vancomycin hcl; (KW) Kuwait: Vancocin | Vancolon | Vancomycin | Voxin; (LB) Lebanon: Vanco | Vancocin | Vancolon | Vancomycin labatec | Vancomycin sandoz | Vancomycine | Vancomycine mylan | Voxin | Zermacin; (LT) Lithuania: Edicin | Vancocin | Vancomicina hikma | Vancomycin | Vancomycin cnp | Vancomycin dr. eberth | Vancomycin Kabi | Vancomycin mylan | Vancomycin teva | Vancosan; (LU) Luxembourg: Vancocin; (LV) Latvia: Edicin | Vancocin | Vancomycin | Vancomycin Kabi | Vancomycin sandoz | Vancomycin teva | Vancosan | Vancotev | Vanmixan; (MA) Morocco: Vancocine; (MX) Mexico: Alvarcin | Averixem | Citerin | Estafimac | Estavam | Loricer | Vacsol | Valcidem | Vanaurus | Vancam | Vancocin | Vancocin cp | Vancomicina | Vancomicina gi | Vancomicina gi lem | Vancomicina gi pre | Vancomicina Probifasa | Vancox | Varamyn | Yectafhal; (MY) Malaysia: Celovan | Dhacocin | Tancofeto | Vancocin cp | Vancoled | Vancomycin | Vancomycin lyomark | Vancorin | Vancotex; (NG) Nigeria: Vancomycin sandoz; (NI) Nicaragua: Vanaurus | Vancomicina | Vancomicina clorhidrato | Varedet; (NL) Netherlands: Vancocin | Vancocin cp | Vancomycin | Vancomycine | Vancomycine erc | Vancomycine hikma | Vancomycine PCH | Vancomycine Sandoz | Vancomycine strides; (NO) Norway: Vancocin | Vancocin matrigel | Vancomycin | Vancomycin axellia | Vancomycin strides | Vancomycin xellia; (NZ) New Zealand: Lyphocin | Vancocin cp; (PA) Panama: Vanaurus | Vancomax | Vancomicina | Vancomicina clorhidrato | Vanconor | Varedet; (PE) Peru: Bactvan | Celovan | Vancobiotic | Vancocina | Vancokem | Vancoliph | Vancomed | Vancomek | Vancomicina | Vancomicina cp | Vanconex | Vanconor | Vancoran | Vancotech cp; (PH) Philippines: Aekovan | Avancomycin | Imvaco | Labvacom | Mersa | Univan | Vancin | Vancocin cp | Vancokem | Vancolon | Vancomet | Vancomycin | Vancophar | Vancorest | Vancosan | Vancouver | Vancowell | Vanosyn | Varsa | Vencusin; (PK) Pakistan: Myvan | Vactrox | Vanbact | Vancare | Vanco | Vancocin | Vancocin cp | Vancolon | Vancom | Vancomark | Vancomycin | Vancosol | Vancotech | Vancotic | Vancowrd | Vanmycin | Vanqid | Vanzy | Vecocin | Vinjec | Zengac; (PL) Poland: Acviscin | Edicin | Lyphocin | Vancocin | Vancoled | Vancomycin AptaPharma | Vancomycin HCL | Vancomycin Kabi | Vancomycin swiss2care | Vancotex; (PR) Puerto Rico: First vancomycin | Firvanq | Vancocin | Vancoled | Vancomycin | Vancomycin HCL; (PT) Portugal: Glipep i p | Vancocina cp | Vancomicina actavis | Vancomicina azevedos | Vancomicina billev | Vancomicina combino | Vancomicina generis | Vancomicina hikma | Vancomicina kabi | Vancomicina labesfal | Vancomicina normon | Vancomicina Pfizer; (PY) Paraguay: Biomicine | Decamil | Vancap | Vancocin | Vancomax | Vancomicina abbott | Vancomicina dallas | Vancomicina dutriec | Vancomicina klonal | Vancomicina libra | Vancomicina nothia | Vancomicina quimfa | Vancomicina richet | Vancotie | Varedet; (QA) Qatar: Firvanq | Vanco | Vancomax | Vancopol | Vantox-CP | Vinkamine | Zermacin; (RO) Romania: Edicin | Levovanox | Vancocin | Vancoled | Vancomicina | Vancomicina actavis | Vancomicina atb | Vancomicina kabi | Vancomicina pharماسwiss | Vancomicina rompharm | Vancomycin teva | Vancotek; (RU) Russian Federation: Edicin | Vancobact | Vancocin | Vancoled | Vancomabol | Vancomycin | Vancomycin j | Vancomycin teva | Vancorus | Vancosin | Vankoter af | Vanmixan | Vero vancomycin; (SA) Saudi Arabia: Colat | Tostaf | Vanco | Vancocin | Vancolon | Vancomycin | Vancomycin alphapharma | Vancomycin HCL | Vancomycin mylan | Vanin; (SE) Sweden: Vancocin | Vancomycin actavis | Vancomycin alpharma | Vancomycin hameln | Vancomycin Hospira | Vancomycin mip | Vancomycin Norcox | Vancomycin orion | Vancomycin sandoz | Vancomycin strides | Vancomycin Tika | Vancomycin viatris | Vancomycin xellia | Vancoscand; (SG) Singapore: Dhacocin | Vancomycin; (SI) Slovenia: Edicin | Vancocyn CP | Vankomicin apta | Vankomicin kabi | Vankomicin mylan | Vankomicin Pfizer | Vankomicin Pharmaswiss; (SK) Slovakia: Edicin | Vancocin | Vancoled | Vancomycin Kabi | Vancomycin mip | Vancomycin mylan | Vancomycin olikla | Vancomycin Pharmaswiss | Vankomicin Actavis; (SR) Suriname: Vancomycine; (SV) El Salvador: Vanaurus; (TH) Thailand: Celovan | Edicin | G vanc | Vancin S | Vancocid | Vancocin cp | Vancogen | Vancolon | Vancomycin | Vancomycin Abbott | Vancomycin dbi | Vancomycin fujisawa | Vangacin; (TN) Tunisia: Vancomed | Vancomycin | Vancomycine; (TR) Turkey: Anko L | Edicin | Vamycin | Vancocin cp | Vancomax | Vancomycin HCL |

Vancorin | Vancotek | Vankopol; (TW) Taiwan: U-Vanco | Vanco | Vancocin cp | Vancomycin | Vancomycin cyh | Vanlyo; (UA) Ukraine: Edicin | Licovanum | Vanco | Vancocin | Vancogen | Vancomek | Vancomycin pharmex | Vancomycin rompharm | Vancomycin vista | Vancomycinum | Vancotex | Vancovan | Vancum | Vanmixan; (UG) Uganda: Stavance | Vamocin | Vanbact | Vancolon | Vinkamine; (UY) Uruguay: Rivervan | Vancomicina | Vancomicina Eron | Vancomicina ion | Vancomicina libra | Vancomicina northia | Varedet; (VE) Venezuela, Bolivarian Republic of: Pravano | Vagram | Vanbiotic | Vancina | Vancobehr | Vancocyn | Vancoled | Vancomicina | Vancowell | Vanlid; (VN) Viet Nam: Novelcin | Valacin | Valbivi | Vecmid; (ZA) South Africa: Aspen Vancomycin | Comavyn | Cytovan | Mylan Vancomycin | Vancocin cp | Vancomycin | Vancomycine fauldin; (ZM) Zambia: Vancocin cp; (ZW) Zimbabwe: Vancomycin HCL

Use of UpToDate is subject to the [Terms of Use](#).

REFERENCES

1. Adane ED, Herald M, Koura F. Pharmacokinetics of vancomycin in extremely obese patients with suspected or confirmed *Staphylococcus aureus* infections. *Pharmacotherapy*. 2015;35(2):127-139. doi:10.1002/phar.1531 [PubMed [25644478](#)]
2. Al-Jafar H, Al-Yousef A, Al-Shatti S, Al-Banwan K. Drug-immune thrombocytopenia with thrombosis versus heparin-induced thrombocytopenia: a critical clinical controversy. *Case Rep Nephrol Dial*. 2015;5(2):152-159. doi:10.1159/000435806 [PubMed [26266247](#)]
3. Aljefri DM, Avedissian SN, Rhodes NJ, Postelnick MJ, Nguyen K, Scheetz MH. Vancomycin area under the curve and acute kidney injury: a meta-analysis. *Clin Infect Dis*. 2019;69(11):1881-1887. doi:10.1093/cid/ciz051 [PubMed [30715208](#)]
4. Al-Jeraisy M, Phelps SJ, Christensen ML, Einhaus S. Intraventricular vancomycin in pediatric patients with cerebrospinal fluid shunt infections. *J Pediatr Pharmacol Ther*. 2004;9(1):36-42.
5. Álvarez R, López Cortés LE, Molina J, Cisneros JM, Pachón J. Optimizing the clinical use of vancomycin. *Antimicrob Agents Chemother*. 2016;60(5):2601-2609. doi: 10.1128/AAC.03147-14. [PubMed [26856841](#)]
6. Alvarez-Arango S, Ogunwole SM, Sequist TD, Burk CM, Blumenthal KG. Vancomycin infusion reaction - moving beyond "Red Man Syndrome". *N Engl J Med*. 2021;384(14):1283-1286. doi:10.1056/NEJMp2031891 [PubMed [33830710](#)]
7. Amaker RD, DiPiro JT, Bhatia J. Pharmacokinetics of Vancomycin in Critically Ill Infants Undergoing Extracorporeal Membrane Oxygenation. *Antimicrob Agents Chemother*. 1996;40(5):1139-1142. [PubMed [8723454](#)]
8. American Academy of Pediatrics (AAP). In: Kimberlin DW, Banerjee R, Barnett ED, Lynfield R, Sawyer MH, eds. *Red Book: 2024-2027 Report of the Committee on Infectious Diseases*. 33rd ed. American Academy of Pediatrics; 2024.
9. American Academy of Pediatrics Committee on Infectious Diseases. *Treatment of bacterial meningitis*. *Pediatrics*. 1988;81(6):904-907. [PubMed [3368290](#)]
10. American College of Obstetricians and Gynecologists (ACOG) Committee on Practice Bulletins-Obstetrics. ACOG practice bulletin no. 199: use of prophylactic antibiotics in labor and delivery. *Obstet Gynecol*. 2018;132(3):e103-e119. [PubMed [30134425](#)]

11. American College of Obstetricians and Gynecologists (ACOG). Prevention of group B streptococcal early-onset disease in newborns: ACOG committee opinion, number 797 [published correction appears in *Obstet Gynecol*. 2020;135(4):978-979]. *Obstet Gynecol*. 2020;135(2):e51-e72. doi:10.1097/AOG.0000000000003668 [PubMed [31977795](#)]
12. American Thoracic Society and Infectious Diseases Society of America, "Guidelines for the Management of Adults With Hospital-Acquired, Ventilator-Associated, and Healthcare-Associated Pneumonia," *Am J Respir Crit Care Med*, 2005, 171(4):388-416. [PubMed [15699079](#)]
13. An SH, Lee EM, Kim JY, Gwak HS. Vancomycin pharmacokinetics in critically ill neonates receiving extracorporeal membrane oxygenation. *Eur J Hosp Pharm*. 2020;27(e1):e25-e29. doi:10.1136/ejhpharm-2018-001720 [PubMed [32296501](#)]
14. An SY, Hwang EK, Kim JH, et al. Vancomycin-associated spontaneous cutaneous adverse drug reactions. *Allergy Asthma Immunol Res*. 2011;3(3):194-198. doi:10.4168/aaair.2011.3.3.194 [PubMed [21738885](#)]
15. Anderson DJ, Podgorny K, Berríos-Torres SI, et al. Strategies to prevent surgical site infections in acute care hospitals: 2014 update. *Infect Control Hosp Epidemiol*. 2014;35(6):605-627. doi: 10.1086/676022. [PubMed [24799638](#)]
16. Anderson DJ, Sexton DJ. Antimicrobial prophylaxis for prevention of surgical site infection in adults. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed June 4, 2020.
17. Anne S, Middleton E Jr, Reisman RE. Vancomycin anaphylaxis and successful desensitization. *Ann Allergy*. 1994;73(5):402-404. [PubMed [7978531](#)]
18. Apisarnthanarak A, Razavi B, Mundy LM. Adjunctive intracolonic vancomycin for severe *Clostridium difficile* colitis: case series and review of the literature. *Clin Infect Dis*. 2002;35(6):690-696. doi:10.1086/342334 [PubMed [12203166](#)]
19. Arnaud FCS, Libório AB. Attributable nephrotoxicity of vancomycin in critically ill patients: a marginal structural model study. *J Antimicrob Chemother*. 2020;75(4):1031-1037. doi:10.1093/jac/dkz520 [PubMed [31904834](#)]
20. Arnold DM, Nazi I, Warkentin TE, et al. Approach to the diagnosis and management of drug-induced immune thrombocytopenia. *Transfus Med Rev*. 2013;27(3):137-145. doi:10.1016/j.tmr.2013.05.005 [PubMed [23845922](#)]
21. Aronoff GR, Bennett WM, Berns JS, et al. *Drug Prescribing in Renal Failure: Dosing Guidelines for Adults and Children*. 5th ed. Philadelphia, PA: American College of Physicians; 2007.
22. Arroyo-Mercado F, Khudyakov A, Chawla GS, Cantres-Fonseca O, McFarlane IM. Red man syndrome with oral vancomycin: a case report. *Am J Med Case Rep*. 2019;7(1):16-17. doi:10.12691/ajmcr-7-1-5 [PubMed [31123702](#)]
23. Aster RH, Bougie DW. Drug-induced immune thrombocytopenia. *N Engl J Med*. 2007;357(6):580-587. doi:10.1056/NEJMra066469 [PubMed [17687133](#)]
24. Austin JP, Foster BA, Empey A. Replace red man syndrome with vancomycin flushing reaction. *Hosp Pediatr*. 2020;10(7):623-624. doi:10.1542/hpeds.2020-0125 [PubMed [32571794](#)]
25. Baddour LM, Chen AF. Periprosthetic joint infection: Treatment. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed May 16, 2022.
26. Baddour LM, Flynn PM. Infections of cerebrospinal fluid shunts. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed October 21, 2019.

27. Baddour LM, Wilson WR, Bayer AS, et al; American Heart Association Committee on Rheumatic Fever, Endocarditis, and Kawasaki Disease of the Council on Cardiovascular Disease in the Young, Council on Clinical Cardiology, Council on Cardiovascular Surgery and Anesthesia, and Stroke Council. Infective endocarditis in adults: diagnosis, antimicrobial therapy, and management of complications: a scientific statement for healthcare professionals from the American Heart Association. *Circulation*. 2015;132(15):1435-1486. doi:10.1161/CIR.0000000000000296 [PubMed [26373316](#)]
28. Baddour LM, Wilson WR, Bayer AS, et al, "Infective Endocarditis. Diagnosis, Antimicrobial Therapy, and Management of Complications: A Statement for Healthcare Professionals From the Committee on Rheumatic Fever, Endocarditis, and Kawasaki Disease, Council on Cardiovascular Disease in the Young, and the Councils on Clinical Cardiology, Stroke, and Cardiovascular Surgery and Anesthesia, American Heart Association - Executive Summary: Endorsed by the Infectious Diseases Society of America," *Circulation*, 2005, 111(23):3167-184. [PubMed [15956145](#)]
29. Bailey P, Gray H. An elderly woman with 'red man syndrome' in association with oral vancomycin therapy: a case report. *Cases J*. 2008;1(1):111. doi:10.1186/1757-1626-1-111 [PubMed [18710566](#)]
30. Bailie GR, Neal D. Vancomycin ototoxicity and nephrotoxicity. A review. *Med Toxicol Adverse Drug Exp*. 1988;3(5):376-386. doi:10.1007/BF03259891 [PubMed [3057327](#)]
31. Baker CJ. Prevention of early-onset group B streptococcal disease in neonates. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed September 8, 2020.
32. Baltimore RS, Gewitz M, Baddour LM, et al; American Heart Association Rheumatic Fever, Endocarditis, and Kawasaki Disease Committee of the Council on Cardiovascular Disease in the Young and the Council on Cardiovascular and Stroke Nursing. Infective endocarditis in childhood: 2015 update: a scientific statement from the American Heart Association. *Circulation*. 2015;132(15):1487-1515. doi:10.1161/CIR.0000000000000298 [PubMed [26373317](#)]
33. Bamogaddam RF, Alamer A, Alqarni S, et al. Incidence and predictors of vancomycin nephrotoxicity and mortality in patients with chronic liver disease: a two-center retrospective cohort study. *BMC Infect Dis*. 2025;25(1):375. doi:10.1186/s12879-025-10763-3 [PubMed [40102766](#)]
34. Bankhead R, Boullata J, Brantley S, et al. Enteral nutrition practice recommendations. *JPEN J Parenter Enteral Nutr*. 2009;33(2):122-167. doi:10.1177/0148607108330314 [PubMed [19171692](#)]
35. Bao H, Lighter J, Dubrovskaya Y, et al. Oral vancomycin as secondary prophylaxis for *Clostridioides difficile* infection. *Pediatrics*. 2021;148(2):e2020031807. doi:10.1542/peds.2020-031807 [PubMed [34330867](#)]
36. Baptista JP, Roberts JA, Sousa E, Freitas R, Devezza N, Pimentel J. Decreasing the time to achieve therapeutic vancomycin concentrations in critically ill patients: developing and testing of a dosing nomogram. *Crit Care*. 2014;18(6):654. doi:10.1186/s13054-014-0654-2 [PubMed [25475123](#)]
37. Barshak MB. Antimicrobial approach to intra-abdominal infections in adults. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed June 8, 2021.
38. Bateman DA, Thomas W, Parravicini E, Polesana E, Locatelli C, Lorenz JM. Serum creatinine concentration in very-low-birth-weight infants from birth to 34-36 wk postmenstrual age. *Pediatr Res*. 2015;77(5):696-702. [PubMed [25675426](#)]
39. Bauer LA, Black DJ, Lill JS. Vancomycin dosing in morbidly obese patients. *Eur J Clin Pharmacol*. 1998;54(8):621-625. doi:10.1007/s002280050524 [PubMed [9860149](#)]
40. Bellón T. Mechanisms of severe cutaneous adverse reactions: recent advances. *Drug Saf*. 2019;42(8):973-992. doi:10.1007/s40264-019-00825-2 [PubMed [31020549](#)]

41. Benner KW, Worthington MA, Kimberlin DW, et al, "Correlation of Vancomycin Dosing to Serum Concentrations in Pediatric Patients: A Retrospective Database Review," *J Pediatr Pharmacol Ther*, 2009, 14:86-93.
42. Benner KW, Worthington MA, Kimberlin DW, Hill K, Buckley K, Tofil NM. Correlation of vancomycin dosing to serum concentrations in pediatric patients: a retrospective database review. *J Pediatr Pharmacol Ther*. 2009;14(2):86-93. doi: 10.5863/1551-6776-14.2.86. [PubMed [23055895](#)]
43. Berbari EF, Kanj SS, Kowalski TJ, et al; Infectious Diseases Society of America. 2015 Infectious Diseases Society of America (IDSA) clinical practice guidelines for the diagnosis and treatment of native vertebral osteomyelitis in adults. *Clin Infect Dis*. 2015;61(6):e26-e46. doi:10.1093/cid/civ482 [PubMed [26229122](#)]
44. Bergman MM, Glew RH, Ebert TH. Acute interstitial nephritis associated with vancomycin therapy. *Arch Intern Med*. 1988;148(10):2139-2140. [PubMed [3178372](#)]
45. Berríos-Torres SI, Umscheid CA, Bratzler DW, et al; Healthcare Infection Control Practices Advisory Committee. Centers for Disease Control and Prevention guideline for the prevention of surgical site infection, 2017. *JAMA Surg*. 2017;152(8):784-791. doi: 10.1001/jamasurg.2017.0904. [PubMed [28467526](#)]
46. Berthaud R, Benaboud S, Hirt D, et al. Early Bayesian dose adjustment of vancomycin continuous infusion in children: a randomized controlled trial. *Antimicrob Agents Chemother*. Published online October 7, 2019. doi:10.1128/AAC.01102-19 [PubMed [31591117](#)]
47. Biggins SW, Angeli P, Garcia-Tsao G, et al. Diagnosis, evaluation, and management of ascites, spontaneous bacterial peritonitis and hepatorenal syndrome: 2021 practice guidance by the American Association for the Study of Liver Diseases. *Hepatology*. 2021;74(2):1014-1048. doi:10.1002/hep.31884 [PubMed [33942342](#)]
48. Bilbao-Meseguer I, Rodríguez-Gascón A, Barrasa H, Isla A, Solinís MÁ. Augmented renal clearance in critically ill patients: a systematic review. *Clin Pharmacokinet*. 2018;57(9):1107-1121. doi:10.1007/s40262-018-0636-7 [PubMed [29441476](#)]
49. Black E, Lau TT, Ensom MH. Vancomycin-induced neutropenia: is it dose- or duration-related? *Ann Pharmacother*. 2011;45(5):629-638. doi:10.1345/aph.1P583 [PubMed [21521866](#)]
50. Blumenthal KG, Peter JG, Trubiano JA, Phillips EJ. Antibiotic allergy. *Lancet*. 2019;393(10167):183-198. doi:10.1016/S0140-6736(18)32218-9 [PubMed [30558872](#)]
51. Blunden M, Zeitlin D, Ashman N, Fan SL. Single UK centre experience on the treatment of PD peritonitis--antibiotic levels and outcomes. *Nephrol Dial Transplant*. 2007;22(6):1714-1719. doi:10.1093/ndt/gfm079 [PubMed [17369615](#)]
52. Bodilsen J, D'Alessandris QG, Humphreys H, et al. European Society of Clinical Microbiology and Infectious Diseases guidelines on diagnosis and treatment of brain abscess in children and adults. *Clin Microbiol Infect*. 2024;30(1):66-89. doi:10.1016/j.cmi.2023.08.016 [PubMed [37648062](#)]
53. Boullata JI. *Guidebook on Enteral Medication Administration*. Silver Spring, MD: American Society for Parenteral and Enteral Nutrition; 2019.
54. Boullata JI, Carrera AL, Harvey L, et al; ASPEN Safe Practices for Enteral Nutrition Therapy Task Force, American Society for Parenteral and Enteral Nutrition. ASPEN safe practices for enteral nutrition therapy. *JPEN J Parenter Enteral Nutr*. 2017;41(1):15-103. doi:10.1177/0148607116673053 [PubMed [27815525](#)]
55. Bose S, Wurm E, Popovich MJ, Silver BJ. Drug-induced immune-mediated thrombocytopenia in the intensive care unit. *J Clin Anesth*. 2015;27(7):602-605. doi:10.1016/j.jclinane.2015.06.021 [PubMed [26260647](#)]
56. Bourget P, Fernandez H, Delouis C, et al, "Transplacental Passage of Vancomycin During the Second Trimester of Pregnancy," *Obstet Gynecol*, 1991, 78(5 Pt 2):908-11. [PubMed [1923224](#)]

57. Bower WA, Yu Y, Person MK, et al. CDC guidelines for the prevention and treatment of anthrax, 2023. *MMWR Recomm Rep*. 2023;72(6):1-47. doi:10.15585/mmwr.rr7206a1 [PubMed [37963097](#)]
58. Bradley JS, Byington CL, Shah SS, et al; Pediatric Infectious Diseases Society and the Infectious Diseases Society of America. The management of community-acquired pneumonia in infants and children older than 3 months of age: clinical practice guidelines by the Pediatric Infectious Diseases Society and the Infectious Diseases Society of America. *Clin Infect Dis*. 2011;53(7):e25-e76. [PubMed [21880587](#)]
59. Bradley JS, Nelson JD, Barnett ED, et al, eds. *Nelson's Pediatric Antimicrobial Therapy*. 30th ed. American Academy of Pediatrics; 2024.
60. Bratzler DW, Dellinger EP, Olsen KM, et al; American Society of Health-System Pharmacists; Infectious Diseases Society of America; Surgical Infection Society; Society for Healthcare Epidemiology of America. Clinical practice guidelines for antimicrobial prophylaxis in surgery. *Am J Health Syst Pharm*. 2013;70(3):195-283. [PubMed [23327981](#)]
61. Britton RA, Young VB. Role of the intestinal microbiota in resistance to colonization by *Clostridium difficile*. *Gastroenterology*. 2014;146(6):1547-1553. doi:10.1053/j.gastro.2014.01.059 [PubMed [24503131](#)]
62. Brockow K, Przybilla B, Aberer W, et al. Guideline for the diagnosis of drug hypersensitivity reactions: S2K-Guideline of the German Society for Allergology and Clinical Immunology (DGAKI) and the German Dermatological Society (DDG) in collaboration with the Association of German Allergologists (AeDA), the German Society for Pediatric Allergology and Environmental Medicine (GPA), the German Contact Dermatitis Research Group (DKG), the Swiss Society for Allergy and Immunology (SGAI), the Austrian Society for Allergology and Immunology (ÖGAI), the German Academy of Allergology and Environmental Medicine (DAAU), the German Center for Documentation of Severe Skin Reactions and the German Federal Institute for Drugs and Medical Products (BfArM). *Allergo J Int*. 2015;24(3):94-105. doi:10.1007/s40629-015-0052-6 [PubMed [26120552](#)]
63. Brouwer MC, Tunkel AR, McKhann GM 2nd, van de Beek D. Brain abscess. *N Engl J Med*. 2014;371(5):447-456. doi:10.1056/NEJMra1301635 [PubMed [25075836](#)]
64. Brown KA, Khanafer N, Daneman N, Fisman DN. Meta-analysis of antibiotics and the risk of community-associated *Clostridium difficile* infection. *Antimicrob Agents Chemother*. 2013;57(5):2326-2332. doi:10.1128/AAC.02176-12 [PubMed [23478961](#)]
65. Brummett RE, Fox KE. Vancomycin- and erythromycin-induced hearing loss in humans. *Antimicrob Agents Chemother*. 1989;33(6):791-796. doi:10.1128/aac.33.6.791 [PubMed [2669623](#)]
66. Brummett RE, Fox KE, Jacobs F, Kempton JB, Stokes Z, Richmond AB. Augmented gentamicin ototoxicity induced by vancomycin in guinea pigs. *Arch Otolaryngol Head Neck Surg*. 1990;116(1):61-64. doi:10.1001/archotol.1990.01870010065019 [PubMed [2294943](#)]
67. Brunetti L, Song JH, Suh D, et al. The risk of vancomycin toxicity in patients with liver impairment. *Ann Clin Microbiol Antimicrob*. 2020;19(1):13. doi:10.1186/s12941-020-00354-2 [PubMed [32234065](#)]
68. Bruniera FR, Ferreira FM, Saviolli LR, et al. The use of vancomycin with its therapeutic and adverse effects: a review. *Eur Rev Med Pharmacol Sci*. 2015;19(4):694-700. [PubMed [25753888](#)]
69. Buck ML, "Pharmacokinetic Changes During Extracorporeal Membrane Oxygenation: Implications for Drug Therapy of Neonates," *Clin Pharmacokinet*, 2003, 42(5):403-17. [PubMed [12739981](#)]
70. Buck ML, "Vancomycin Pharmacokinetics in Neonates Receiving Extracorporeal Membrane Oxygenation," *Pharmacotherapy*, 1998, 18(5):1082-6. [PubMed [9758319](#)]

71. Butler M, Olson A, Drekonja D, et al; Agency for Healthcare Research and Quality (US). Early diagnosis, prevention, and treatment of *Clostridium difficile*: update. Comparative effectiveness review No. 172. <https://www.ncbi.nlm.nih.gov/books/NBK361173/>. Published March 2016. [PubMed 27148613]
72. Cacoub P, Musette P, Descamps V, et al. The DRESS syndrome: a literature review. *Am J Med*. 2011;124(7):588-597. doi:10.1016/j.amjmed.2011.01.017 [PubMed 21592453]
73. Capparelli EV, Lane JR, Romanowski GL, et al, "The Influences of Renal Function and Maturation on Vancomycin Elimination in Newborns and Infants," *J Clin Pharmacol*, 2001, 41(9):927-34. [PubMed 11549096]
74. Carmichael AJ, Al-Zahawi MF. Pancytopenia associated with vancomycin. *Br Med J*. 1986;293:1103.
75. Carreno JJ, Kenney RM, Lomaestro B. Vancomycin-associated renal dysfunction: where are we now?. *Pharmacotherapy*. 2014;34(12):1259-1268. doi:10.1002/phar.1488 [PubMed 25220436]
76. Centers for Disease Control and Prevention (CDC). Prevention of perinatal group B streptococcal disease, revised guidelines from CDC, 2010. *MMWR Recomm Rep*. 2010; 59(RR-10):1-32. [PubMed 21088663]
77. Centers for Disease Control and Prevention (CDC). FAQs for clinicians about *C. diff*. <https://www.cdc.gov/c-diff/hcp/clinical-overview/>. March 27, 2020. Accessed May 6, 2020.
78. Chang D, "Influence of Malignancy on the Pharmacokinetics of Vancomycin in Infants and Children," *Pediatr Infect Dis J*, 1995, 14(8):667-73. [PubMed 8532423]
79. Chang D, Liem L, and Malogolowkin M, "A Prospective Study of Vancomycin Pharmacokinetics and Dosage Requirements in Pediatric Cancer Patients," *Pediatr Infect Dis J*, 1994, 13(11):969-74. [PubMed 7845750]
80. Changela A, Javaiya H, Rickenback K, Elnawawi A, Changela K. Toxic epidermal necrolysis after vancomycin use: a case report and discussion of management. *Am J Ther*. 2013;20(2):223-225. doi:10.1097/MJT.0b013e3181f94c27 [PubMed 21273948]
81. Chenhsu RY, Hall BA, Tran H, et al. Vancomycin area under the curve to minimum inhibitory concentration ratio for treatment effectiveness in pediatric and neonatal staphylococcal infections: a systematic review. *J Pediatr Pharmacol Ther*. 2025;30(1):52-64. doi:10.5863/1551-6776-30.1.52 [PubMed 39935555]
82. Chopra N, Oppenheimer J, Derimanov GS, Fine PL. Vancomycin anaphylaxis and successful desensitization in a patient with end stage renal disease on hemodialysis by maintaining steady antibiotic levels. *Ann Allergy Asthma Immunol*. 2000;84(6):633-635. doi:10.1016/S1081-1206(10)62416-7 [PubMed 10875494]
83. Chu VH. Staphylococcal toxic shock syndrome. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed March 23, 2021.
84. Cies JJ, Moore WS 2nd, Nichols K, Knoderer CA, Carella DM, Chopra A. Population pharmacokinetics and pharmacodynamic target attainment of vancomycin in neonates on extracorporeal life support. *Pediatr Crit Care Med*. 2017;18(10):977-985. [PubMed 28650363]
85. Collins CE, Ayturk MD, Flahive JM, Emhoff TA, Anderson FA Jr, Santry HP. Epidemiology and outcomes of community-acquired *Clostridium difficile* infections in Medicare beneficiaries. *J Am Coll Surg*. 2014;218(6):1141-1147.e1. doi:10.1016/j.jamcollsurg.2014.01.053 [PubMed 24755188]
86. Comce MH, Weersink RA, Beuers U, van Hest RM, Lantinga MA. Pharmacokinetics of ceftriaxone, gentamicin, meropenem and vancomycin in liver cirrhosis: a systematic review. *J Antimicrob Chemother*. 2024;79(11):2750-2761. doi:10.1093/jac/dkac310 [PubMed 39289819]
87. Cook AM, Mieure KD, Owen RD, Pesaturo AB, Hatton J. Intracerebroventricular administration of drugs. *Pharmacotherapy*. 2009;29(7):832-845. [PubMed 19558257]

88. Covvey JR, Erickson O, Fiumara D, et al. Comparison of vancomycin area-under-the-curve dosing versus trough target-based dosing in obese and nonobese patients with methicillin-resistant *Staphylococcus aureus* bacteremia. *Ann Pharmacother*. 2020;54(7):644-651. doi:10.1177/1060028019897100 [PubMed [31888350](#)]
89. Crass RL, Dunn R, Hong J, Krop LC, Pai MP. Dosing vancomycin in the super obese: less is more. *J Antimicrob Chemother*. 2018;73(11):3081-3086. doi:10.1093/jac/dky310 [PubMed [30203073](#)]
90. Craycraft ME, Arunakul VL, Humeniuk JM. Probable vancomycin-associated toxic epidermal necrolysis. *Pharmacotherapy*. 2005;25(2):308-312. doi:10.1592/phco.25.2.308.56953 [PubMed [15767246](#)]
91. Crew P, Heintz SJ, Heintz BH. Vancomycin dosing and monitoring for patients with end-stage renal disease receiving intermittent hemodialysis. *Am J Health Syst Pharm*. 2015;72(21):1856-1864. doi:10.2146/ajhp150051 [PubMed [26490819](#)]
92. Cristallini S, Hites M, Kabtouri H, et al. New regimen for continuous infusion of vancomycin in critically ill patients. *Antimicrob Agents Chemother*. 2016;60(8):4750-4756. doi:10.1128/AAC.00330-16 [PubMed [27216073](#)]
93. Dao K, Guidi M, André P, et al. Optimisation of vancomycin exposure in neonates based on the best level of evidence. *Pharmacol Res*. 2020;154:104278. doi:10.1016/j.phrs.2019.104278 [PubMed [31108184](#)]
94. David V, Christou N, Etienne P, et al. Extravasation of noncytotoxic drugs. *Ann Pharmacother*. 2020;54(8):804-814. doi:10.1177/1060028020903406 [PubMed [32054312](#)]
95. de Hoog M, Mouton JW, van den Anker JN. Vancomycin: pharmacokinetics and administration regimens in neonates. *Clin Pharmacokinet*. 2004;43(7):417-440. [PubMed [15139793](#)]
96. De Vriese AS, Vandecasteele SJ. Vancomycin: the tale of the vanquisher and the pyrrhic victory. *Perit Dial Int*. 2014;34(2):154-161. doi:10.3747/pdi.2014.00001 [PubMed [24676741](#)]
97. Debast SB, Bauer MP, Kuijper EJ; European Society of Clinical Microbiology and Infectious Diseases. European Society of Clinical Microbiology and Infectious Diseases: update of the treatment guidance document for *Clostridium difficile* infection. *Clin Microbiol Infect*. 2014;20(suppl 2):1-26. doi: 10.1111/1469-0691.12418 [PubMed [24118601](#)]
98. Delgado V, Ajmone Marsan N, de Waha S, et al; ESC Scientific Document Group. 2023 ESC guidelines for the management of endocarditis. *Eur Heart J*. 2023;44(39):3948-4042. doi:10.1093/eurheartj/ehad193 [PubMed [37622656](#)]
99. Denaburg M, Patel S. Salvage antibiotic-lock therapy in critically ill pediatric patients: a pharmacological review for pediatric intensive care unit nurses. *AACN Adv Crit Care*. 2013;24(3):233-240. [PubMed [23880743](#)]
100. Deshpande A, Pasupuleti V, Thota P, et al. Community-associated *Clostridium difficile* infection and antibiotics: a meta-analysis. *J Antimicrob Chemother*. 2013;68(9):1951-1961. doi:10.1093/jac/dkt129 [PubMed [23620467](#)]
101. di Fonzo H, Villegas Gutsch M, Castroagudin A, Cabrera MV, Mazzei ME, Rueda D. Agranulocytosis induced by vancomycin. Case report and literature review. *Am J Case Rep*. 2018;19:1053-1056. doi:10.12659/AJCR.909956 [PubMed [30174327](#)]
102. Domis MJ, Moritz ML. Red man syndrome following intraperitoneal vancomycin in a child with peritonitis. *Front Pediatr*. 2014;2:55. doi:10.3389/fped.2014.00055 [PubMed [24926475](#)]
103. Donaldson N, Sanders J, Child J, Parker S. Acute hematogenous bacterial osteoarticular infections in children. *Pediatr Rev*. 2020;41(3):120-136. doi:10.1542/pir.2018-0201 [PubMed [32123023](#)]

104. Dos Santos ACML, Hernandez RT, Montelli AC, et al. Clinical and microbiological factors predicting outcomes of nonfermenting gram-negative bacilli peritonitis in peritoneal dialysis. *Sci Rep*. 2021;11(1):12248. doi:10.1038/s41598-021-91410-0 [PubMed [34112833](#)]
105. Downes KJ, Hayes M, Fitzgerald JC, et al. Mechanisms of antimicrobial-induced nephrotoxicity in children. *J Antimicrob Chemother*. 2020;75(1):1-13. doi:10.1093/jac/dkz325 [PubMed [31369087](#)]
106. Drew RH, Sakoulas G. Vancomycin: Parenteral dosing, monitoring, and adverse effects in adults. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed December 28, 2020.
107. Duff JM, Moreb JS, Muwalla F. Severe neutropenia following a prolonged course of vancomycin that progressed to agranulocytosis with drug reexposure. *Ann Pharmacother*. 2012;46(1):e1. doi:10.1345/aph.1Q467 [PubMed [22170976](#)]
108. Durand M. Bacterial endophthalmitis. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed May 7, 2020.
109. Earwood JS, Walker TR, Sue GJC. Septic arthritis: diagnosis and treatment. *Am Fam Physician*. 2021;104(6):589-597. [PubMed [34913662](#)]
110. Elyasi S, Khalili H, Dashti-Khavidaki S, Mohammadpour A. Vancomycin-induced nephrotoxicity: mechanism, incidence, risk factors and special populations. A literature review. *Eur J Clin Pharmacol*. 2012;68(9):1243-1255. doi:10.1007/s00228-012-1259-9 [PubMed [22411630](#)]
111. Endophthalmitis Vitrectomy Study Group. Results of the Endophthalmitis Vitrectomy Study. A randomized trial of immediate vitrectomy and of intravenous antibiotics for the treatment of postoperative bacterial endophthalmitis. *Arch Ophthalmol*. 1995;113(12):1479-1496. [PubMed [7487614](#)]
112. Ensom MH, Decarie D, and Lakhani A, "Stability of Vancomycin 25 mg/mL in Ora-Sweet and Water in Unit-Dose Cups and Plastic Bottles at 4°C and 25°C," *Can J Hosp Pharm* 2010, 63(5):366-72. [PubMed [22479004](#)]
113. Evans L, Rhodes A, Alhazzani W, et al. Surviving sepsis campaign: international guidelines for management of sepsis and septic shock 2021. *Intensive Care Med*. 2021;47(11):1181-1247. doi:10.1007/s00134-021-06506-y [PubMed [34599691](#)]
114. Expert opinion. Senior Enteral Feeding Tube Editorial Team: Joseph I. Boullata, PharmD, RPh, CNS-S, FASPEN, FACN; Peggi A. Guenter PhD, RN, FASPEN; Kathleen Gura, PharmD, BCNSP, FASHP, FASPEN, FPPA, FMSHP; Mark G. Klang MS, RPh, BCNSP, PhD, FASPEN; Linda Lord, NP, ACNP-BC, CNSC, FASPEN.
115. Expert opinion. Senior Hepatic Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.
116. Expert opinion. Senior Obesity Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.
117. Expert opinion. Senior Renal Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.
118. File TM Jr. Treatment of community-acquired pneumonia in adults who require hospitalization. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed September 28, 2021.
119. Filippone EJ, Kraft WK, Farber JL. The nephrotoxicity of vancomycin. *Clin Pharmacol Ther*. 2017;102(3):459-469. doi:10.1002/cpt.726 [PubMed [28474732](#)]

120. Fiorito TM, Luther MK, Dennehy PH, LaPlante KL, Matson KL. Nephrotoxicity with vancomycin in the pediatric population: a systematic review and meta-analysis. *Pediatr Infect Dis J*. 2018;37(7):654-661. doi:10.1097/INF.0000000000001882 [PubMed [29280786](#)]
121. *Firvanq (vancomycin hydrochloride) [prescribing information]*. Wilmington, MA: Azurity Pharmaceuticals; January 2021.
122. *Firvanq (vancomycin hydrochloride) [prescribing information]*. Wilmington, MA: Azurity Pharmaceuticals; December 2021.
123. Flume PA, Mogayzel PJ Jr, Robinson KA, et al; Clinical Practice Guidelines for Pulmonary Therapies Committee. Cystic fibrosis pulmonary guidelines: treatment of pulmonary exacerbations. *Am J Respir Crit Care Med*. 2009;180(9):802-808. doi: 10.1164/rccm.200812-1845PP. [PubMed [19729669](#)]
124. Forouzesh A, Moise PA, Sakoulas G. Vancomycin ototoxicity: a reevaluation in an era of increasing doses. *Antimicrob Agents Chemother*. 2009;53(2):483-486. doi:10.1128/AAC.01088-08 [PubMed [19001107](#)]
125. Forrence EA, Goldman MP. Vancomycin-associated exfoliative dermatitis. *DICP*. 1990;24(4):369-371. doi:10.1177/106002809002400405 [PubMed [2139270](#)]
126. Freiman JP, Graham DJ, Reed TG, McGoodwin EB. Chemical peritonitis following the intraperitoneal administration of vancomycin. *Perit Dial Int*. 1992;12(1):57-60. [PubMed [1543783](#)]
127. Frymoyer A, Guglielmo BJ, Hersh AL. Desired vancomycin trough serum concentration for treating invasive methicillin-resistant Staphylococcal infections. *Pediatr Infect Dis J*. 2013;32(10):1077-1079. [PubMed [23652479](#)]
128. Frymoyer A, Hersh AL, Benet LZ, Guglielmo BJ. Current recommended dosing of vancomycin for children with invasive methicillin-resistant Staphylococcus aureus infections is inadequate. *Pediatr Infect Dis J*. 2009;28(5):398-402. doi: 10.1097/INF.0b013e3181906e40. [PubMed [19295465](#)]
129. Frymoyer A, Hersh AL, El-Komy MH, et al. Association between vancomycin trough concentration and area under the concentration-time curve in neonates. *Antimicrob Agents Chemother*. 2014;58(11):6454-6461. [PubMed [25136027](#)]
130. Frymoyer A, Stockmann C, Hersh AL, Goswami S, Keizer RJ. Individualized empiric vancomycin dosing in neonates using a model-based approach. *J Pediatric Infect Dis Soc*. 2019;8(2):97-104. [PubMed [29294072](#)]
131. Gan IM, van Dissel JT, Beekhuis H, Swart W, van Meurs JC. Intravitreal vancomycin and gentamicin concentrations in patients with postoperative endophthalmitis. *Br J Ophthalmol*. 2001;85(11):1289-1293. [PubMed [11673290](#)]
132. Genuini M, Oualha M, Bouazza N, et al. Achievement of therapeutic vancomycin exposure with continuous infusion in critically ill children. *Pediatr Crit Care Med*. 2018;19(6):e263-e269. doi:10.1097/PCC.0000000000001474 [PubMed [29394210](#)]
133. Gilmore ES, Friedman JS, Morrell DS. Extensive fixed drug eruption secondary to vancomycin. *Pediatr Dermatol*. 2004;21(5):600-602. doi:10.1111/j.0736-8046.2004.21516.x [PubMed [15461771](#)]
134. Girand HL. Continuous infusion vancomycin in pediatric patients: a critical review of the evidence. *J Pediatr Pharmacol Ther*. 2020;25(3):198-214. doi:10.5863/1551-6776-25.3.198 [PubMed [32265603](#)]
135. Goldenberg DL, Sexton DJ. Septic arthritis of native joints in adults. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <https://www.uptodate.com>. Accessed August 13, 2025.
136. Golightly LK, Teitelbaum I, Kiser TH, et al, eds. *Renal Pharmacotherapy*. Springer Science; 2013.

137. Goodpasture HC, Dolan PJ Jr, Jacobs ER, Meredith WT. Pseudomembranous colitis & antibiotics. *Kans Med.* 1986;87(5):133-146. [PubMed [3712916](#)]
138. Goss CH, Heltshe SL, West NE, et al; STOP2 Investigators. A randomized clinical trial of antimicrobial duration for cystic fibrosis pulmonary exacerbation treatment. *Am J Respir Crit Care Med.* 2021;204(11):1295-1305. doi:10.1164/rccm.202102-0461OC [PubMed [34469706](#)]
139. Gottlieb M, Long B, Koyfman A. The evaluation and management of toxic shock syndrome in the emergency department: a review of the literature. *J Emerg Med.* 2018;54(6):807-814. doi: 10.1016/j.jemermed.2017.12.048. [PubMed [29366615](#)]
140. Gould FK, Denning DW, Elliott TS, et al; Working Party of the British Society for Antimicrobial Chemotherapy. Guidelines for the diagnosis and antibiotic treatment of endocarditis in adults: a report of the Working Party of the British Society for Antimicrobial Chemotherapy [published correction appears in *J Antimicrob Chemother.* 2012;67(5):1304]. *J Antimicrob Chemother.* 2012;67(2):269-289. [PubMed [22086858](#)]
141. Guilhaumou R, Marsot A, Dupouey J, et al. Pediatric patients with solid or hematological tumor disease: vancomycin population pharmacokinetics and dosage optimization. *Ther Drug Monit.* 2016;38(5):559-566. doi:10.1097/FTD.0000000000000318 [PubMed [27631462](#)]
142. Gupta S, Sharma S, Menon N, Ahuja S, Dahdouh M. Case report of vancomycin-induced pancytopenia. *Rev Soc Bras Med Trop.* 2016;49(2):258-259. doi:10.1590/0037-8682-0263-2015 [PubMed [27192600](#)]
143. Gwee A, Cranswick N, McMullan B, et al. Defining target vancomycin trough concentrations for treating *Staphylococcus aureus* infection in infants aged 0 to 90 days. *JAMA Pediatr.* 2019;173(8):791-793. doi:10.1001/jamapediatrics.2019.1488 [PubMed [31180478](#)]
144. Gwee A, Duffull SB, Daley AJ, et al. Identifying a therapeutic target for vancomycin against staphylococci in young infants. *J Antimicrob Chemother.* 2022;77(3):704-710. doi:10.1093/jac/dkab469 [PubMed [35037934](#)]
145. Hanberger H, Nilsson LE, Maller R, Isaksson B. Pharmacodynamics of daptomycin and vancomycin on *Enterococcus faecalis* and *Staphylococcus aureus* demonstrated by studies of initial killing and postantibiotic effect and influence of Ca²⁺ and albumin on these drugs. *Antimicrob Agents Chemother.* 1991;35(9):1710-1716. doi:10.1128/aac.35.9.1710 [PubMed [1659305](#)]
146. Hassaballa H, Mallick N, Orlowski J. Vancomycin anaphylaxis in a patient with vancomycin-induced red man syndrome. *Am J Ther.* 2000;7(5):319-320. doi:10.1097/00045391-200007050-00010 [PubMed [11317179](#)]
147. Healy DP, Sahai JV, Fuller SH, Polk RE. Vancomycin-induced histamine release and "red man syndrome": comparison of 1- and 2-hour infusions. *Antimicrob Agents Chemother.* 1990;34(4):550-554. [PubMed [1693055](#)]
148. Hecht SM, Ardura MI, Yildiz VO, Ouellette CP. Central venous catheter management in high-risk children with bloodstream infections. *Pediatr Infect Dis J.* 2020;39(1):17-22. [PubMed [31725118](#)]
149. Hecht JR, Olinger EJ. Clostridium difficile colitis secondary to intravenous vancomycin. *Dig Dis Sci.* 1989;34(1):148-149. doi:10.1007/BF01536172 [PubMed [2910675](#)]
150. Hedge DD, Strain JD, Heins JR, Farver DK. New advances in the treatment of Clostridium difficile infection (CDI). *Ther Clin Risk Manag.* 2008;4(5):949-964. doi:10.2147/tcrm.s3145 [PubMed [19209277](#)]
151. Heil EL, Claeys KC, Mynatt RP, et al. Making the change to area under the curve-based vancomycin dosing. *Am J Health Syst Pharm.* 2018;75(24):1986-1995. doi: 10.2146/ajhp180034. [PubMed [30333114](#)]
152. Heintz BH, Matzke GR, Dager WE. Antimicrobial dosing concepts and recommendations for critically ill adult patients receiving continuous renal replacement therapy or intermittent hemodialysis. *Pharmacotherapy.* 2009;29(5):562-577. [PubMed [19397464](#)]

153. Hensgens MP, Goorhuis A, Dekkers OM, Kuijper EJ. Time interval of increased risk for *Clostridium difficile* infection after exposure to antibiotics. *J Antimicrob Chemother.* 2012;67(3):742-748. doi:10.1093/jac/dkr508 [PubMed [22146873](#)]
154. Hepner DL, Castells MC. Anaphylaxis during the perioperative period. *Anesth Analg.* 2003;97(5):1381-1395. doi:10.1213/01.ane.0000082993.84883.7d [PubMed [14570656](#)]
155. Hill LF, Clements MN, Turner MA, et al. Optimised versus standard dosing of vancomycin in infants with Gram-positive sepsis (NeoVanc): a multicentre, randomised, open-label, phase 2b, non-inferiority trial. *Lancet Child Adolesc Health.* 2022;6(1):49-59. doi:10.1016/S2352-4642(21)00305-9 [PubMed [34843669](#)]
156. Holmes NE, Turnidge JD, Munckhof WJ, et al. Vancomycin AUC/MIC ratio and 30-day mortality in patients with *staphylococcus aureus* bacteremia. *Antimicrob Agents Chemother.* 2013;57(4):1654-1663. doi:10.1128/AAC.01485-12 [PubMed [23335735](#)]
157. Hong J, Krop LC, Johns T, Pai MP. Individualized vancomycin dosing in obese patients: a two-sample measurement approach improves target attainment. *Pharmacotherapy.* 2015;35(5):455-463. doi:10.1002/phar.1588 [PubMed [26011138](#)]
158. Hsiao SH, Chou CH, Lin WL, et al. High risk of cross-reactivity between vancomycin and sequential teicoplanin therapy. *J Clin Pharm Ther.* 2012;37(3):296-300. doi:10.1111/j.1365-2710.2011.01291.x [PubMed [22017186](#)]
159. Hung YP, Lee NY, Chang CM, et al. Tolerability of teicoplanin in 117 hospitalized adults with previous vancomycin-induced fever, rash, or neutropenia: a retrospective chart review. *Clin Ther.* 2009;31(9):1977-1986. doi:10.1016/j.clinthera.2009.09.010 [PubMed [19843487](#)]
160. Hurst AL, Baumgartner C, MacBrayne CE, Child J. Experience with continuous infusion vancomycin dosing in a large pediatric hospital. *J Pediatric Infect Dis Soc.* 2019;8(2):174-179. doi:10.1093/jpids/piy032 [PubMed [29718415](#)]
161. Iwata K, Doi A, Fukuchi T, et al. A systematic review for pursuing the presence of antibiotic associated enterocolitis caused by methicillin resistant *Staphylococcus aureus*. *BMC Infect Dis.* 2014;14:247-259. [PubMed [24884581](#)]
162. Jacqz-Aigrain E, Leroux S, Thomson AH, et al. Population pharmacokinetic meta-analysis of individual data to design the first randomized efficacy trial of vancomycin in neonates and young infants. *J Antimicrob Chemother.* 2019;74(8):2128-2138. doi:10.1093/jac/dkz158 [PubMed [31049551](#)]
163. Jha P, Swanson K, Stromich J, Michalski BM, Olasz E. A rare case of vancomycin-induced linear immunoglobulin a bullous dermatosis. *Case Rep Dermatol Med.* 2017;2017:7318305. doi:10.1155/2017/7318305 [PubMed [28168063](#)]
164. Johnson S, Lavergne V, Skinner AM, et al. Clinical practice guideline by the Infectious Diseases Society of America (IDSA) and Society for Healthcare Epidemiology of America (SHEA): 2021 focused update guidelines on management of *Clostridioides difficile* infection in adults. *Clin Infect Dis.* 2021;73(5):e1029-e1044. doi:10.1093/cid/ciab549 [PubMed [34164674](#)]
165. Jumah MTB, Vasoo S, Menon SR, De PP, Neely M, Teng CB. Pharmacokinetic/pharmacodynamic determinants of vancomycin efficacy in Enterococcal bacteremia. *Antimicrob Agents Chemother.* 2018;62(3). doi:10.1128/AAC.01602-17 [PubMed [29263057](#)]
166. Justo JA, Bookstaver PB. Antibiotic lock therapy: review of technique and logistical challenges. *Infect Drug Resist.* 2014;7:343-363. [PubMed [25548523](#)]

167. Kalil AC, Metersky ML, Klompas M, et al. Management of adults with hospital-acquired and ventilator-associated pneumonia: 2016 clinical practice guidelines by the Infectious Diseases Society of America and the American Thoracic Society [published online July 14, 2016]. *Clin Infect Dis*. doi: 10.1093/cid/ciw353. [PubMed [27418577](#)]
168. Kelly CP, Lamont JT, Bakken JS. *Clostridioides difficile* in adults: treatment and prevention. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <https://www.uptodate.com>. Accessed May 28, 2025.
169. Kelly CR, Fischer M, Allegretti JR, et al. ACG clinical guidelines: prevention, diagnosis, and treatment of *Clostridioides difficile* infections. *Am J Gastroenterol*. 2021;116(6):1124-1147. doi:10.14309/ajg.0000000000001278 [PubMed [34003176](#)]
170. Khicher S, Weinberger JJ. Vancomycin-associated erythema multiforme. *Am J Ther*. 2019. doi:10.1097/MJT.0000000000001038 [PubMed [31356340](#)]
171. Kim AJ, Lee JY, Choi SA, Shin WG. Comparison of the pharmacokinetics of vancomycin in neurosurgical and non-neurosurgical patients. *Int J Antimicrob Agents*. 2016;48(4):381-387. doi:10.1016/j.ijantimicag.2016.06.022 [PubMed [27546217](#)]
172. Kim BK, Kim JH, Sohn KH, Kim JY, Chang YS, Kim SH. Incidence of teicoplanin adverse drug reactions among patients with vancomycin-associated adverse drug reactions and its risk factors. *Korean J Intern Med*. 2020;35(3):714-722. doi:10.3904/kjim.2018.404 [PubMed [31722513](#)]
173. Kim YR, Chun HJ, Heo JY, et al. Optimizing vancomycin area under the concentration-time curve targets in enterococcal bacteremia: balancing efficacy and nephrotoxicity. *Yonsei Med J*. 2025;66(9):609-617. doi:10.3349/ymj.2024.0513. [PubMed [40873148](#)]
174. King MA, Cross SJ, Morton TH, et al. Evaluation of continuous infusion vancomycin in a pediatric hematology/oncology population. *Pediatr Infect Dis J*. 2024;43(6):520-524. doi:10.1097/INF.0000000000004278 [PubMed [38359358](#)]
175. Klang MG. Developing guidance for feeding tube administration of oral medications. *JPEN J Parenter Enteral Nutr*. 2023;47(4):519-540. doi:10.1002/jpen.2490 [PubMed [36847617](#)]
176. Klibanov OM, Filicko JE, DeSimone JA Jr, Tice DS. Sensorineural hearing loss associated with intrathecal vancomycin. *Ann Pharmacother*. 2003;37(1):61-65. doi:10.1345/aph.1C145 [PubMed [12503934](#)]
177. Konvinse KC, Trubiano JA, Pavlos R, et al. HLA-A*32:01 is strongly associated with vancomycin-induced drug reaction with eosinophilia and systemic symptoms. *J Allergy Clin Immunol*. 2019;144(1):183-192. doi:10.1016/j.jaci.2019.01.045 [PubMed [30776417](#)]
178. Kullar R, Davis SL, Levine DP, Rybak MJ. Impact of vancomycin exposure on outcomes in patients with methicillin-resistant *Staphylococcus aureus* bacteremia: support for consensus guidelines suggested targets. *Clin Infect Dis*. 2011;52(8):975-981. doi:10.1093/cid/cir124 [PubMed [21460309](#)]
179. Kupstaite R, Baranauskaite A, Pileckyte M, Sveikata A, Kadusevicius E, Muckiene G. Severe vancomycin-induced anaphylactic reaction. *Medicina (Kaunas)*. 2010;46(1):30-33. [PubMed [20234160](#)]
180. Langton MM, Ahern JW, MacDougall J. An AUC target simulation for vancomycin in patients with class III obesity. *J Pharm Pract*. Published online November 10, 2019. doi:10.1177/0897190019885241 [PubMed [31709893](#)]
181. LaPlante KL, Mermel LA. In vitro activity of daptomycin and vancomycin lock solutions on staphylococcal biofilms in a central venous catheter model. *Nephrol Dial Transplant*. 2007;22(8):2239-2246. [PubMed [17403700](#)]

182. Launay-Vacher V, Izzedine H, Mercadal L, Deray G. Clinical review: use of vancomycin in haemodialysis patients. *Crit Care*. 2002;6(4):313-316. [PubMed [12225605](#)]
183. Le J, Bradley JS, Murray W, et al. Improved vancomycin dosing in children using area under the curve exposure. *Pediatr Infect Dis J*. 2013;32(4):e155-163. [PubMed [23340565](#)]
184. Le J, Ny P, Capparelli E, et al. Pharmacodynamic characteristics of nephrotoxicity associated with vancomycin use in children. *J Pediatric Infect Dis Soc*. 2015;4(4):e109-e116. doi:10.1093/jpids/piu110 [PubMed [26582878](#)]
185. Lee JY, Ko KS, Peck KR, Oh WS, Song JH. In vitro evaluation of the antibiotic lock technique (ALT) for the treatment of catheter-related infections caused by staphylococci. *J Antimicrob Chemother*. 2006;57(6):1110-1115. [PubMed [16556639](#)]
186. Leonard MB, Koren G, Stevenson DK, et al. "Vancomycin Pharmacokinetics in Very Low Birth Weight Neonates," *Pediatr Infect Dis J*, 1989, 8(5):282-6. [PubMed [2657617](#)]
187. Levy JH, Marty AT. Vancomycin and adverse drug reactions. *Crit Care Med*. 1993;21(8):1107-1108. doi:10.1097/00003246-199308000-00002 [PubMed [7687945](#)]
188. Li PK, Chow KM, Cho Y, et al. ISPD peritonitis guideline recommendations: 2022 update on prevention and treatment. *Perit Dial Int*. 2022;42(2):110-153. doi:10.1177/08968608221080586 [PubMed [35264029](#)]
189. Limelette A, Tromeur T, Rhaïem R, Bonnet M, N'Guyen Y. No impact of vancomycin MIC, AUC, or AUC/MIC in enterococcus faecium bacteremia. *Fundam Clin Pharmacol*. 2025;39(5):e70039. doi:10.1111/fcp.70039 [PubMed [40808241](#)]
190. Lin Wu FL, Liu SS, Yang TY, et al. A larger dose of vancomycin is required in adult neurosurgical intensive care unit patients due to augmented clearance. *Ther Drug Monit*. 2015;37(5):609-618. doi:10.1097/FTD.0000000000000187 [PubMed [25627406](#)]
191. Lin YF, Yang CH, Sindy H, et al. Severe cutaneous adverse reactions related to systemic antibiotics. *Clin Infect Dis*. 2014;58(10):1377-1385. doi:10.1093/cid/ciu126 [PubMed [24599767](#)]
192. Lin Z, Kotler DP, Schlievert PM, Sordillo EM. Staphylococcal enterocolitis: forgotten but not gone? *Dig Dis Sci*. 2010;55(5):1200-1207. [PubMed [19609675](#)]
193. Liu C, Bayer A, Cosgrove SE, et al. Clinical practice guidelines by the Infectious Diseases Society of America for the treatment of methicillin-resistant *Staphylococcus aureus* infections in adults and children: executive summary. *Clin Infect Dis*. 2011;52(3):285-292. doi:10.1093/cid/cir034 [PubMed [21217178](#)]
194. Lodise TP, Drusano GL, Zasowski E, et al. Vancomycin exposure in patients with methicillin-resistant *Staphylococcus aureus* bloodstream infections: how much is enough? *Clin Infect Dis*. 2014;59(5):666-675. doi:10.1093/cid/ciu398 [PubMed [24867791](#)]
195. Lodise TP, Lomaestro B, Graves J, Drusano GL. Larger vancomycin doses (at least four grams per day) are associated with an increased incidence of nephrotoxicity. *Antimicrob Agents Chemother*. 2008;52(4):1330-1336. doi: 10.1128/AAC.01602-07. [PubMed [18227177](#)]
196. Lodise TP, Patel N, Lomaestro BM, Rodvold KA, Drusano GL. Relationship between initial vancomycin concentration-time profile and nephrotoxicity among hospitalized patients. *Clin Infect Dis*. 2009;49(4):507-514. doi:10.1086/600884 [PubMed [19586413](#)]
197. Lodise TP, Rosenkranz SL, Finnemeyer M, et al. The emperor's new clothes: prospective observational evaluation of the association between initial vancomycin exposure and failure rates among adult hospitalized patients with methicillin-resistant *Staphylococcus aureus* bloodstream infections (PROVIDE). *Clin Infect Dis*. 2020;70(8):1536-1545. doi:10.1093/cid/ciz460 [PubMed [31157370](#)]

198. Löwdin E, Odenholt I, Cars O. *In vitro* studies of pharmacodynamic properties of vancomycin against *Staphylococcus aureus* and *Staphylococcus epidermidis*. *Antimicrob Agents Chemother*. 1998;42(10):2739-2744. doi:10.1128/AAC.42.10.2739 [PubMed [9756787](#)]
199. Lucksiri A, Scott MK, Mueller BA, Hamburger RJ, Sowinski KM. CAHP-210 dialyzer influence on intra-dialytic vancomycin removal. *Nephrol Dial Transplant*. 2002;17(9):1649-1654. doi:10.1093/ndt/17.9.1649 [PubMed [12198218](#)]
200. Mandl DL, Garrison MW, Palpant SD. Agranulocytosis induced by vancomycin or ticarcillin/clavulanate. *Ann Pharmacother*. 1997;31(11):1321-1324. doi:10.1177/106002809703101109 [PubMed [9391687](#)]
201. Marik PE, Ferris N. Delayed hypersensitivity reaction to vancomycin. *Pharmacotherapy*. 1997;17(6):1341-1344. [PubMed [9399623](#)]
202. Marsot A, Boulamery A, Bruguerolle B, et al. Vancomycin: a review of population pharmacokinetic analyses. *Clin Pharmacokinet*. 2012;51(1):1-13. [PubMed [22149255](#)]
203. Martí R, Rosell M, Pou L, Garcia L, Pascual C. Influence of biochemical parameters of liver function on vancomycin pharmacokinetics. *Pharmacol Toxicol*. 1996;79(2):55-59. [PubMed [8878246](#)]
204. Mathews CJ, Coakley G. Septic arthritis: current diagnostic and therapeutic algorithm. *Curr Opin Rheumatol*. 2008;20(4):457-462. doi:10.1097/BOR.0b013e3283036975 [PubMed [18525361](#)]
205. Matsunaga N, Hisata K, Shimizu T. An investigation into the vancomycin concentration in the cerebrospinal fluid due to vancomycin intraventricular administration in newborns: a study of 13 cases. *Medicine (Baltimore)*. 2015;94(22):e922. [PubMed [26039127](#)]
206. Matzke GR, McGory RW, Halstenson CE, Keane WF. Pharmacokinetics of vancomycin in patients with various degrees of renal function. *Antimicrob Agents Chemother*. 1984;25(4):433-437. doi:10.1128/aac.25.4.433 [PubMed [6732213](#)]
207. Matzke GR, Zhanel GG, and Guay DRP, "Clinical Pharmacokinetics of Vancomycin," *Clin Pharmacokinet*, 1986, 11(4):257-82. [PubMed [3530582](#)]
208. Mawri S, Jain T, Shah J, Hurst G, Swiderek J. Vancomycin-induced acute generalized exanthematous pustulosis (AGEP) masquerading septic shock-an unusual presentation of a rare disease. *J Intensive Care*. 2015;3:47. doi:10.1186/s40560-015-0114-3 [PubMed [26561525](#)]
209. Mayhew JF, Deutsch S. Cardiac arrest following administration of vancomycin. *Can Anaesth Soc J*. 1985;32(1):65-66. doi:10.1007/BF03008541 [PubMed [3971208](#)]
210. Mazuski JE, Tessier JM, May AK, et al. The Surgical Infection Society revised guidelines on the management of intra-abdominal infection. *Surg Infect (Larchmt)*. 2017;18(1):1-76. doi:10.1089/sur.2016.261 [PubMed [28085573](#)]
211. McDonald LC, Gerding DN, Johnson S, et al. Clinical practice guidelines for *Clostridium difficile* infection in adults and children: 2017 update by the Infectious Diseases Society of America (IDSA) and Society for Healthcare Epidemiology of America (SHEA). *Clin Infect Dis*. 2018;66(7):987-994. doi:10.1093/cid/ciy149 [PubMed [29562266](#)]
212. McKamy S, Chen T, Lee M, Ambrose PJ. Evaluation of a pediatric continuous-infusion vancomycin therapy guideline. *Am J Health Syst Pharm*. 2012;69(23):2066-2071. doi:10.2146/ajhp120072 [PubMed [23172265](#)]
213. McNeil JC. Lock therapy for treatment and prevention of intravascular non-hemodialysis catheter-related infection. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed September 9, 2019.

214. Men P, Li HB, Zhai SD, Zhao RS. Association between the AUC₀₋₂₄/MIC ratio of vancomycin and its clinical effectiveness: a systematic review and meta-analysis. *PLoS One*. 2016;11(1):e0146224. doi: 10.1371/journal.pone.0146224. [PubMed [26731739](#)]
215. Merli M, Lucidi C, Di Gregorio et al. An empirical broad spectrum antibiotic therapy in health-care-associated infections improves survival in patients with cirrhosis: a randomized trial. *Hepatology*. 2016;63(5):1632-1639. doi:10.1002/hep.28332 [PubMed [26529126](#)]
216. Mermel LA, Allon M, Bouza E, et al. Clinical practice guidelines for the diagnosis and management of intravascular catheter-related infection: 2009 update by the Infectious Diseases Society of America. *Clin Infect Dis*. 2009;49(1):1-45. doi:10.1086/599376 [PubMed [19489710](#)]
217. Metlay JP, Waterer GW, Long AC, et al. Diagnosis and treatment of adults with community-acquired pneumonia. An official clinical practice guideline of the American Thoracic Society and Infectious Diseases Society of America. *Am J Respir Crit Care Med*. 2019;200(7):e45-e67. doi: 10.1164/rccm.201908-1581ST. [PubMed [31573350](#)]
218. Middleton PG, Gade EJ, Aguilera C, et al. ERS/TSANZ Task Force Statement on the management of reproduction and pregnancy in women with airways diseases. *Eur Respir J*. 2020;55(2):1901208. doi:10.1183/13993003.01208-2019 [PubMed [31699837](#)]
219. Miller EM, Peters HM. Outcomes of continuous enteral vancomycin (2022) study [written communication]. Indiana University Health; April 29, 2024.
220. Miller WR, Arias CA. Treatment of enterococcal infections. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <https://www.uptodate.com>. Accessed April 29, 2026.
221. Min Z, Garcia RR, Murillo M, Uchin JM, Bhanot N. Vancomycin-associated Henoch-Schönlein purpura. *J Infect Chemother*. 2017;23(3):180-184. doi:10.1016/j.jiac.2016.08.012 [PubMed [27681233](#)]
222. Minhas JS, Wickner PG, Long AA, Banerji A, Blumenthal KG. Immune-mediated reactions to vancomycin: A systematic case review and analysis. *Ann Allergy Asthma Immunol*. 2016;116(6):544-553. doi:10.1016/j.anai.2016.03.030 [PubMed [27156746](#)]
223. Miyazu D, Kodama N, Yamashita D, et al. DRESS syndrome caused by cross-reactivity between vancomycin and subsequent teicoplanin administration: a case report. *Am J Case Rep*. 2016;17:625-631. doi:10.12659/ajcr.899149 [PubMed [27572807](#)]
224. Moffett BS, Morris J, Galati M, Munoz F, Arian AA. Population pharmacokinetics of vancomycin in pediatric extracorporeal membrane oxygenation. *Pediatr Crit Care Med*. 2018;19(10):973-980. [PubMed [30063652](#)]
225. Mohammadi M, Jahangard-Rafsanjani Z, Sarayani A, Hadjibabaei M, Taghizadeh-Ghehi M. Vancomycin-induced thrombocytopenia: a narrative review. *Drug Saf*. 2017;40(1):49-59. doi:10.1007/s40264-016-0469-y [PubMed [27848200](#)]
226. Moise-Broder PA, Forrest A, Birmingham MC, Schentag JJ. Pharmacodynamics of vancomycin and other antimicrobials in patients with *Staphylococcus aureus* lower respiratory tract infections. *Clin Pharmacokinet*. 2004;43(13):925-942. doi:10.2165/00003088-200443130-00005 [PubMed [15509186](#)]
227. Mulla H, Pooboni S. Population pharmacokinetics of vancomycin in patients receiving extracorporeal membrane oxygenation. *Br J Clin Pharmacol*. 2005;60(3):265-275. [PubMed [16120065](#)]
228. Mulubwa M, Griesel HA, Mugabo P, Dippenaar R, van Wyk L. Assessment of vancomycin pharmacokinetics and dose regimen optimisation in preterm neonates. *Drugs R D*. 2020;20(2):105-113. doi:10.1007/s40268-020-00302-7 [PubMed [32266599](#)]

229. Murray BE, Arias CA, Nannini EC. Glycopeptides (vancomycin and teicoplanin), streptogramins (quinupristin-dalfopristin), lipopeptides (daptomycin), and lipoglycopeptides (telavancin). In: Bennett JE, Dolin R, Blaser MJ, eds. *Mandell, Douglas, and Bennett's Principles and Practice of Infectious Diseases*. 8th ed. Philadelphia, PA: Elsevier Saunders;2015:377.
230. Nagahama Y, VanBeek MJ, Greenlee JDW. Red man syndrome caused by vancomycin powder. *J Clin Neurosci*. 2018;50:149-150. doi:10.1016/j.jocn.2018.01.044 [PubMed [29398192](#)]
231. Nathanson DR, Sheahan M, Chao L, Wallack MK. Intracolonic use of vancomycin for treatment of *Clostridium difficile* colitis in a patient with a diverted colon: report of a case. *Dis Colon Rectum*. 2001;44(12):1871-1872. doi:10.1007/BF02234471 [PubMed [11742178](#)]
232. National Institute for Health and Care Excellence (NICE). Drug allergy: diagnosis and management. <https://www.nice.org.uk/guidance/cg183>. September 3, 2014. Accessed May 6, 2020.
233. Navinés-Ferrer A, Serrano-Candelas E, Lafuente A, Muñoz-Cano R, Martín M, Gastaminza G. MRGPRX2-mediated mast cell response to drugs used in perioperative procedures and anaesthesia. *Sci Rep*. 2018;8(1):11628. doi:10.1038/s41598-018-29965-8 [PubMed [30072729](#)]
234. Neely MN, Kato L, Youn G, et al. Prospective trial on the use of trough concentration versus area under the curve to determine therapeutic vancomycin dosing. *Antimicrob Agents Chemother*. 2018;62(2):e02042-17. doi:10.1128/AAC.02042-17. [PubMed [29203493](#)]
235. Ng K, Mabasa VH, Chow I, Ensom MH. Systematic review of efficacy, pharmacokinetics, and administration of intraventricular vancomycin in adults. *Neurocrit Care*. 2014;20(1):158-171. doi: 10.1007/s12028-012-9784-z. [PubMed [23090839](#)]
236. Nielsen HE, Sorensen I, and Hansen HE, "Peritoneal Transport of Vancomycin During Peritoneal Dialysis," *Nephron*, 1979, 24(6):274-7. [PubMed [514426](#)]
237. Nousari HC, Costarangos C, Anhalt GJ. Vancomycin-associated linear IgA bullous dermatosis. *Ann Intern Med*. 1998;129(6):507-508. doi:10.7326/0003-4819-129-6-199809150-00021 [PubMed [9735094](#)]
238. Nyman HA, Agarwal A, Senekjian HO, Leypoldt JK, Cheung AK. Removal of vancomycin administered during dialysis by a high-flux dialyzer. *Hemodial Int*. 2018;22(3):383-387. doi:10.1111/hdi.12637 [PubMed [29380499](#)]
239. Ocampos-Martinez E, Penaccini L, Scolletta S, et al. Determinants of early inadequate vancomycin concentrations during continuous infusion in septic patients. *Int J Antimicrob Agents*. 2012;39(4):332-337. doi:10.1016/j.ijantimicag.2011.12.008 [PubMed [22333933](#)]
240. O'Donnell E, Shepherd C, Neff A. Immune thrombocytopenia from vancomycin in orthopedic cement. *Am J Hematol*. 2007;82(12):1122. doi:10.1002/ajh.20916 [PubMed [17665503](#)]
241. Ong J, Van Gerpen R. Recommendations for management of noncytotoxic vesicant extravasations. *J Infus Nurs*. 2020;43(6):319-343. doi:10.1097/NAN.0000000000000392 [PubMed [33141794](#)]
242. Onwuchuruba CN, Towers CV, Howard BC, Hennessy MD, Wolfe L, Brown MS. Transplacental passage of vancomycin from mother to neonate. *Am J Obstet Gynecol*. 2014;210(4):352.e1-352.e4. doi: 10.1016/j.ajog.2014.01.019. [PubMed [24679944](#)]
243. Osler T, Lott D, Bordley J 4th, Lynch F, Ellsworth C, Kozak A. Cefazolin-induced pseudomembranous colitis resulting in perforation of the sigmoid colon. *Dis Colon Rectum*. 1986;29(2):140-143. doi:10.1007/BF02555402 [PubMed [3510836](#)]

244. Osmon DR, Berbari EF, Berendt AR, et al; Infectious Diseases Society of America. Diagnosis and management of prosthetic joint infection: clinical practice guideline by the Infectious Diseases Society of America. *Clin Infect Dis*. 2013;56(1):e1-e25. [PubMed [23223583](#)]
245. Otani IM, Kuhlen JL Jr, Blumenthal KG, Guyer A, Banerji A. A role for vancomycin epicutaneous skin testing in the evaluation of perioperative anaphylaxis. *J Allergy Clin Immunol Pract*. 2015;3(6):984-985. doi:10.1016/j.jaip.2015.06.017 [PubMed [26246124](#)]
246. Packer M, Chang DF, Dewey SH, et al; ASCRS Cataract Clinical Committee. Prevention, diagnosis, and management of acute postoperative bacterial endophthalmitis. *J Cataract Refract Surg*. 2011;37(9):1699-1714. [PubMed [21782382](#)]
247. Parasuraman JM, Albur M, Fellows G, Heep A. Monitoring intraventricular vancomycin for ventriculostomy access device infection in preterm infants. *Childs Nerv Syst*. 2018;34(3):473-479. [PubMed [29067501](#)]
248. Pasic M, Jost R, Carrel T, Von Segesser L, Turina M. Intracolonic vancomycin for pseudomembranous colitis. *N Engl J Med*. 1993;329(8):583. doi:10.1056/NEJM199308193290819 [PubMed [8336766](#)]
249. Pea F, Furlanut M, Negri C, et al. Prospectively validated dosing nomograms for maximizing the pharmacodynamics of vancomycin administered by continuous infusion in critically ill patients. *Antimicrob Agents Chemother*. 2009;53(5):1863-1867. doi:10.1128/AAC.01149-08 [PubMed [19223642](#)]
250. Pediatric Infectious Diseases Society (PIDS). Pediatric Infectious Diseases Society endorses new terminology: vancomycin flushing syndrome. https://pids.org/wp-content/uploads/2021/04/PIDS-Endorses-New-Terminology_Vanc-Flushing-Syn_Final_4.27.21-1.pdf. April 27, 2021. Accessed May 5, 2021.
251. Peel TN. Vertebral osteomyelitis: treatment. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <https://www.uptodate.com>. Accessed October 16, 2025.
252. Peters H, Iqbal A, Miller E, Khalid S, Rahman O. Outcomes of continuous enteral vancomycin infusion in intensive care unit patients: a novel treatment modality for severe *Clostridium difficile* colitis. *Cureus*. 2022;14(3):e22872. doi:10.7759/cureus.22872 [PubMed [35392442](#)]
253. Pettit NN, DePestel DD, Fohl AL, et al. Risk factors for systemic vancomycin exposure following administration of oral vancomycin for the treatment of *Clostridium difficile* infection. *Pharmacotherapy*. 2015;35(2):119-126. doi: 10.1002/phar.1538. [PubMed [25689243](#)]
254. Pettit RS, Peters SJ, McDade EJ, et al. Vancomycin dosing and monitoring in the treatment of cystic fibrosis: results of a National Practice Survey. *J Pediatr Pharmacol Ther*. 2017;22(6):406-411. doi: 10.5863/1551-6776-22.6.406. [PubMed [29290740](#)]
255. Pfausler B, Haring HP, Kampfl A, Wissel J, Schober M, Schmutzhard E. Cerebrospinal fluid (CSF) pharmacokinetics of intraventricular vancomycin in patients with staphylococcal ventriculitis associated with external CSF drainage. *Clin Infect Dis*. 1997;25(3):733-735. [PubMed [9314470](#)]
256. Pingili CS, Okon EE. Vancomycin-induced leukocytoclastic vasculitis and acute renal failure due to tubulointerstitial nephritis. *Am J Case Rep*. 2017;18:1024-1027. doi:10.12659/ajcr.905214 [PubMed [28943633](#)]
257. Pogue JM, DePestel DD, Kaul DR, Khaled Y, Frame DG. Systemic absorption of oral vancomycin in a peripheral blood stem cell transplant patient with severe graft-versus-host disease of the gastrointestinal tract. *Transpl Infect Dis*. 2009;11(5):467-470. doi:10.1111/j.1399-3062.2009.00426.x [PubMed [19638004](#)]
258. Radu L, Bengry T, Akierman A, Alshaikh B, Yusuf K, Dersch-Mills D. Evolution of empiric vancomycin dosing in a neonatal population. *J Perinatol*. 2018;38(12):1702-1707. [PubMed [30341404](#)]

259. Rainkie D, Ensom MH, Carr R. Pediatric assessment of vancomycin empiric dosing (PAVED): a retrospective review. *Paediatr Drugs*. 2015;17(3):245-253. [PubMed [25813682](#)]
260. Reardon J, Lau TT, Ensom MH. Vancomycin loading doses: a systematic review. *Ann Pharmacother*. 2015;49(5):557-565. doi: 10.1177/1060028015571163. [PubMed [25712445](#)]
261. Refer to manufacturer's labeling.
262. Regal RE, Ren SP, Paige G, Alaniz C. Evaluation of vancomycin dosing in patients with cirrhosis: beginning deliveries about a new nomogram. *Hosp Pharm*. 2019;54(2):125-129. doi:10.1177/0018578718772266 [PubMed [30923406](#)]
263. Reyes MP and Ostrea EM Jr, "Toxicity of Vancomycin Given During Pregnancy (Reply)," *Am J Obstet Gynecol*, 1990, 163(4 Pt 1):1376.
264. Reyes MP, Ostrea EM Jr, Cabinian AE, et al. Vancomycin during pregnancy: does it cause hearing loss or nephrotoxicity in the infant? *Am J Obstet Gynecol*. 1989;161(4):977-981. [PubMed [2801848](#)]
265. Rocha JL, Kondo W, Baptista MI, Da Cunha CA, Martins LT. Uncommon vancomycin-induced side effects. *Braz J Infect Dis*. 2002;6(4):196-200. doi:10.1590/s1413-86702002000400007 [PubMed [12204187](#)]
266. Rodvold KA, Everett JA, Pryka RD, and Kraus DM, "Pharmacokinetics and Administration Regimens of Vancomycin in Neonates, Infants and Children," *Clin Pharmacokinet*, 1997, 33(1):32-51. [PubMed [9250422](#)]
267. Rokas KE, Johnson JW, Beardsley JR, Ohl CA, Luther VP, Williamson JC. The addition of intravenous metronidazole to oral vancomycin is associated with improved mortality in critically ill patients with *Clostridium difficile* infection. *Clin Infect Dis*. 2015;61(6):934-941. doi: 10.1093/cid/civ409. [PubMed [26024909](#)]
268. Ross JJ. Septic arthritis of native joints. *Infect Dis Clin North Am*. 2017;31(2):203-218. doi:10.1016/j.idc.2017.01.001 [PubMed [28366221](#)]
269. Rybak MJ, Albrecht LM, Boike SC, et al, "Nephrotoxicity of Vancomycin, Alone and With an Aminoglycoside," *J Antimicrob Chemother*, 1990, 25(4):679-87. [PubMed [2351627](#)]
270. Rybak MJ, Boike SC. Monitoring vancomycin therapy. *Drug Intell Clin Pharm*. 1986;20(10):757-761. [PubMed [3769763](#)]
271. Rybak MJ, Le J, Lodise TP, et al. Therapeutic monitoring of vancomycin for serious methicillin-resistant *Staphylococcus aureus* infections: a revised consensus guideline and review by the American Society of Health-System Pharmacists, the Infectious Diseases Society of America, the Pediatric Infectious Diseases Society, and the Society of Infectious Diseases Pharmacists (ASHP/IDSA/SIDP). *Am J Health Syst Pharm*. 2020;77(11):835-864. doi:10.1093/ajhp/zxaa036 [PubMed [32191793](#)]
272. Rybak M, Lomaestro B, Rotschafer JC, et al. Therapeutic monitoring of vancomycin in adult patients: a consensus review of the American Society of Health-System Pharmacists, the Infectious Diseases Society of America, and the Society of Infectious Diseases Pharmacists (ASHP/IDSA/SIDP). *Am J Health Syst Pharm*. 2009;66(1):82-98. doi:10.2146/ajhp080434 [PubMed [19106348](#)]
273. Saavedra-Lozano J, Falup-Pecurariu O, Faust SN, et al. Bone and joint infections. *Pediatr Infect Dis J*. 2017;36(8):788-799. doi:10.1097/INF.0000000000001635 [PubMed [28708801](#)]
274. Saffouri G, Khanna S, Estes L, Pardi D. Outcomes from rectal vancomycin therapy in patients with *Clostridium difficile* infection. *Am J Gastroenterol*. 2014;109(6):924-925. doi:10.1038/ajg.2014.80 [PubMed [24896763](#)]
275. Sampson JH, Gea-Banacloche JC. Intracranial epidural abscess. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed April 30, 2026.

276. Sawada A, Kawanishi K, Morikawa S, et al. Biopsy-proven vancomycin-induced acute kidney injury: a case report and literature review. *BMC Nephrol.* 2018;19(1):72. doi:10.1186/s12882-018-0845-1 [PubMed [29587650](#)]
277. Sawyer RG, Claridge JA, Nathens AB, et al; STOP-IT Trial Investigators. Trial of short-course antimicrobial therapy for intraabdominal infection. *N Engl J Med.* 2015;372(21):1996-2005. doi:10.1056/NEJMoa1411162 [PubMed [25992746](#)]
278. Schaad UB, McCracken GH Jr, Nelson JD. Clinical pharmacology and efficacy of vancomycin in pediatric patients. *J Pediatr.* 1980;96(1):119-126. [PubMed [7350291](#)]
279. Schmelzer TM, Christmas AB, Norton HJ, Heniford BT, Sing RF. Vancomycin intermittent dosing versus continuous infusion for treatment of ventilator-associated pneumonia in trauma patients. *Am Surg.* 2013;79(11):1185-1190. doi:10.1177/000313481307901123 [PubMed [24165255](#)]
280. Schutze GE, Willoughby RE, Committee on Infectious Diseases, American Academy of Pediatrics. Clostridium difficile infection in infants and children. *Pediatrics.* 2013;131(1):196-200. [PubMed [23277317](#)]
281. Scott MK, Macias WL, Kraus MA, Clark WR, Carfagna MA, Mueller BA. Effects of dialysis membrane on intradialytic vancomycin administration. *Pharmacotherapy.* 1997;17(2):256-262. [PubMed [9085316](#)]
282. Sel EK, Tufan B, Atagun GK, et al. Association of vancomycin trough levels, AUC and AUC/MIC ratios with clinical outcomes in patients with enterococcal bacteremia: a prospective cohort study. *BMC Infect Dis.* 2025;25(1):958. doi:10.1186/s12879-025-11400-9. [PubMed [40731327](#)]
283. Senneville É, Albalawi Z, van Asten SA, et al. IWGDF/IDSA guidelines on the diagnosis and treatment of diabetes-related foot infections (IWGDF/IDSA 2023). *Diabetes Metab Res Rev.* 2024;40(3):e3687. doi:10.1002/dmrr.3687 [PubMed [37779323](#)]
284. Sexton DJ, Sampson JH. Spinal epidural abscess. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed April 30, 2026.
285. Shah-Khan F, Scheetz MH, Ghossein C. Biopsy-proven acute tubular necrosis due to vancomycin toxicity. *Int J Nephrol.* 2011;2011:436856. doi:10.4061/2011/436856 [PubMed [21716699](#)]
286. Shaker MS, Wallace DV, Golden DBK, et al. Anaphylaxis-a 2020 practice parameter update, systematic review, and Grading of Recommendations, Assessment, Development and Evaluation (GRADE) analysis. *J Allergy Clin Immunol.* 2020;145(4):1082-1123. doi:10.1016/j.jaci.2020.01.017 [PubMed [32001253](#)]
287. Shane AL, Mody RK, Crump JA, et al. 2017 Infectious Diseases Society of America clinical practice guidelines for the diagnosis and management of infectious diarrhea. *Clin Infect Dis.* 2017;65(12):e45-e80. doi:10.1093/cid/cix669 [PubMed [29053792](#)]
288. Sherman DS, Fish DN, Teitelbaum I. Assessing renal function in cirrhotic patients: problems and pitfalls. *Am J Kidney Dis.* 2003;41(2):269-278. doi:10.1053/ajkd.2003.50035 [PubMed [12552488](#)]
289. Shokouhi S, Alavi Darazam I. Determination of vancomycin trough level in serum and cerebrospinal fluid of patients with acute community-acquired meningitis: a prospective study. *J Infect.* 2014;69(5):424-429. doi:10.1016/j.jinf.2014.06.010 [PubMed [24973553](#)]
290. Simon RH. Cystic fibrosis: Management of pulmonary exacerbations. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed January 20, 2022.
291. Singapura P, Ma TW, Sarmast N, et al. Estimating glomerular filtration rate in cirrhosis using creatinine-based and cystatin C-based equations: systematic review and meta-analysis. *Liver Transpl.* 2021;27(11):1538-1552. doi:10.1002/lt.26216 [PubMed [34143570](#)]

292. Sivagnanam S, Deleu D. Red man syndrome. *Crit Care*. 2003;7(2):119-120. doi:10.1186/cc1871 [PubMed [12720556](#)]
293. Smetana KS, Cook AM. Cerebrospinal fluid vancomycin dosing and monitoring: the quandary posed by guideline recommendations. *Clin Infect Dis*. 2018;67(6):980-981. doi: 10.1093/cid/ciy189. [PubMed [29518176](#)]
294. Smith PF, Taylor CT. Vancomycin-induced neutropenia associated with fever: similarities between two immune-mediated drug reactions. *Pharmacotherapy*. 1999;19(2):240-244. doi:10.1592/phco.19.3.240.30912 [PubMed [10030777](#)]
295. Solomkin JS, Mazuski JE, Bradley JS, et al. Diagnosis and management of complicated intra-abdominal infections in adults and children: guidelines by the Surgical Infection Society and the Infectious Diseases Society of America [published correction appears in *Clin Infect Dis*. 2010;50(12):1695]. *Clin Infect Dis*. 2010;50(2):133-164. [PubMed [20034345](#)]
296. Southwick FS. Treatment and prognosis of bacterial brain abscess. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <https://www.uptodate.com>. Accessed April 30, 2026.
297. Spadaro S, Berselli A, Fogagnolo A, et al. Evaluation of a protocol for vancomycin administration in critically patients with and without kidney dysfunction. *BMC Anesthesiol*. 2015;15:95. doi:10.1186/s12871-015-0065-1 [PubMed [26116239](#)]
298. Spitzer PG, Eliopoulos GM. Systemic absorption of enteral vancomycin in a patient with pseudomembranous colitis. *Ann Intern Med*. 1984;100(4):533-534. doi:10.7326/0003-4819-100-4-533 [PubMed [6703548](#)]
299. Stefanos SS, Kiser TH, MacLaren R, Mueller SW, Reynolds PM. Management of noncytotoxic extravasation injuries: a focused update on medications, treatment strategies, and peripheral administration of vasopressors and hypertonic saline. *Pharmacotherapy*. 2023;43(4):321-337. doi:10.1002/phar.2794 [PubMed [36938775](#)]
300. Stevens DL, Bisno AL, Chambers HF, et al. Practice guidelines for the diagnosis and management of skin and soft tissue infections: 2014 update by the Infectious Diseases Society of America [published online ahead of print June 18, 2014]. *Clin Infect Dis*. 2014;59(2):147-159. doi: 10.1093/cid/ciu296. [PubMed [24947530](#)]
301. Stockmann C, Hersh AL, Roberts JK, et al. Predictive performance of a vancomycin population pharmacokinetic model in neonates. *Infect Dis Ther*. 2015;4(2):187-198. [PubMed [25998107](#)]
302. Surawicz CM, Brandt LJ, Binion DG, et al. Guidelines for diagnosis, treatment, and prevention of *Clostridium difficile* infections. *Am J Gastroenterol*. 2013;108(4):478-498. doi:10.1093/cid/cix1085 [PubMed [23439232](#)]
303. Suzuki Y, Kawasaki K, Sato Y, et al. Is peak concentration needed in therapeutic drug monitoring of vancomycin? A pharmacokinetic-pharmacodynamic analysis in patients with methicillin-resistant staphylococcus aureus pneumonia. *Chemotherapy*. 2012;58(4):308-312. doi:10.1159/000343162 [PubMed [23147106](#)]
304. Symons NL, Hobbes AF, Leaver HK. Anaphylactoid reactions to vancomycin during anaesthesia: two clinical reports. *Can Anaesth Soc J*. 1985;32(2):178-181. doi:10.1007/BF03010047 [PubMed [3986655](#)]
305. Szymusiak-Mutnick BA, Ross MB. Minimizing the occurrence of red-man syndrome. *Am J Health Syst Pharm*. 1996;53(17):2098. [PubMed [8870903](#)]
306. Tande AJ. Osteomyelitis in the absence of hardware: approach to treatment in adults. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed October 16, 2025.
307. Tashima S, Konishi K, Koga H, Hashimoto T. A case of vancomycin-induced linear IgA bullous dermatosis with circulating IgA antibodies to the NC16a domain of BP180. *Int J Dermatol*. 2014;53(3):e207-e209. doi:10.1111/ijd.12047 [PubMed [23829415](#)]

308. Tetsuka S, Suzuki T, Ogawa T, Hashimoto R, Kato H. Spinal epidural abscess: a review highlighting early diagnosis and management. *JMA J.* 2020;3(1):29-40. doi:10.31662/jmaj.2019-0038 [PubMed [33324773](#)]
309. Traber PG, Levine DP. Vancomycin Ototoxicity in patient with normal renal function. *Ann Intern Med.* 1981;95(4):458-460. doi:10.7326/0003-4819-95-4-458 [PubMed [7283300](#)]
310. Trotman RL, Williamson JC, Shoemaker DM, Salzer WL. Antibiotic dosing in critically ill adult patients receiving continuous renal replacement therapy. *Clin Infect Dis.* 2005;41(8):1159-1166. [PubMed [16163635](#)]
311. Trubiano JA, Cheng AC, Korman TM, et al. Australasian Society of Infectious Diseases updated guidelines for the management of *Clostridium difficile* infection in adults and children in Australia and New Zealand. *Intern Med J.* 2016;46(4):479-493. [PubMed [27062204](#)]
312. Tsai HC, Huang LM, Chang LY, et al. Central venous catheter-associated bloodstream infections in pediatric hematology-oncology patients and effectiveness of antimicrobial lock therapy. *J Microbiol Immunol Infect.* 2015;48(6):639-646. [PubMed [25311403](#)]
313. Tseng SH, Lim CP, Chen Q, Tang CC, Kong ST, Ho PC. Evaluating the relationship between vancomycin trough concentration and 24-hour area under the concentration-time curve in neonates. *Antimicrob Agents Chemother.* 2018;62(4):e01647-17. doi:10.1128/AAC.01647-17 [PubMed [29358290](#)]
314. Tufariello JM. Infection due to coagulase-negative staphylococci: treatment. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed December 17, 2020.
315. Tunkel AR, Hartman BJ, Kaplan SL, et al. Practice guidelines for the management of bacterial meningitis. *Clin Infect Dis.* 2004;39(9):1267-1284. [PubMed [15494903](#)]
316. Tunkel AR, Hasbun R, Bhimraj A, et al. 2017 Infectious Diseases Society of America's clinical practice guidelines for healthcare-associated ventriculitis and meningitis. *Clin Infect Dis.* 2017;64(6):e34-e65. doi:10.1093/cid/ciw861 [PubMed [28203777](#)]
317. Tyzavan (vancomycin) [prescribing information]. Berkeley Heights, NJ: Hikma Pharmaceuticals USA Inc; August 2025.
318. Udy AA, Roberts JA, Boots RJ, Paterson DL, Lipman J. Augmented renal clearance: implications for antibacterial dosing in the critically ill. *Clin Pharmacokinet.* 2010;49(1):1-16. doi:10.2165/11318140-000000000-00000 [PubMed [20000886](#)]
319. University of Oklahoma Health Science Center (OUHSC). Thrombotic thrombocytopenic purpura hemolytic uremic syndrome (TTP-HUS). <https://www.ouhsc.edu/platelets/ditp.html>. May 4, 2015. Accessed May 6, 2020.
320. van Hal SJ, Paterson DL, Lodise TP. Systematic review and meta-analysis of vancomycin-induced nephrotoxicity associated with dosing schedules that maintain troughs between 15 and 20 milligrams per liter. *Antimicrob Agents Chemother.* 2013;57(2):734-744. doi:10.1128/AAC.01568-12 [PubMed [23165462](#)]
321. Van Hise NW, Bryant AM, Hennessey EK, Crannage AJ, Khoury JA, Manian FA. Efficacy of oral vancomycin in preventing recurrent *Clostridium difficile* infection in patients treated with systemic antimicrobial agents. *Clin Infect Dis.* 2016;63(5):651-653. doi:10.1093/cid/ciw401 [PubMed [27318333](#)]
322. Vancocin (vancomycin hydrochloride) [prescribing information]. Baudette, MN: ANI Pharmaceuticals Inc; October 2025.
323. Vancomycin hydrochloride 1.25 g and 1.5 g lyophilized vial [prescribing information]. Berkeley Heights, NJ: Hikma Pharmaceuticals USA Inc; February 2023.

324. *Vancomycin hydrochloride capsule [prescribing information]*. Baudette, MN: ANI Pharmaceuticals Inc; December 2021.
325. *Vancomycin hydrochloride fliptop vial 5 g and 10 g pharmacy bulk package [prescribing information]*. Berkeley Heights, NJ: Hikma Pharmaceuticals USA Inc; March 2023.
326. *Vancomycin hydrochloride fliptop vial 5 g and 10 g pharmacy bulk package [prescribing information]*. Lake Forest, IL: Hospira; October 2021.
327. *Vancomycin hydrochloride injection in galaxy container [prescribing information]*. Deerfield, IL: Baxter Healthcare Corporation; May 2025.
328. *Vancomycin hydrochloride injection [prescribing information]*. Lake Zurich, IL: Fresenius Kabi; February 2018.
329. *Vancomycin hydrochloride injection [prescribing information]*. Lake Zurich, IL: Fresenius Kabi; January 2019.
330. *Vancomycin hydrochloride injection [prescribing information]*. Berkeley Heights, NJ: Hikma Pharmaceuticals USA Inc; May 2025.
331. *Vancomycin hydrochloride injection [prescribing information]*. Buffalo Grove, IL: Xellia Pharmaceuticals USA, LLC; January 2020.
332. *Vancomycin hydrochloride injection [prescribing information]*. Berkeley Heights, NJ: Hikma Pharmaceuticals USA Inc; June 2025.
333. *Vancomycin hydrochloride injection [prescribing information]*. Cherry Hill, NJ: Samson Medical Technologies, LLC; December 2018.
334. *Vancomycin hydrochloride injection [prescribing information]*. Columbus, OH: Slate Run Pharmaceuticals; April 2020.
335. *Vancomycin hydrochloride injection [prescribing information]*. Morgantown, WV: Mylan Institutional LLC; June 2024.
336. *Vancomycin hydrochloride injection [prescribing information]*. Rockford, IL: Myland Institutional LLC; August 2019.
337. *Vancomycin hydrochloride injection [prescribing information]*. Schaumburg, IL: Sagent Pharmaceuticals; January 2018.
338. *Vancomycin hydrochloride lyophilized vial [prescribing information]*. Morgantown, WV: Mylan Pharmaceuticals; January 2021.
339. *Vancomycin hydrochloride oral solution [prescribing information]*. Bridgewater, NJ: Amneal Pharmaceuticals; February 2022.
340. Vardakas KZ, Trigkidis KK, Boukouvala E, Falagas ME. Clostridium difficile infection following systemic antibiotic administration in randomised controlled trials: a systematic review and meta-analysis. *Int J Antimicrob Agents*. 2016;48(1):1-10. doi:10.1016/j.ijantimicag.2016.03.008 [PubMed [27216385](#)]
341. Vermont Oxford Network (VON). Neonatal drug concentrations. Updated November 2022. Accessed August 10, 2023. <https://public.vtoxford.org/neonatal-drug-concentrations>
342. Verrall AJ, Llorin R, Tam VH, et al. Efficacy of continuous infusion of vancomycin for the outpatient treatment of methicillin-resistant Staphylococcus aureus infections. *J Antimicrob Chemother*. 2012;67(12):2970-2973. doi: 10.1093/jac/dks328. [PubMed [22915464](#)]

343. Visentainer L, Massuda JY, Cintra ML, Magalhães RF. Vancomycin-induced linear IgA bullous dermatosis (LABD)-an atypical presentation. *Clin Case Rep*. 2019;7(5):1091-1093. doi:10.1002/ccr3.2039 [PubMed [31110752](#)]
344. Von Drygalski A, Curtis BR, Bougie DW, et al. Vancomycin-induced immune thrombocytopenia. *N Engl J Med*. 2007;356(9):904-910. doi:10.1056/NEJMoa065066 [PubMed [17329697](#)]
345. Vu DH, Nguyen DA, Delattre IK, et al. Determination of optimal loading and maintenance doses for continuous infusion of vancomycin in critically ill patients: population pharmacokinetic modelling and simulations for improved dosing schemes. *Int J Antimicrob Agents*. 2019;54(6):702-708. doi:10.1016/j.ijantimicag.2019.09.018 [PubMed [31600554](#)]
346. Waineo MF, Kuhn TC, Brown DL. The pharmacokinetic/pharmacodynamic rationale for administering vancomycin via continuous infusion. *J Clin Pharm Ther*. 2015;40(3):259-265. doi: 10.1111/jcpt.12270. [PubMed [25865426](#)]
347. Waldman MA, Black DR, Callen JP. Vancomycin-induced linear IgA bullous disease presenting as toxic epidermal necrolysis. *Clin Exp Dermatol*. 2004;29(6):633-636. doi:10.1111/j.1365-2230.2004.01649.x [PubMed [15550142](#)]
348. Wang JT, Fang CT, Chen YC, Chang SC. Necessity of a loading dose when using vancomycin in critically ill patients. *J Antimicrob Chemother*. 2001;47(2):246. [PubMed [11157921](#)]
349. Warady BA, Bakkaloglu S, Newland J, et al. Consensus Guidelines for the Prevention and Treatment of Catheter-Related Infections and Peritonitis in Pediatric Patients Receiving Peritoneal Dialysis: 2012 Update. *Perit Dial Int*. 2012;(32)(suppl 2):S32-S86. [PubMed [22851742](#)]
350. Warady BA, Same R, Borzych-Duzalka D, et al. Clinical practice guideline for the prevention and management of peritoneal dialysis associated infections in children: 2024 update. *Perit Dial Int*. 2024;44(5):303-364. doi:10.1177/08968608241274096 [PubMed [39313225](#)]
351. Watson T, Hickok J, Fraker S, Korwek K, Poland RE, Septimus E. Evaluating the risk factors for hospital-onset *Clostridium difficile* infections in a large healthcare system. *Clin Infect Dis*. 2018;66(12):1957-1959. doi:10.1093/cid/cix1112 [PubMed [29272341](#)]
352. Wazny LD, Daghigh B. Desensitization protocols for vancomycin hypersensitivity. *Ann Pharmacother*. 2001;35(11):1458-1464. doi:10.1345/aph.1A002 [PubMed [11724099](#)]
353. White R, Bradnam V; British Pharmaceutical Nutrition Group. *Handbook of Drug Administration via Enteral Feeding Tubes*. Third ed. London, England: Pharmaceutical Press; 2015.
354. Wilhem MP, Estes L. Symposium on antimicrobial agents--Part XII. Vancomycin. *Mayo Clin Proc*. 1999;74(9):928-935. [PubMed [10488798](#)]
355. Wolf J, Allison KJ, Tang L, Sun Y, Hayden RT, Flynn PM. No evidence of benefit from antibiotic lock therapy in pediatric oncology patients with central line-related bloodstream infection: results of a retrospective matched cohort study and review of the literature. *Pediatr Blood Cancer*. 2014;61(10):1811-1815. [PubMed [24923808](#)]
356. Wolfson AR, Zhou L, Li Y, Phadke NA, Chow OA, Blumenthal KG. Drug reaction with eosinophilia and systemic symptoms (DRESS) syndrome identified in the electronic health record allergy module. *J Allergy Clin Immunol Pract*. 2019;7(2):633-640. doi:10.1016/j.jaip.2018.08.013 [PubMed [30176295](#)]
357. Woods CR, Bradley JS, Chatterjee A, et al. Clinical practice guideline by the Pediatric Infectious Diseases Society and the Infectious Diseases Society of America: 2021 guideline on diagnosis and management of acute hematogenous osteomyelitis in pediatrics. *J Pediatric Infect Dis Soc*. Published online August 5, 2021. doi:10.1093/jpids/piab027 [PubMed [34350458](#)]

358. Woods CR, Bradley JS, Chatterjee A, et al. Clinical practice guideline by the Pediatric Infectious Diseases Society (PIDS) and the Infectious Diseases Society of America (IDSA): 2023 guideline on diagnosis and management of acute bacterial arthritis in pediatrics. *J Pediatric Infect Dis Soc.* Published online November 6, 2023. doi:10.1093/jpids/piad089 [PubMed [37941444](#)]
359. World Health Organization (WHO). Breastfeeding and maternal medication, recommendations for drugs in the eleventh WHO model list of essential drugs. <https://apps.who.int/iris/handle/10665/62435>. Published 2002. [PubMed [23215911](#)]
360. Zasowski EJ, Murray KP, Trinh TD, et al. Identification of vancomycin exposure-toxicity thresholds in hospitalized patients receiving intravenous vancomycin. *Antimicrob Agents Chemother.* 2017;62(1):e01684-17. doi:10.1128/AAC.01684-17 [PubMed [29084753](#)]
361. Zylbersztajn BL, Izquierdo G, Santana RC, et al. Therapeutic drug monitoring of vancomycin in pediatric patients with extracorporeal membrane oxygenation support. *J Pediatr Pharmacol Ther.* 2018;23(4):305-310. [PubMed [30181721](#)]

Topic 10037 Version 965.0

→